LOCAL TYPE I NAVIER–STOKES REDUCTION: CONCENTRATION, SEREGIN LIMITS, AND LIOUVILLE-CLASS CLOSURES

GUILLEM DURAN BALLESTER

Abstract. We prove the concentration, compactness, and Liouville reductions used in the exclusion of local pointwise Type I Navier–Stokes singularities. Starting from a finite-energy local pointwise Type I singular point, we obtain a raw extracted bounded centered Seregin ancient profile with positive compact local L^3 velocity mass, then pass to the raw generated Seregin state space with invariant local nonvanishing. We define the small-amplitude, stationary L^3, uniformly L^3-tight, and structure and decay classes, and prove their exclusion by perturbative, stationary, endpoint L^3, and axisymmetric or weighted-vorticity Liouville theorems. The exclusion of the uniformly L^3-tight class rests on the Albritton–Barker endpoint ancient Liouville theorem after a physical pullback on truncated ancient intervals. The companion residual paper proves the remaining raw-class closure Theorem 17.1 for the exact generated Seregin state space $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ defined below. Inserting that theorem into the final local assembly gives the unconditional Type I exclusion.

Contents

2020 *Mathematics Subject Classification.* 35Q30, 35B44, 35B40, 76D05.

Key words and phrases. Navier–Stokes equations, Type I singularities, ancient solutions, Seregin limits, local concentration, Liouville theorems.

A note on human–AI collaboration 102

Introduction and Structure of the Argument

The local Type I reduction sends a hypothetical local pointwise Type I singularity to a raw extracted bounded centered ancient solution, then to a raw generated Seregin state with invariant local nonvanishing, and excludes the non-residual Liouville classes obtained from standard local compactness and explicit Liouville theorems. The residual class is closed by the companion residual theorem Theorem 17.1.

The generated state space used in the final assembly is a raw local Seregin state space. The projected mild formulation is not imposed on every raw generated state. Instead, mildness is treated as a branch condition in those Liouville classes whose exclusion theorems require it. Generated states not known to satisfy the full hypotheses of one of the explicit Liouville branches belong to the remaining class $\mathcal{R}(\mathcal{S})$, and are removed by the companion residual-closure theorem.

The reduction has four components. First, the local concentration theorem shows that a singular point carries positive Caffarelli–Kohn–Nirenberg critical mass at arbitrarily small scales and gives the local Type I/local Type II dichotomy. It also identifies the endpoint weak-Serrin estimate used to obtain terminal velocity no-escape for local pointwise Type I rescalings. Second, the Type I blow-up analysis gives the admissible Seregin extraction: an admissible Type I extraction produces a raw smooth bounded ancient solution of the centered self-similar equation with positive compact local L^3 velocity mass. The compact mass is converted into invariant local nonvanishing before forming the raw generated descendant hull. The paper then defines the Liouville classes and the residual class on that raw hull.

Third, the uniformly tight analysis excludes nonzero uniformly L^3-tight generated raw Seregin states that lie in the mild stratum. Boundedness plus uniform L^3-tightness gives a backward sequence of finite critical norm after physical pullback, so the Albritton–Barker endpoint ancient Liouville theorem applies. Fourth, the structured part gives the axisymmetric and perturbative weighted-vorticity exclusions used in the Liouville-class analysis. These results turn the corresponding structured hypotheses into the zero ancient solution, contradicting the invariant local nonvanishing retained by the raw generated Seregin state space.

Consequently, a generated raw Seregin state in $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ must either belong to one of the excluded Liouville classes or to the residual class. The companion theorem closes the second alternative on the same incoming profile and ledger. Thus every branch from the hypothetical singular profile ends in a contradiction, which proves the Type I exclusion unconditionally.

Dependency table.

Step	Status/source	Role
Velocity epsilon criterion	imported from Gustafson–Kang–Tsai	singularity gives L^3 concentration
Weak-Serrin Type I bridge	imported from Albritton–Barker, Lemma 2.5; used here	local Type I gives A, C, D, E bounds
Terminal no-escape	proved here using Albritton–Barker	prevents mass from escaping into $t = T$
Seregin extraction	proved here	produces a raw extracted centered ancient profile
Raw generated closure	defined here	gives the ambient state space for residual closure
Mild stratum	defined here	supplies the branch condition for mild Liouville theorems
Tight L^3 branch	proved here using Albritton–Barker, Theorem 1.2	excludes a non-residual class
Structured Liouville branches	imported theorem by theorem plus Type I constant removal	exclude the structured classes
Residual closure	theorem proved in the companion residual paper	removes $\mathcal{R}(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*))$
Contradiction	unconditional two-paper assembly	rules out local pointwise Type I singularities

How to read this paper. Readers interested only in the construction of the ancient limit should read Parts I and II together with the final assembly Theorem 34.1. Readers interested in the non-residual Liouville exclusions should read Parts III and IV. The exact companion residual-closure theorem is recorded in Theorem 17.1.

Pressure notation. Lowercase p denotes the source pressure of a suitable weak solution. Q denotes a physical ancient pressure obtained as a local distributional limit after blow-up. Π denotes a centered pressure obtained from Q by the self-similar change of variables. All pressures are considered modulo functions of time only; this gauge ambiguity is invisible to the equations and to the local energy inequality (Lemmas 11.9 and 12.4 and Remarks 12.5 and 14.1).

1. Architecture and reading guide for the local Type I proof

This section is a referee-facing guide to the proof of Theorem 34.1. The definitions, statements, and proofs in the remainder of the manuscript are the single source of mathematical truth; the guide only records their dependency order. Its input is a finite-energy suitable weak solution and a terminal point satisfying the local pointwise Type I bound. The contradiction branch supposes that this point is singular, extracts a nonzero raw centered Seregin profile, and keeps the profile's ancestry and invariant-normalization data fixed while forming the raw generated descendant hull.

The extraction is local on compact negative-time cylinders. No whole-space critical L^3 estimate for the original solution is introduced. The whole-space L^3 norms in the uniformly tight branch appear only after that branch has been selected and after physical pullback, exactly as in Theorem 26.8. Likewise, projected mildness is not imposed on the whole raw hull: it is used only in the classes whose definitions include membership in the mild stratum. A non-mild generated state not otherwise classified reaches the residual complement.

Rectangles below are states or interfaces with one incoming and one outgoing live edge; node [1] is the unique root. Diamonds are exhaustive tests. Solid ellipses are terminal closures and have no outgoing edge. Facts already entered into the ancestry and normalization ledger remain available after subsequence extraction, time translation, and locally smooth limits, and are therefore not redrawn as backward arrows. Every cross-part continuation has a matching unnumbered incoming arrow in the part containing its target.

The four explicit Liouville classes need not be disjoint. Nodes [14]–[17] therefore test membership in a fixed order. If a state belongs to more than one class, the first applicable exclusion already gives the contradiction; if all four tests fail, the definition of the remaining class places the state in $\mathcal{R}(\mathcal{S})$. This ordering is only a diagrammatic realization of the set-theoretic exhaustion in Proposition 15.11; it adds no hypothesis. The residual closure at [36]–[39] acts on the same incoming profile, raw hull, ancestry, and invariant-normalization ledger. It is the imported theorem Theorem 17.1, not an assumption and not a separate branch created later in the proof.

1.1. **Diagram map.** The two parts below form one directed acyclic graph. Bracketed node numbers are unique and consecutive in source order. An arrow marked "continue at [n]" and the unnumbered incoming arrow marked "from [m]" in the target part are the same graph edge.

Table 1: Map of the two parts of the local Type I proof-dependency diagram.

Part	Nodes	Branch resolved	Principal sources
I	[1]–[13]	Turns a suspected singular point into a concrete object to argue about. Assuming the point is not regular, the local Type I bound forces a sequence of rescalings along which velocity concentrates and cannot escape; compactness then extracts an ancient profile that is bounded, centred and not identically zero. Normalizing that profile fixes the data every later part reasons about.	Corollary 6.8, Theorem 9.4, Lemma 9.7, and Definition 14.13

II	[14]–[39]	Puts that profile through an exhaustive list of classes and empties all but one. Small-amplitude, stationary, uniformly tight and the five structured classes each close by their own Liouville theorem, leaving a single residual family; the companion paper is what shows the residual family is empty too, and the contradiction follows.	Proposition 15.11 and Theorems 16.7, 17.1, 26.8 and 33.1

1.2. **Proof-dependency diagram.** The live proof begins at [1]. If the terminal point is already regular, the theorem is complete at [3]. Otherwise Parts I–II construct one normalized generated state and assign it to an exhaustive class. Part II expands every class and residual closure in place. There is no common assembly node after the terminal ellipses: each nonregular path ends by forcing the normalized state to vanish or by proving that its residual row is empty, contradicting the invariant nonvanishing retained at [12]–[13].

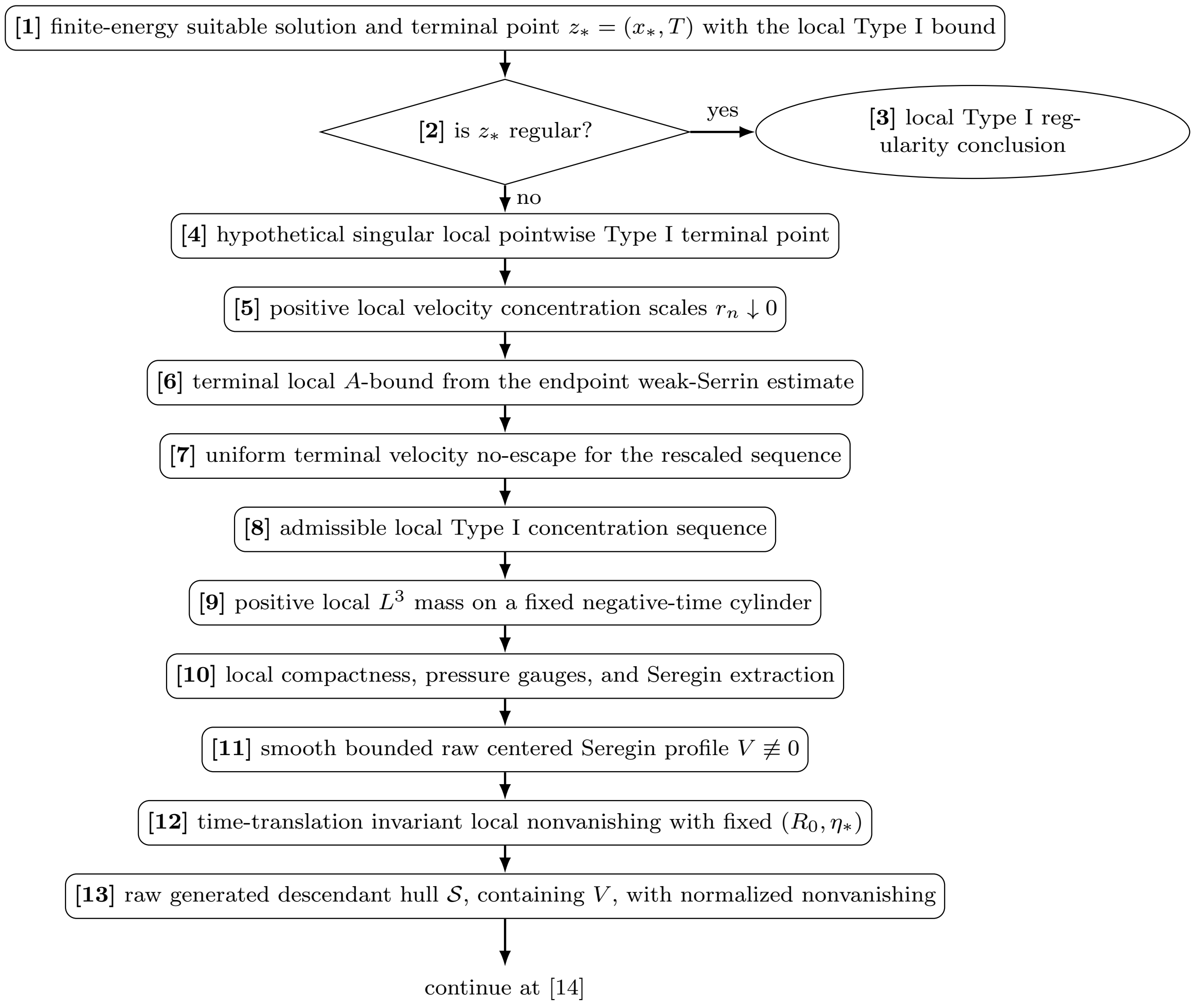

FIGURE 1. Proof-dependency diagram, Part I. The theorem input is [1]. The contradiction branch at [4] produces the same velocity concentration sequence used in the final proof. Nodes [6]–[9] display the local endpoint weak-Serrin and no-escape interface. Compactness and extraction give the nonzero centered profile [11], whose invariant normalization is fixed before the raw generated hull is formed at [13].

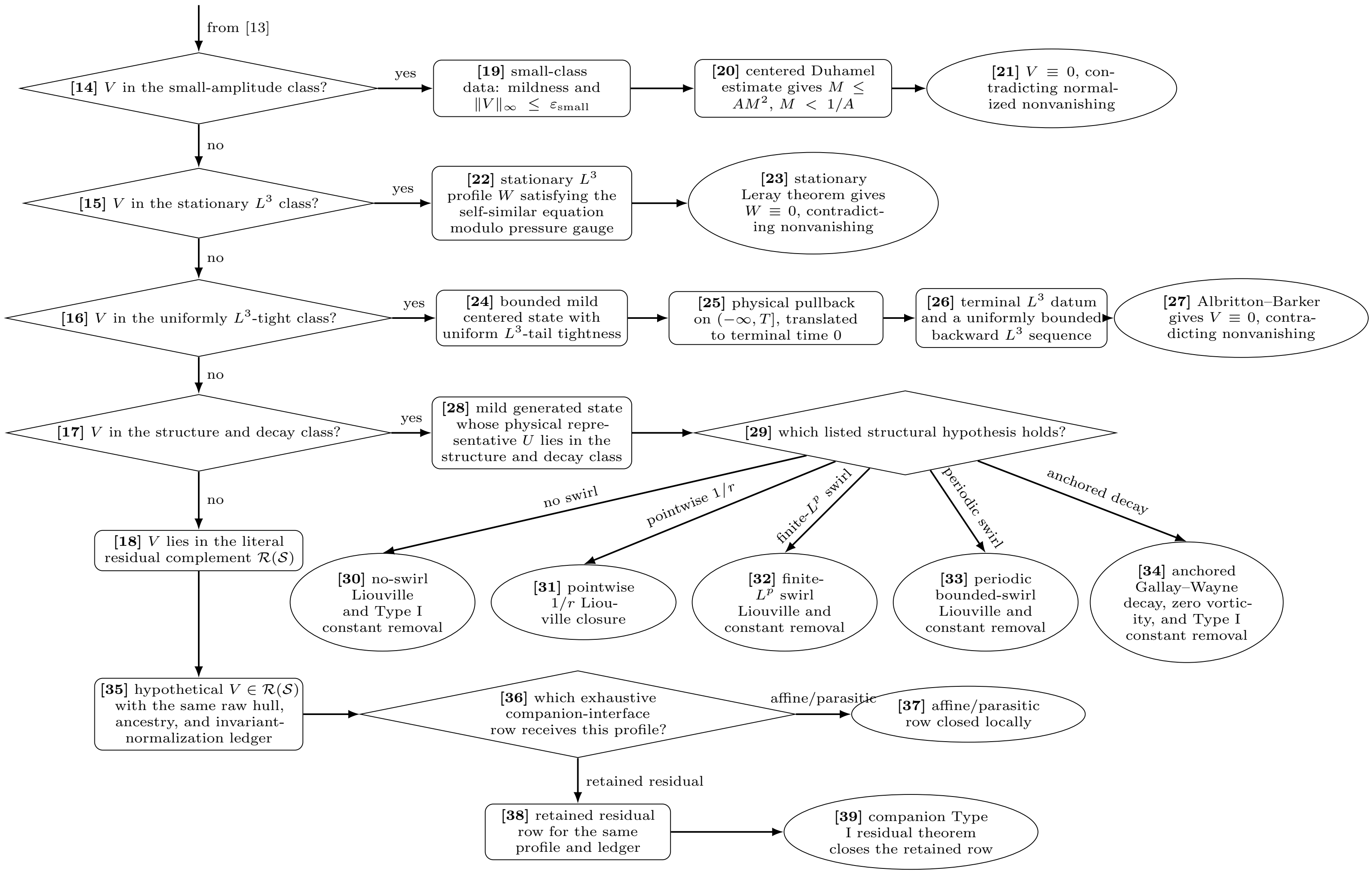

FIGURE 2. Proof-dependency diagram, Part II. The ordered tests [14]–[18] realize Proposition 15.11 without asserting disjointness. The small, stationary, tight, five structured, and two residual rows are all expanded here to the terminal closures [21], [23], [27], [30]–[34], [37], and [39].

1.3. **Node-by-node audit table.** The following table identifies the statement in the body of the manuscript that supports every numbered node. Its successor column records the live state transition; a terminal entry is a proof closure, not merely a named alternative.

Table 2: Node-by-node audit ledger for the local Type I proof-dependency diagram.

Node	State or exhaustive test	Source in the proof	Successor or closure
[1]	Finite-energy suitable solution and a terminal local pointwise Type I point	Theorem 34.1	Test regularity at [2]
[2]	Regular-versus-singular test at the terminal point	Proof of Theorem 34.1	Regular gives [3]; singular gives [4]
[3]	Local Type I regularity conclusion	Theorem 34.1	Terminal theorem conclusion
[4]	Hypothetical singular local Type I point	Proof of Theorem 34.1	Extract velocity concentration at [5]
[5]	Positive velocity concentration sequence	Corollary 6.8	Apply the terminal A-estimate at [6]
[6]	Terminal local A-bound	Lemma 9.3	Derive no-escape at [7]
[7]	Terminal velocity no-escape	Theorem 9.4	Verify admissibility at [8]
[8]	Admissible local Type I concentration sequence	Definition 9.1 and Corollary 9.5	Retain negative-time mass at [9]
[9]	Positive L^3 mass away from the terminal slice	Lemma 9.6	Run local compactness at [10]
[10]	Local compactness, pressure gauges, and Seregin extraction	Proposition 9.2 and Lemma 9.7	Obtain [11]
[11]	Smooth bounded nonzero raw centered Seregin profile	Definition 14.10 and Corollary 14.11	Convert the normalization at [12]
[12]	Time-translation invariant local nonvanishing	Lemma 14.12	Form the fixed raw hull at [13]
[13]	Raw generated descendant hull containing V and retaining nonvanishing	Definition 14.13, Lemma 14.16, and Corollary 14.19	Continue to [14]
[14]	Small-amplitude membership test	Definition 15.1 and Proposition 15.11	Yes gives [19]; no gives [15]
[15]	Stationary L^3 membership test	Definition 15.2 and Proposition 15.11	Yes gives [22]; no gives [16]
[16]	Uniform L^3-tightness membership test	Definition 15.3 and Proposition 15.11	Yes gives [24]; no gives [17]
[17]	Structure and decay membership test	Definition 15.4 and Proposition 15.11	Yes gives [28]; no gives [18]
[18]	Literal residual complement	Definition 15.7 and Proposition 15.11	Continue to [35]

Continued on next page

Table 2 continued

Node	State or exhaustive test	Source in the proof	Successor or closure
[19]	Mild small-amplitude class data	Definition 15.1 and Lemma 16.3	Apply the Duhamel estimate at [20]
[20]	Small-data Duhamel inequality	Lemmas 16.1 and 16.4	Force the closure [21]
[21]	Small profile vanishes	Lemma 16.4 and Theorem 16.7	Terminal contradiction to [12]–[13]
[22]	Stationary L^3 centered profile	Definition 15.2 and Lemma 16.5	Apply stationary rigidity at [23]
[23]	Stationary profile vanishes	Theorems 16.7 and 22.4	Terminal contradiction to [12]–[13]
[24]	Bounded mild uniformly L^3-tight centered state	Definition 15.3 and Theorem 26.8	Pull back physically at [25]
[25]	Bounded mild physical pullback on a truncated ancient interval	Lemmas 26.1 and 26.4	Establish the endpoint data at [26]
[26]	Terminal L^3 datum and bounded backward L^3 sequence	Lemmas 26.2 and 26.3	Apply endpoint rigidity at [27]
[27]	Uniformly tight profile vanishes	Theorems 16.8, 26.5 and 26.8	Terminal contradiction to [12]–[13]
[28]	Mild physical Type I representative in the structure and decay class	Definition 15.4 and Theorem 16.9	Classify the listed hypothesis at [29]
[29]	Five-way structural hypothesis test	Definition 15.4 and Theorem 33.1	At least one applicable closure [30]–[34]
[30]	Axisymmetric no-swirl closure	Proposition 31.1 and Theorem 33.1	Terminal zero contradiction
[31]	Axisymmetric pointwise $1/r$ closure	Proposition 31.2 and Theorem 33.1	Terminal zero contradiction
[32]	Finite-L^p swirl closure	Proposition 31.4 and Theorem 33.1	Terminal zero contradiction
[33]	Periodic bounded-swirl closure	Proposition 31.5 and Theorem 33.1	Terminal zero contradiction
[34]	Anchored Gallay–Wayne perturbative closure	Proposition 32.7 and Theorem 33.1	Terminal zero contradiction
[35]	Hypothetical residual profile with unchanged raw hull and ledger	Definition 15.7 and Theorem 17.1	Run the companion interface at [36]
[36]	Exhaustive companion residual-interface row	Proof of Theorem 17.1	Affine/parasitic gives [37]; retained residual gives [38]
[37]	Affine/parasitic residual-interface closure	Theorem 17.1	Terminal local companion closure
[38]	Retained residual row on the same profile and ledger	Theorem 17.1 and Proposition 17.5	Apply the companion residual theorem at [39]
[39]	Retained residual closure	Theorem 17.1 and Corollary 17.4	Terminal emptiness of the residual row

1.4. **Monotone proof ledger.** The diagram draws only the live state. The following facts remain available after the indicated extractions and translations; their persistence is what allows the class and residual closures to act on the original contradiction branch without duplicating it.

Table 3: Monotone facts retained by the local Type I proof.

Fact	Established by	Retained through	Terminal use
Finite-energy suitability and the local pointwise Type I bound	Input of Theorem 34.1	Rescaling and local extraction [1]–[11]	Supplies the local endpoint and compactness interfaces
Positive local velocity signal	Corollary 6.8	No-escape, negative-time selection, and strong local L^3 convergence [5]–[11]	Prevents the extracted profile from vanishing
Local pressure gauges and suitable compactness	Proposition 9.2	The ancient extraction on compact negative-time cylinders [10]–[13]	Supplies the centered local-pressure formulation
Fixed ancestry and invariant data (R_0, η_*)	Lemma 14.12 and Definition 14.13	Time translations and permitted locally smooth generated limits [12]–[39]	Contradicts every zero closure and fixes the residual interface
Projected mildness only on its stated stratum	Definition 14.14	Small, tight, and structure/decay branches only	Non-mild unclassified states remain available to [18], not silently excluded
Exact raw residual complement	Definition 15.7 and Theorem 17.1	Companion routing of the same incoming profile [35]–[39]	Gives literal emptiness without a new global hypothesis

1.5. **Branch-closure audit ledger.** Each nonregular path from [1] reaches exactly one of the closures below. The table distinguishes the class-producing statement from the theorem that actually eliminates the class.

Table 4: Closure audit for every terminal class in the local Type I diagram.

State or exit	Diagram route	Closure source	Terminal output
Point already regular	[1]–[3]	Theorem 34.1	[3]
Small-amplitude mild state	[14], [19]–[21]	Lemmas 16.3 and 16.4 and Theorem 16.7	[21]
Stationary L^3 state	[15], [22]–[23]	Lemma 16.5 and Theorems 16.7 and 22.4	[23]
Uniformly L^3-tight mild state	[16], [24]–[27]	Lemmas 26.3 and 26.4 and Theorems 26.5 and 26.8	[27]
Axisymmetric no-swirl state	[17], [28]–[30]	Proposition 31.1 and Theorem 33.1	[30]
Axisymmetric pointwise $1/r$ state	[17], [28]–[31]	Proposition 31.2 and Theorem 33.1	[31]
Finite-L^p swirl state	[17], [28]–[32]	Proposition 31.4 and Theorem 33.1	[32]
Periodic bounded-swirl state	[17], [28]–[33]	Proposition 31.5 and Theorem 33.1	[33]
Anchored perturbative-vorticity state	[17], [28]–[34]	Proposition 32.7 and Theorem 33.1	[34]
Affine/parasitic residual-interface row	[18], [35]–[37]	Theorem 17.1	[37]
Retained residual row	[18], [35]–[39]	Theorem 17.1, Proposition 17.5, and Corollary 17.4	[39]

Every path from [1] therefore terminates at [3], [21], [23], [27], one of [30]–[34], [37], or [39]. This leaf exhaustion is the diagrammatic reading of the contradiction in Theorem 34.1. The theorem's unconditional conclusion is not used to justify any earlier node.

Part I. Local Concentration and the Type I/Type II Dichotomy

Overview. This part proves a local analytic reduction for possible finite-time singularities of the three-dimensional Navier–Stokes equations. For a suitable weak solution, every singular point at time T has arbitrarily small backward parabolic cylinders on which both the standard mean-subtracted Caffarelli–Kohn–Nirenberg scale-invariant quantity $C + D$ and the scale-invariant velocity quantity C are bounded below by positive constants. If these local scale-invariant quantities tend to zero at each point of the singular set, the solution is regular at time T by the ε-regularity theorem. Along any such positive concentration sequence, the solution either satisfies a local Type I scale-invariant bound or belongs to the local Type II alternative. For local pointwise Type I terminal singularities of suitable weak solutions, the endpoint weak-Serrin argument gives the velocity no-escape condition used in the Type I blow-up analysis.

2. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0.$$

The viscosity is normalized to one. The weak solution theory goes back to Leray [7]. Suitable weak solutions and their partial regularity theory were introduced and developed by Scheffer [9] and by Caffarelli, Kohn, and Nirenberg [3]; see also Lin [8]. The local reduction below connects a possible singularity at time T with the Type I/Type II blow-up rate dichotomy.

Scale-invariant regularity and blow-up criteria provide the standard background for this formulation; see Serrin [11], Giga [5], Escauriaza–Seregin–Šverak [4], and Seregin [10]. The principal regularity theorem is the Caffarelli–Kohn–Nirenberg ε-regularity theorem.

The proof has four components. First, the local scale-invariant quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg ε-regularity theorem implies that vanishing of $C + D$ at a putative singular point gives local regularity, while the velocity ε-regularity criterion gives positive concentration in the scale-invariant $L^3_{x,t}$ quantity. Third, a finite-time singularity therefore produces sequences of shrinking cylinders on which $C + D$, and separately C, are bounded below by positive constants. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or belongs to the local Type II alternative.

The concentration and Type I/Type II dichotomy proofs are local in spacetime. Apart from the standard Caffarelli–Kohn–Nirenberg ε-regularity theorem, all quantities and alternatives used in that local analysis are defined in this part. The local Type I reduction section shows how the endpoint weak-Serrin estimate upgrades the local pointwise Type I envelope to terminal velocity no-escape through the local endpoint weak-Serrin Type I estimate.

3. Suitable Weak Solutions and Singular Points

Definition 3.1 (Suitable weak solution)**.** Let $I \subset \mathbb{R}$ be an open interval. A pair (u, p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|u(x,t_2)|^2}{2} \phi(x,t_2)\, dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \phi \, dx \, dt \\
&\le \int_{\mathbb{R}^3} \frac{|u(x,t_1)|^2}{2} \phi(x,t_1)\, dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \phi + \Delta \phi)\, dx \, dt \\
&\quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \phi \, dx \, dt.
\end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

Also write $Q_r(x_0, t_0)$ for $Q_r((x_0, t_0))$.

Definition 3.2 (Singular set at a fixed time)**.** For a time T, define

$$\Sigma(T) = \{x \in \mathbb{R}^3 : \ u \text{ is not locally bounded in any } Q_r(x, T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x,T))} \le M$.

If a finite-time singularity occurs at time T, then $\Sigma(T) \neq \emptyset$ in this local sense.

Temporal cutoff convention.

Lemma 3.3 (Temporal cutoff convention)**.** *For a suitable weak solution, every local energy inequality used below with endpoint times or compact time slabs is obtained by applying the standard distributional local energy inequality to a test function of the form $\phi(x,t)\chi_\varepsilon(t)$, where χ_ε is a smooth nonincreasing temporal cutoff with $\chi_\varepsilon = 1$ on a closed sub-interval and supported in a slightly larger open interval, and then letting $\varepsilon \downarrow 0$ along an admissible set of times where the L^2 trace exists.*

We invoke this convention silently in subsequent proofs.

4. Parabolic Rescaling and Scale-Invariant Quantities

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y,s) = r\, u(x_0 + ry, T + r^2 s), \qquad p^{(r)}(y,s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y, s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 4.1 (Invariance under parabolic rescaling)**.** *If (u, p) is a suitable weak solution in a cylinder containing (x_0, T), then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. The local energy inequality is verified to fix the normalization.

Let $\psi \in C_c^\infty$ be nonnegative and compactly supported in the rescaled domain. Define

$$\phi(x,t) = \psi\left(\frac{x - x_0}{r}, \frac{t - T}{r^2} \right).$$

If $y=(x-x_0)/r$ and $s=(t-T)/r^2$, then

$$\partial_t\phi = r^{-2}(\partial_s\psi)(y,s), \qquad \nabla_x\phi = r^{-1}(\nabla_y\psi)(y,s), \qquad \Delta_x\phi = r^{-2}(\Delta_y\psi)(y,s).$$

For almost every $s_1<s_2$, the corresponding times $T+r^2s_1<T+r^2s_2$ are admissible in the original local energy inequality. Apply that inequality for (u,p) to ϕ on $[T+r^2s_1, T+r^2s_2]$. After the change of variables $x=x_0+ry$, $t=T+r^2s$, using

$$u(x_0+ry, T+r^2s) = r^{-1}u^{(r)}(y,s), \qquad p(x_0+ry,T+r^2s) = r^{-2}p^{(r)}(y,s),$$

each term contains the same common factor r. Dividing by this factor gives the rescaled inequality

$$\begin{aligned}
&\int_{\mathbb{R}^3}\frac{|u^{(r)}(y,s_2)|^2}{2}\psi(y,s_2)\,dy + \int_{s_1}^{s_2}\int_{\mathbb{R}^3}|\nabla_y u^{(r)}|^2\psi\,dy\,ds\\
&\le \int_{\mathbb{R}^3}\frac{|u^{(r)}(y,s_1)|^2}{2}\psi(y,s_1)\,dy + \int_{s_1}^{s_2}\int_{\mathbb{R}^3}\frac{|u^{(r)}|^2}{2}(\partial_s\psi+\Delta_y\psi)\,dy\,ds\\
&\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3}\left(\frac{|u^{(r)}|^2}{2}+p^{(r)}\right)u^{(r)}\cdot\nabla_y\psi\,dy\,ds.
\end{aligned}$$

Thus $(u^{(r)},p^{(r)})$ is suitable on the rescaled domain. The pressure transforms according to the Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s. $\square$

Definition 4.2 (Scale-invariant local quantities)**.** For $z_0=(x_0,t_0)$ and $r>0$, set

$$C(z_0,r) = r^{-2}\int_{Q_r(z_0)}|u|^3\,dx\,dt$$

and

$$D(z_0,r) = r^{-2}\int_{t_0-r^2}^{t_0}\int_{B_r(x_0)}|p(x,t)-(p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot,t)$ on $B_r(x_0)$.

The subtraction of the spatial mean removes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 4.3 (Local finiteness and scaling)**.** *Let (u,p) be a suitable weak solution in a cylinder containing $z_0=(x_0,T)$. Then $C(z_0,r)$ and $D(z_0,r)$ are finite for every sufficiently small r. Moreover, for every fixed $R<\infty$,*

$$\begin{aligned}
&\int_{-R^2}^0\int_{B_R}\left(|u^{(r)}|^3+|p^{(r)}-(p^{(r)})_{B_R}(s)|^{3/2}\right)\,dy\,ds\\
&\qquad = R^2\big(C(z_0,rR)+D(z_0,rR)\big).
\end{aligned}$$

Proof. Local finiteness of the velocity term follows from

$$u\in L_t^\infty L^2_{x,\mathrm{loc}}\cap L_t^2H^1_{x,\mathrm{loc}}\subset L^3_{\mathrm{loc}}$$

on finite cylinders. Indeed, if $B\Subset\mathbb{R}^3$ and $(a,b)\Subset I$, Sobolev and interpolation give

$$\int_a^b\|u(t)\|_{L^3(B)}^3\,dt \le C\left(\operatorname*{ess\,sup}_{a<t<b}\|u(t)\|_{L^2(B)}\right)^{3/2}\left(\int_a^b\|u(t)\|^2_{H^1(B)}\,dt\right)^{3/4}(b-a)^{1/4}.$$

Thus $u\in L^3_{\mathrm{loc}}$. The pressure term is finite because $p\in L^{3/2}_{\mathrm{loc}}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability. More explicitly, for a ball B,

$$|p-(p)_B(t)|^{3/2}\le C\left(|p|^{3/2}+|(p)_B(t)|^{3/2}\right),$$

while Jensen's inequality gives

$$|(p)_B(t)|^{3/2} \le \frac{1}{|B|}\int_B |p(x,t)|^{3/2}\,dx.$$

After integration over B and over time, the mean-subtracted pressure term is controlled by the local $L^{3/2}$ norm of p.

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2 s$. The velocity term transforms as

$$\int_{-R^2}^{0}\int_{B_R} |u^{(r)}|^3\,dy\,ds = r^{-2}\int_{T-(rR)^2}^{T}\int_{B_{rR}(x_0)} |u|^3\,dx\,dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2 (p)_{B_{rR}(x_0)}(T + r^2 s),$$

because

$$\frac{1}{|B_R|}\int_{B_R} r^2 p(x_0 + ry, T + r^2 s)\,dy = \frac{r^2}{|B_{rR}|}\int_{B_{rR}(x_0)} p(x, T + r^2 s)\,dx.$$

Therefore

$$\begin{aligned}\int_{-R^2}^{0}\int_{B_R} &|p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2}\,dy\,ds \\ &= r^{-2}\int_{T-(rR)^2}^{T}\int_{B_{rR}(x_0)} |p - (p)_{B_{rR}(x_0)}(t)|^{3/2}\,dx\,dt.\end{aligned}$$

Adding the velocity and pressure identities gives the displayed formula. □

5. ε-Regularity and the Vanishing Case

Theorem 5.1 (Caffarelli–Kohn–Nirenberg ε-regularity)**.** *There exists a universal constant $\varepsilon_{\mathrm{CKN}} > 0$ with the following property. Let (u,p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_{\mathrm{CKN}},$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. We use the local ε-regularity theorem of Caffarelli–Kohn–Nirenberg [3]; see also Lin [8]. The statement above is written with the spatial mean subtraction used in D. Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. □

Theorem 5.2 (Velocity ε-regularity)**.** *There exists a universal constant $\varepsilon_v > 0$ with the following property. Let (u,p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) < \varepsilon_v,$$

then u is locally bounded in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. We use the one-scale velocity criterion of Gustafson–Kang–Tsai [6] in the case $p = q = 3$. In their notation the scaled $L^{p,q}_{x,t}$ velocity norm is small under the range $3/p + 2/q \le 2$; taking $p = q = 3$ gives the scale-invariant quantity $C(z_0, r)$ used here, up to the harmless normalization convention. Thus a singular point must satisfy the concentration consequence

$$\limsup_{r\downarrow 0} C(z_0, r) \ge \varepsilon_v.$$

The statement is scale invariant because C is invariant under the parabolic Navier–Stokes scaling. □

Definition 5.3 (Vanishing local scale-invariant quantities at time T)**.** We say that the local scale-invariant quantities vanish at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r\downarrow 0}\big(C((x_0,T),r) + D((x_0,T),r)\big) = 0.$$

If $\Sigma(T) = \emptyset$, there is no condition to impose.

Proposition 5.4 (Equivalent rescaled formulation)**.** *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r\downarrow 0} \int_{-R^2}^{0}\int_{B_R} \Big(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2}\Big)\, dy\, ds = 0.$$

Proof. By Proposition 4.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2\big(C((x_0,T),rR) + D((x_0,T),rR)\big).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. □

Theorem 5.5 (Regularity under vanishing local quantities)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta,T)$ for some $\delta > 0$. If the local scale-invariant quantities vanish at time T, then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that

$$C((x_0,T),r) + D((x_0,T),r) < \varepsilon_{\mathrm{CKN}}.$$

Theorem 5.1 implies that u is bounded in $Q_{r/2}(x_0,T)$, hence $x_0 \notin \Sigma(T)$. This contradicts the choice of x_0. Hence $\Sigma(T) = \emptyset$. □

Corollary 5.6 (Vanishing local quantities exclude a terminal singularity)**.** *Let (u,p) be suitable near (x_0,T). If*

$$\limsup_{r\downarrow 0}\big(C((x_0,T),r) + D((x_0,T),r)\big) = 0,$$

then u is bounded in some backward cylinder $Q_\rho(x_0,T)$, and hence $x_0 \notin \Sigma(T)$. Equivalently, membership in $\Sigma(T)$ requires that $C+D$ have a positive lower bound along a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C+D$ is below the universal threshold in Theorem 5.1. The ε-regularity theorem gives local boundedness in a backward cylinder ending at T, so $x_0 \notin \Sigma(T)$. □

6. Positive Local Concentration

Lemma 6.1 (Failure of vanishing gives a positive concentration sequence)**.** *Suppose the local scale-invariant quantities do not vanish at time T. Then there exist $x_0 \in \Sigma(T)$, a constant $\eta > 0$, and radii $r_n \downarrow 0$ such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \textit{for all } n.$$

Equivalently, the rescaled sequence around (x_0,T) has a positive lower bound for the corresponding local scale-invariant quantities on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of

$$C((x_0,T),r) + D((x_0,T),r)$$

as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 4.3. □

Definition 6.2 (Positive local concentration sequence)**.** Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0, T) if there is $\eta > 0$ such that

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \text{for all } n.$$

The corresponding parabolic rescalings are those defined by

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 6.3 (Persistence of positive concentration)**.** *Let (u_n, p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2, 0)$. Suppose that*

$$\int_{-R^2}^0 \int_{B_R} \Big(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2} \Big)\, dy\, ds \geq \eta > 0$$

for all n. Suppose further that, after passing to a subsequence,

$$u_n \to U \quad \text{in } L^3(B_R \times (-R^2,0))$$

and

$$p_n - (p_n)_{B_R}(s) \to \Pi \quad \text{in } L^{3/2}(B_R \times (-R^2,0)).$$

Then

$$\int_{-R^2}^0 \int_{B_R} \Big(|U|^3 + |\Pi|^{3/2} \Big)\, dy\, ds \geq \eta.$$

In particular, the limiting pair retains the same positive lower bound for the mean-subtracted local scale-invariant quantity. If $\Pi = P - (P)_{B_R}(s)$, this is the corresponding Caffarelli–Kohn–Nirenberg quantity of (U,P) on $B_R \times (-R^2,0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^0 \int_{B_R} |u_n|^3\, dy\, ds \longrightarrow \int_{-R^2}^0 \int_{B_R} |U|^3\, dy\, ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the mean-subtracted pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. □

Remark 6.4. Proposition 6.3 is an auxiliary compactness observation, independent of the proof of the dichotomy below. The Seregin extraction in Part II uses only the velocity-only version below; strong $L^{3/2}$-pressure convergence is not needed for nontriviality.

Proposition 6.5 (Persistence of velocity concentration)**.** *If $u_n \to U$ strongly in $L^3(B_R \times (-R^2,0))$ and*

$$\int_{-R^2}^0 \int_{B_R} |u_n|^3\, dy\, ds \geq \eta,$$

then

$$\int_{-R^2}^0 \int_{B_R} |U|^3\, dy\, ds \geq \eta.$$

Proof. Immediate from continuity of $\|\cdot\|_{L^3}^3$ under strong L^3-convergence. □

Proposition 6.6 (Local alternative)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$ for some $\delta > 0$. Exactly one of the following alternatives holds.*

(i) *For every $x_0 \in \Sigma(T)$,*

$$\limsup_{r \downarrow 0} \big(C((x_0,T),r) + D((x_0,T),r) \big) = 0.$$

(ii) *There exist* $x_0 \in \Sigma(T)$, $\eta > 0$, *and* $r_n \downarrow 0$ *such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \textit{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set, and the second is its failure expressed by a sequence. If the first alternative fails, then for some $x_0 \in \Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a sequence along which the lower bound holds gives the second alternative. Conversely, the second alternative contradicts the first. Hence precisely one alternative holds. □

Theorem 6.7 (Finite-time singularity yields positive local scale-invariant concentration)**.** *Let* (u,p) *be a suitable weak solution on* $\mathbb{R}^3 \times (T-\delta,T)$*. If* $\Sigma(T) \neq \emptyset$*, then there are* $x_0 \in \Sigma(T)$*,* $\eta > 0$*, and* $r_n \downarrow 0$ *such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \textit{for all } n.$$

Proof. If the first alternative in Proposition 6.6 held, then Theorem 5.5 would imply $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore the second alternative in Proposition 6.6 holds. □

Corollary 6.8 (Singular points carry positive velocity concentration)**.** *Let* (u,p) *be a suitable weak solution on* $\mathbb{R}^3 \times (T-\delta,T)$*. If* $x_0 \in \Sigma(T)$*, then there are* $\eta_v > 0$ *and* $r_n \downarrow 0$ *such that*

$$C((x_0,T),r_n) \geq \eta_v \qquad \textit{for all } n.$$

Proof. If the conclusion failed, then

$$\limsup_{r\downarrow 0} C((x_0,T),r) = 0.$$

Choose $r > 0$ so small that

$$C((x_0,T),r) < \varepsilon_v,$$

where ε_v is the constant in Theorem 5.2. The velocity ε-regularity theorem gives boundedness of u in $Q_{r/2}(x_0,T)$, contradicting $x_0 \in \Sigma(T)$. Hence the limsup is positive; taking any η_v below it and passing to a sequence with $C((x_0,T),r_n) \geq \eta_v$ gives the result. □

Proposition 6.9 (Concentration consequences at a terminal singular point)**.** *Let* (u,p) *be a suitable weak solution on* $\mathbb{R}^3 \times (T-\delta,T)$ *and let* $x_* \in \Sigma(T)$*. Then:*

(1) $\limsup_{r\downarrow 0}\big(C+D\big)((x_*,T),r) > 0$ *by Theorem 5.5;*
(2) $\limsup_{r\downarrow 0} C((x_*,T),r) > 0$ *by Corollary 6.8;*
(3) *if the local pointwise Type I bound* (9.1) *holds at* $z_* = (x_*,T)$ *(or the global pointwise Type I bound holds), then Theorem 9.4 (respectively Theorem 7.5) supplies an admissible local Type I concentration sequence in the sense of Definition 9.1 suitable for the Seregin extraction.*

Proof. Items (1) and (2) are restatements of the cited results at the singular point x_*. Item (3) combines (2) with the no-escape conclusion in Theorem 9.4 or Theorem 7.5, together with Corollary 9.5 for the local entry and Corollary 7.6 for the global entry. □

7. No-Escape for Finite-Energy Type I Singularities

The preceding concentration result gives the positive velocity concentration sequence used in the Type I blow-up analysis. In the Seregin extraction theorem Theorem 10.1, the additional compactness hypothesis is a no-escape condition excluding the possibility that all of the selected scale-invariant velocity mass collapses into the final rescaled time slice. The next theorem shows that this no-escape condition is forced by the finite-energy Type I hypotheses.

For this section we use, in addition to C and D, the standard local kinetic-energy and dissipation quantities

$$A(z_0, r) = r^{-1} \operatorname*{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x,t)|^2 \, dx,$$

and

$$E(z_0, r) = r^{-1} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |\nabla u(x,t)|^2 \, dx \, dt.$$

These quantities are invariant under the Navier–Stokes scaling.

Definition 7.1 (Finite-energy Type I terminal singularity)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (0,T)$. We say that $z_* = (x_*, T)$ is a finite-energy Type I terminal singularity if $x_* \in \Sigma(T)$,

$$u \in L^\infty(0,T; L^2(\mathbb{R}^3)) \cap L^2(0,T; \dot{H}^1(\mathbb{R}^3)),$$

and there are $T_0 < T$ and $M < \infty$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{T-t}}, \qquad T_0 < t < T. \tag{7.1}$$

The main estimate is a uniformly local energy propagation bound. It is stated at unit scale because the application is obtained by parabolic rescaling from (x_*, T). The estimate is a local differential inequality: the unknown is the uniformly-local kinetic energy, and the Type I envelope turns the nonlinear pressure contribution into an integrable linear coefficient. No bound below depends on the size of the global L^2 energy or the global dissipation; finite energy is used to place the pressure in the standard whole-space Leray gauge and to provide fixed-scale terminal L^3 absolute continuity for each fixed rescaling.

The standard Leray pressure gauge is used throughout this section. At almost every terminal time on which the Type I bound holds, $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$. The whole-space pressure

$$p^L(t) = \mathcal{R}_i \mathcal{R}_j (u_i(t) u_j(t))$$

is then defined by the Calderon–Zygmund theory. The distributional pressure equation gives $-\Delta p = \partial_i \partial_j (u_i u_j) = -\Delta p^L$, and the whole-space de Rham argument applied to the momentum equation shows that $p - p^L$ is a function of time. Thus replacing p by p^L is only a pressure-gauge change and does not alter suitability. This gauge choice uses the finite-energy setting to define the representative; the uniform constants below, in particular B_{uloc} and B_A, do not contain $\|u\|_{L^\infty(0,T;L^2(\mathbb{R}^3))}$ or $\int_0^T \|\nabla u(t)\|^2_{L^2(\mathbb{R}^3)} \, dt$.

Lemma 7.2 (Uniformly-local energy propagation under a Type I envelope)**.** *Let (v, π) be a suitable weak solution on $\mathbb{R}^3 \times (-4, 0)$ with $v(s) \in L^2(\mathbb{R}^3)$ for a.e. s, and take π to be the whole-space Leray pressure representative, modulo functions of time. Assume that*

$$|v(x,s)| \le M(-s)^{-1/2}, \qquad \text{for a.e. } (x,s) \in \mathbb{R}^3 \times (-4,0), \tag{7.2}$$

and that for some admissible initial time $s_0 \in (-4,-3)$,

$$\alpha_0 := \sup_{a \in \mathbb{R}^3} \int_{B_2(a)} |v(x,s_0)|^2 \, dx \le K. \tag{7.3}$$

Then there is a constant $B_{\mathrm{uloc}} = B_{\mathrm{uloc}}(M,K) < \infty$ such that

$$\operatorname*{ess\,sup}_{-1 < s < 0} \sup_{a \in \mathbb{R}^3} \int_{B_1(a)} |v(x,s)|^2 \, dx + \sup_{a \in \mathbb{R}^3} \int_{-1}^0 \int_{B_1(a)} |\nabla v|^2 \, dx \, ds \le B_{\mathrm{uloc}}. \tag{7.4}$$

Proof. The estimate supplies the terminal no-escape bound for Theorem 7.5. Let

$$\alpha(s) = \sup_{a\in\mathbb{R}^3} \int_{B_1(a)} |v(x,s)|^2\,dx.$$

By (7.2),

$$\alpha(s)^{1/2} \le CM(-s)^{-1/2}, \qquad -4 < s < 0. \tag{7.5}$$

Fix $a \in \mathbb{R}^3$ and choose $\phi_a \in C_c^\infty(B_2(a))$ with $0 \le \phi_a \le 1$, $\phi_a \equiv 1$ on $B_1(a)$, and $|\nabla\phi_a|+|\Delta\phi_a| \le C$, uniformly in a. Applying the local energy inequality on (s_0,t), using the standard monotone approximation of temporal cutoffs and with the pressure gauge chosen below, gives for a.e. $t \in (s_0, 0)$

$$\begin{aligned}
&\int |v(t)|^2\phi_a\,dx + 2\int_{s_0}^t\int |\nabla v|^2\phi_a\,dx\,ds \\
&\quad\le \int |v(s_0)|^2\phi_a\,dx + C\int_{s_0}^t\int_{B_2(a)} |v|^2\,dx\,ds + C\int_{s_0}^t\int_{B_2(a)} |v|^3\,dx\,ds \\
&\qquad + C\int_{s_0}^t\int_{B_2(a)} |\pi - c_a(s)|\,|v|\,dx\,ds.
\end{aligned} \tag{7.6}$$

Here $c_a(s)$ is an arbitrary function of time; adding or subtracting such a function does not change the local energy inequality, since $\nabla\cdot v = 0$ and ϕ_a is compactly supported.

The velocity terms are controlled by the uniformly local kinetic energy. Since $B_2(a)$ is covered by finitely many unit balls,

$$\int_{B_2(a)} |v|^2\,dx \le C\alpha(s),$$

and by (7.2),

$$\int_{B_2(a)} |v|^3\,dx \le M(-s)^{-1/2}\int_{B_2(a)} |v|^2\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{7.7}$$

It remains to estimate the pressure term uniformly in a.

Use the whole-space pressure representative $\pi = \mathcal{R}_i\mathcal{R}_j(v_iv_j)$, modulo functions of time. Let $\chi_a \in C_c^\infty(B_8(a))$, $\chi_a \equiv 1$ on $B_4(a)$, and write inside $B_4(a)$

$$\pi = q_a + h_a, \qquad q_a = \mathcal{R}_i\mathcal{R}_j(\chi_a v_iv_j).$$

Then h_a is harmonic in $B_4(a)$. Calderon–Zygmund boundedness [12] and (7.2) give, for a.e. s,

$$\begin{aligned}
\|q_a(s)\|_{L^2(B_4(a))} &\le C\|\chi_a v(s)\otimes v(s)\|_{L^2(\mathbb{R}^3)} \\
&\le CM(-s)^{-1/2}\|v(s)\|_{L^2(B_8(a))} \\
&\le CM(-s)^{-1/2}\alpha(s)^{1/2},
\end{aligned}$$

where the last step again uses a finite covering by unit balls. Hence

$$\int_{B_2(a)} |q_a|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{7.8}$$

For the harmonic part, set $c_a(s) = h_a(a,s)$. The kernel estimate

$$|K_{ij}(x-z) - K_{ij}(a-z)| \le C|x-a|\,|z-a|^{-4}, \qquad x\in B_2(a), \quad |z-a|>4,$$

where K_{ij} is the kernel of $\mathcal{R}_i\mathcal{R}_j$, gives

$$|h_a(x,s) - c_a(s)| \le C\sum_{m=2}^\infty 2^{-4m}\int_{2^m<|z-a|<2^{m+1}} |v(z,s)|^2\,dz.$$

The annulus $\{2^m < |z-a| < 2^{m+1}\}$ is covered by at most $C2^{3m}$ unit balls. Therefore

$$\|h_a(s) - c_a(s)\|_{L^\infty(B_2(a))} \le C \sum_{m=2}^{\infty} 2^{-m}\alpha(s) \le C\alpha(s). \tag{7.9}$$

Consequently,

$$\begin{aligned}\int_{B_2(a)} |h_a - c_a|\,|v|\,dx &\le C\alpha(s) \int_{B_2(a)} |v|\,dx \\ &\le C\alpha(s)^{3/2} \\ &\le CM(-s)^{-1/2}\alpha(s),\end{aligned} \tag{7.10}$$

where the last inequality is precisely the Type I absorbing estimate (7.5). Combining (7.8) and (7.10),

$$\int_{B_2(a)} |\pi - c_a(s)|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{7.11}$$

Insert (7.7) and (7.11) into (7.6), then take the supremum over a. Since $(-s)^{-1/2} \in L^1(-4,0)$, Gronwall's inequality yields

$$\operatorname*{ess\,sup}_{s_0<s<0} \alpha(s) \le CK \exp\left(C \int_{s_0}^{0} \left(1 + M(-s)^{-1/2}\right) ds \right) \le B_1(M,K).$$

Returning to (7.6), using this bound, and then letting $t \uparrow 0$ through admissible times gives the corresponding uniformly-local dissipation estimate on $(-1,0)$. This proves (7.4). □

Remark 7.3 (Local energy estimate as an absorbing bound). The preceding proof can be read as a three-term estimate for the local quantity $\alpha(s)$. The diffusion and transport terms are linear in α, with coefficient $1 + M(-s)^{-1/2}$. The local pressure term is also linear after Calderon–Zygmund and the Type I envelope. The remaining nonlinear term is the harmonic pressure flux, which is $O(\alpha^{3/2})$; the pointwise Type I envelope implies $\alpha^{1/2} \le CM(-s)^{-1/2}$, reducing it to the same integrable linear coefficient. Thus large values of α cannot persist: the resulting differential inequality bounds α in terms of constants depending only on M and the initial uniformly local energy level.

Proposition 7.4 (Terminal kinetic tightness)**.** *Let (u,p,z_*), $z_* = (x_*,T)$, be a finite-energy Type I terminal singularity in the sense of Definition 7.1. Then there exist $r_0 > 0$ and $B_A < \infty$, with B_A depending only on the Type I constant M, such that*

$$A(z_*, r) \le B_A, \qquad 0 < r < r_0. \tag{7.12}$$

Moreover,

$$r^{-1} \int_{T-r^2}^{T} \int_{B_r(x_*)} |\nabla u|^2\,dx\,dt \le B_A, \qquad 0 < r < r_0. \tag{7.13}$$

Proof. Choose $r_0 > 0$ so small that $T - 4r^2 > T_0$ for $0 < r < r_0$. For such r, use the whole-space Leray pressure gauge fixed above and define the rescaled solution

$$v^{(r)}(y,s) = ru(x_* + ry, T + r^2 s), \qquad \pi^{(r)}(y,s) = r^2 p(x_* + ry, T + r^2 s),$$

on $\mathbb{R}^3 \times (-4,0)$. The Type I bound (7.1) gives

$$|v^{(r)}(y,s)| \le M(-s)^{-1/2}, \qquad -4 < s < 0.$$

For any admissible $s_0 \in (-4,-3)$ at which the Type I bound holds, and any $a \in \mathbb{R}^3$,

$$\begin{aligned}\int_{B_2(a)} |v^{(r)}(y,s_0)|^2\,dy &= r^{-1}\int_{B_{2r}(x_*+ra)} |u(x,T+r^2 s_0)|^2\,dx \\ &\le r^{-1}|B_{2r}|\frac{M^2}{r^2|s_0|} \le CM^2.\end{aligned}$$

Such times form a full-measure subset of $(-4,-3)$. The preceding estimate is uniform in this choice, so Lemma 7.2 applies with $K = CM^2$, uniformly in r. In particular K depends only on M, so $B_A = B_{\mathrm{uloc}}(M, CM^2)$ depends only on M. Hence

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_1} |v^{(r)}(y,s)|^2\,dy + \int_{-1}^{0}\int_{B_1} |\nabla v^{(r)}|^2\,dy\,ds \le B_A.$$

Undoing the parabolic change of variables gives precisely (7.12) and (7.13). The constant B_A depends only on M; no global energy norm of u has been used in the estimate. □

Theorem 7.5 (Global pointwise Type I implies finite-energy velocity no-escape)**.** *Let (u,p,z_*), $z_* = (x_*,T)$, be a finite-energy Type I terminal singularity. Let $\lambda_k \downarrow 0$ be any sequence of scales, and set*

$$u_k(x,t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then

$$\lim_{\sigma\downarrow 0} \sup_k \int_{-\sigma}^{0}\int_{B_1} |u_k(x,t)|^3\,dx\,dt = 0. \tag{7.14}$$

Proof. By Proposition 7.4, after discarding at most finitely many indices,

$$\operatorname*{ess\,sup}_{-1<t<0} \int_{B_1} |u_k(x,t)|^2\,dx \le B_A.$$

The Type I bound gives, for the same tail of the sequence,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \le M(-t)^{-1/2}, \qquad -1<t<0.$$

Therefore, for $0<\sigma<1$,

$$\begin{aligned}\int_{-\sigma}^{0}\int_{B_1} |u_k|^3\,dx\,dt &\le \int_{-\sigma}^{0} \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x,t)|^2\,dx\,dt \\ &\le MB_A \int_{-\sigma}^{0} (-t)^{-1/2}\,dt = 2MB_A\sigma^{1/2}.\end{aligned}$$

This bound is uniform over the tail. For each of the finitely many discarded indices k, choose $\sigma_k > 0$ so that $T + \lambda_k^2 t > T_0$ for $-\sigma_k < t < 0$. On this terminal subinterval the Type I envelope and the finite-energy assumption imply

$$\int_{-\sigma_k}^{0}\int_{B_1} |u_k|^3\,dx\,dt \le \int_{-\sigma_k}^{0} \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x,t)|^2\,dx\,dt < \infty.$$

Indeed, on $(-\sigma_k, 0)$,

$$\|u_k(t)\|_{L^\infty(B_1)} \le M(-t)^{-1/2}, \qquad \int_{B_1} |u_k(x,t)|^2\,dx \le \lambda_k^{-1}\|u(T+\lambda_k^2 t)\|_{L^2(\mathbb{R}^3)}^2,$$

and the right-hand side is finite for this fixed k by finite energy. Thus each discarded index has absolute continuity of its L^3-integral at the terminal slice. Taking the maximum over this finite set gives the same limit for the full supremum over k. This proves (7.14). □

Corollary 7.6 (Admissibility of finite-energy Type I concentration sequences)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Then there exist $\eta_v > 0$ and $\lambda_k \downarrow 0$ such that*

$$C(z_*, \lambda_k) \geq \eta_v \qquad \text{for every } k,$$

and the corresponding rescaled velocities satisfy the no-escape condition (7.14). Hence the concentration and no-escape admissibility condition used in the Type I blow-up limit theorem Theorem 10.1 follows for finite-energy Type I singularities.

Proof. Since $x_* \in \Sigma(T)$, Corollary 6.8 gives $\eta_v > 0$ and $\lambda_k \downarrow 0$ with $C(z_*, \lambda_k) \geq \eta_v$ for all k. The no-escape condition for this same sequence follows from Theorem 7.5. □

8. The Type I/Type II Blow-Up Rate Dichotomy

Let $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ be given by Theorem 6.7. Define

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) are suitable weak solutions on expanding backward cylinders, and by Proposition 4.3 they retain a positive lower bound for the mean-subtracted local scale-invariant quantity on a fixed bounded cylinder.

Definition 8.1 (Type I concentration sequence)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T). The sequence is called Type I if the concentration point (x_0, T) satisfies a local Type I scale-invariant bound: there are $\rho > 0$ and $M < \infty$, with $\rho^2 < \delta$, such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_0))} \leq M.$$

This is the Type I alternative.

Definition 8.2 (Local Type II alternative)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T). The sequence belongs to the local Type II alternative if the concentration point (x_0, T) satisfies no local Type I scale-invariant bound. Equivalently, for every $\rho > 0$ with $\rho^2 < \delta$,

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_0))} = \infty.$$

Theorem 8.3 (Local blow-up rate dichotomy)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

The sequence r_n is a positive local concentration sequence. The corresponding concentration regime satisfies either the local Type I bound or the local Type II alternative. Consequently, once the vanishing case has been excluded by ε-regularity, every finite-time singular case belongs to precisely one of the two blow-up rate alternatives.

Proof. If the local scale-invariant quantities vanished at every point of $\Sigma(T)$, then Theorem 5.5 would give $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore Theorem 6.7 gives $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ with the stated lower bound. The rescaled sequence has a positive local scale-invariant lower bound by Proposition 4.3. By definition, either the Type I bound in Definition 8.1 holds, or the local Type II alternative in Definition 8.2 holds. □

Remark 8.4. This theorem is local. Its conclusion is the existence of a positive mean-subtracted scale-invariant concentration sequence and the dichotomy between a local Type I bound and failure of every such bound at the concentration point.

9. Local Pointwise Type I Reduction to Seregin Extraction

The following consequence of the weak-Serrin endpoint argument is used. The local pointwise Type I alternative in Definition 8.1 is strong enough, after rescaling at the singular point, to give compactness on every compact negative-time cylinder. The endpoint weak-Serrin implication reads: for suitable weak solutions the local pointwise Type I envelope belongs to $L_t^{2,\infty} L_x^\infty$, hence the Albritton–Barker weak Serrin local Type I estimate gives the scale-invariant A-bound on smaller cylinders and therefore terminal velocity no-escape. Thus positive local Type I concentration sequences are admissible for the local Seregin extraction in Lemma 9.7.

Throughout this section $z_* = (x_*, T)$, and

$$Q_1(0,0) = B_1(0) \times (-1, 0).$$

Definition 9.1 (Admissible local Type I concentration sequence)**.** For a local pointwise Type I point $z_* = (x_*, T)$, a sequence $r_n \downarrow 0$ is an admissible local Type I concentration sequence if the rescaled velocities

$$u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t)$$

satisfy, for some $\eta_v > 0$,

$$\int_{Q_1(0,0)} |u_n|^3 \, dx \, dt \geq \eta_v \qquad \text{for every } n,$$

and

$$\lim_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 \, dx \, dt = 0.$$

Proposition 9.2 (Local pointwise Type I rescalings have terminal compactness)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3 \times (0,T)$ satisfying the finite-energy bounds*

$$u \in L^\infty(0,T; L^2(\mathbb{R}^3)) \cap L^2(0,T; \dot{H}^1(\mathbb{R}^3)).$$

Let $z_* = (x_*, T)$ be a singular point, and assume that z_* satisfies the local pointwise Type I bound: there exist $\rho > 0$ and $M < \infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M. \tag{9.1}$$

For any sequence $r_n \downarrow 0$, define

$$u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t), \qquad p_n(x,t) = r_n^2 p(x_* + r_n x, T + r_n^2 t). \tag{9.2}$$

Then the following hold.

(i) *If*

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \qquad R, S < \infty, \quad \sigma > 0,$$

then, for all sufficiently large n,

$$\|u_n\|_{L^\infty(K)} \leq M \sigma^{-1/2}. \tag{9.3}$$

(ii) *On each such compact cylinder K, after subtracting time-dependent local pressure gauges $a_{n,K}(t)$, the sequence satisfies uniform local Seregin bounds*

$$\|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \leq C_K. \tag{9.4}$$

Here C_K may depend on K, M, ρ, and on the finite Leray energy level of the original solution, but it does not depend on n and does not use a terminal no-escape assumption.

(iii) *After passing to a subsequence,* $(u_n, p_n - a_{n,K})$ *converges on every compact* $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ *to an ancient suitable weak solution* (U, Q)*, with*

$$u_n \to U \quad \text{strongly in } L^q(K), \qquad 1 \le q < \infty,$$
$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}(K),$$

and

$$|U(x,t)| \le M(-t)^{-1/2}, \qquad t < 0.$$

Proof. Fix $K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0)$. For large n,

$$r_n R < \rho/2, \qquad r_n^2 S < \rho^2/2.$$

Thus $x_* + r_n x \in B_\rho(x_*)$ and $T + r_n^2 t \in (T - \rho^2, T)$ for a.e. $(x,t) \in K$. The local pointwise Type I bound (9.1) gives

$$|u_n(x,t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \le r_n M(T - (T + r_n^2 t))^{-1/2} = M(-t)^{-1/2} \le M\sigma^{-1/2}.$$

This proves (9.3). It also gives uniform local $L^\infty_t L^2_x$ and $L^3_{x,t}$ bounds on K.

It remains to explain the pressure and dissipation bounds. At this point the ambient solution is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, not merely a local suitable weak solution. Hence the whole-space Leray pressure representative may be chosen, modulo functions of time, as $p = \mathcal{R}_i \mathcal{R}_j (u_i u_j)$. All local pressure estimates below are made for this representative, followed by local time-dependent mean subtraction. Choose $\chi_R \in C^\infty_c(B_{8R})$, $\chi_R \equiv 1$ on B_{4R}, and write inside B_{4R}

$$p_n = q_n + h_n, \qquad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

Calderon–Zygmund boundedness and the local L^3-bound for u_n give a uniform $L^{3/2}$-bound for q_n. For h_n, take the gauge $a_{n,K}(t) = h_n(0, t)$. Let

$$A_m = \{2^m R < |z| < 2^{m+1} R\}, \qquad m \ge 2.$$

For $x \in B_{2R}$, the harmonic pressure kernel estimate gives

$$|h_n(x,t) - a_{n,K}(t)| \le CR \sum_{m \ge 2} (2^m R)^{-4} \int_{A_m} |u_n(z,t)|^2 \, dz.$$

Choose m_n by

$$2^{m_n} R r_n \le \rho < 2^{m_n + 1} R r_n.$$

After discarding finitely many indices, assume $m_n \ge 2$. If $m \le m_n - 1$, then the physical image of A_m is contained in $B_\rho(x_*)$. On the time slab $[-S, -\sigma]$, the local Type I bound therefore gives

$$\int_{A_m} |u_n(z,t)|^2 \, dz \le CM^2 \sigma^{-1} (2^m R)^3.$$

Consequently the inner part of the harmonic tail is bounded by

$$\begin{aligned} CR \sum_{m \le m_n - 1} (2^m R)^{-4} \int_{A_m} |u_n(z,t)|^2 \, dz &\le CRM^2 \sigma^{-1} \sum_{m \le m_n - 1} (2^m R)^{-1} \\ &\le CM^2 \sigma^{-1}. \end{aligned}$$

For $m \ge m_n$, the finite Leray energy gives

$$\int_{A_m} |u_n(z,t)|^2 \, dz \le \int_{\mathbb{R}^3} |u_n(z,t)|^2 \, dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 \, dy.$$

Since $2^{m_n}R \simeq \rho/r_n$, the outer part satisfies

$$CR \sum_{m\geq m_n} (2^m R)^{-4} \int_{A_m} |u_n(z,t)|^2\,dz \leq CRr_n^{-1}\|u\|^2_{L^\infty(0,T;L^2)} \sum_{m\geq m_n}(2^m R)^{-4}$$
$$\leq C_{\rho,R}\, r_n^3 \|u\|^2_{L^\infty(0,T;L^2)}.$$

Thus $h_n - a_{n,K}$ is bounded in $L^\infty([-S,-\sigma];L^\infty(B_{2R}))$, uniformly in n. Together with the $L^{3/2}$-bound for q_n, this gives the $L^{3/2}(K)$ pressure estimate in (9.4).

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder. The local kinetic, cubic velocity, and pressure-flux terms are bounded by the preceding estimates, so $\|\nabla u_n\|_{L^2(K)} \leq C_K$. The Navier–Stokes equation then bounds $\partial_t u_n$ locally in a negative Sobolev space. Aubin–Lions compactness and the pressure bound give the asserted Seregin subsequence convergence. Passing to the limit in the equations and local energy inequality gives an ancient suitable weak solution, and the pointwise Type I bound passes to the limit. ☐

Lemma 9.3 (Terminal A-bound from the local weak-Serrin estimate)**.** *Let (u,p) be a suitable weak solution in a backward cylinder $Q_\rho(z_*)$, $z_* = (x_*,T)$, and assume the local pointwise Type I bound (9.1). Then, for every $0<\theta<1$, there is $K_\theta<\infty$ such that, for all sufficiently small r,*

$$r^{-1} \operatorname*{ess\,sup}_{T-r^2<t<T} \int_{B_r(x_*)} |u(x,t)|^2\,dx \leq K_\theta.$$

Proof. After translating and scaling, assume $z_* = (0,0)$ and work in

$$Q_\rho = B_\rho \times (-\rho^2, 0).$$

Put

$$f(t) = \|u(t)\|_{L^\infty(B_\rho)}.$$

The pointwise Type I bound gives $f(t) \leq M(-t)^{-1/2}$. Hence, for every $\alpha>0$,

$$|\{t\in(-\rho^2,0) : f(t)>\alpha\}| \leq \min\{\rho^2, M^2\alpha^{-2}\},$$

and therefore

$$\|f\|_{L^{2,\infty}(-\rho^2,0)} \leq CM.$$

Thus

$$u \in L_t^{2,\infty} L_x^\infty(Q_\rho).$$

Fix $0<\theta<1$. Apply the Albritton–Barker weak-Serrin Type I lemma [2, Lemma 2.5] with $p=\infty$ and $q=2$. In their notation,

$$I(\omega) = \sup_{Q'\Subset\omega} \big(A(Q') + C(Q') + D(Q') + E(Q')\big),$$

and the endpoint weak-Serrin hypothesis above gives a finite bound

$$I(Q_{\theta\rho}) \leq K_\theta.$$

Let $r < (\theta\rho)/\sqrt{2}$ and $0<\varepsilon<r^2$. Then

$$Q_r((0,-\varepsilon)) = B_r \times (-r^2-\varepsilon, -\varepsilon) \Subset Q_{\theta\rho},$$

so the Albritton–Barker estimate gives

$$r^{-1} \operatorname*{ess\,sup}_{-r^2-\varepsilon<t<-\varepsilon} \int_{B_r} |u(x,t)|^2\,dx \leq K_\theta.$$

Since $(-r^2, -\varepsilon) \subset (-r^2 - \varepsilon, -\varepsilon)$, it follows that

$$r^{-1} \operatorname*{ess\,sup}_{-r^2<t<-\varepsilon} \int_{B_r} |u(x,t)|^2 \, dx \le K_\theta.$$

As $\varepsilon \downarrow 0$, the intervals $(-r^2, -\varepsilon)$ increase to $(-r^2, 0)$. By the definition of essential supremum over nested measurable sets,

$$r^{-1} \operatorname*{ess\,sup}_{-r^2<t<0} \int_{B_r} |u(x,t)|^2 \, dx \le K_\theta.$$

Undoing the translation and scaling gives the stated estimate at z_*. □

Theorem 9.4 (Local endpoint weak-Serrin Type I estimate)**.** *Let (u,p) be a suitable weak solution in a backward cylinder $Q_\rho(z_*)$, $z_* = (x_*, T)$, and assume the local pointwise Type I bound* (9.1)

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))} \le M.$$

Then, after possibly shrinking ρ, the rescaled velocities $u_n(y,s) = r_n u(x_* + r_n y, T + r_n^2 s)$ along every scale sequence $r_n \downarrow 0$ satisfy the terminal velocity no-escape condition

$$\lim_{\sigma\downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n(y,s)|^3 \, dy \, ds = 0.$$

Proof. After translating and scaling, assume $z_* = (0,0)$ and work in

$$Q_\rho = B_\rho \times (-\rho^2, 0).$$

Let $r_n \downarrow 0$ and

$$u_n(y,s) = r_n u(r_n y, r_n^2 s).$$

Fix $0 < \theta < 1$. By Lemma 9.3, the terminal A-quantity is bounded uniformly along the chosen scales. For all sufficiently large n,

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_1} |u_n(y,s)|^2 \, dy = r_n^{-1} \operatorname*{ess\,sup}_{-r_n^2<t<0} \int_{B_{r_n}} |u(x,t)|^2 \, dx \le K_\theta.$$

Moreover,

$$\|u_n(s)\|_{L^\infty(B_1)} \le M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_n|^3 \, dy \, ds &\le \int_{-\sigma}^0 \|u_n(s)\|_{L^\infty(B_1)} \int_{B_1} |u_n(y,s)|^2 \, dy \, ds \\ &\le M K_\theta \int_{-\sigma}^0 (-s)^{-1/2} \, ds = 2 M K_\theta \sigma^{1/2}. \end{aligned}$$

This proves the required terminal velocity no-escape for all sufficiently large n. The finitely many remaining indices are harmless by absolute continuity of the local L^3 integral, since suitable weak solutions are locally in L^3 on terminal subcylinders. Taking the supremum in n and then letting $\sigma \downarrow 0$ gives the claim. □

Corollary 9.5 (Local pointwise Type I concentration sequences are admissible)**.** *Under the hypotheses of Proposition 9.2, let $r_n \downarrow 0$ be any sequence such that the rescaled velocities satisfy*

$$\int_{Q_1(0,0)} |u_n|^3 \, dx \, dt \ge \eta_v > 0 \qquad \textit{for all } n.$$

Then r_n is an admissible local Type I concentration sequence in the sense of Definition 9.1.

Proof. The displayed positive concentration lower bound is the first requirement in Definition 9.1. The Albritton–Barker local weak-Serrin Type I estimate [2] gives terminal velocity no-escape from the local pointwise Type I envelope for every rescaled sequence; the precise statement used here is Theorem 9.4. This gives the second requirement in Definition 9.1. □

Lemma 9.6 (Local Type I concentration enters the Seregin extraction)**.** *Under the hypotheses of Proposition 9.2, let $r_n \downarrow 0$ be a velocity concentration sequence satisfying*

$$\int_{Q_1(0,0)} |u_n|^3 \, dx \, dt \geq \eta_v > 0 \qquad \text{for all } n.$$

Then r_n is admissible by Corollary 9.5. There are $\sigma_0 \in (0,1)$ and $\eta_1 > 0$ such that

$$\int_{-1}^{-\sigma_0} \int_{B_1} |u_n|^3 \, dx \, dt \geq \eta_1 \qquad \text{for all } n. \tag{9.5}$$

Consequently, the rescaled sequence retains positive local L^3 mass on a compact negative-time cylinder. In particular the associated Seregin extraction has no loss of the local L^3 mass needed for nontriviality.

Proof. The no-escape condition in Definition 9.1 gives $\sigma_0 > 0$ such that the terminal layer $B_1 \times (-\sigma_0, 0)$ carries at most half of the fixed Q_1 velocity mass. This proves (9.5). The compactness part follows from Proposition 9.2 on the compact negative-time cylinder $B_1 \times [-1, -\sigma_0]$. Strong local L^3 convergence on this cylinder passes the positive mass to the ancient limit. Terminal no-escape is used only at this step.

Thus the local Type I concentration sequence is admissible for the Seregin extraction and retains positive localized L^3-mass away from the terminal slice. □

Lemma 9.7 (Local Seregin extraction from an admissible local Type I sequence)**.** *Under the hypotheses of Proposition 9.2, let $r_n \downarrow 0$ be an admissible local Type I concentration sequence in the sense of Definition 9.1. Then, after passing to a subsequence, the rescaled velocities produce a smooth bounded centered solution V of (10.3) on $\mathbb{R}^3 \times \mathbb{R}$. The solution V satisfies the normalization used for Seregin ancient limits: there are a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$ such that*

$$\iint_K |V|^3 \, dy \, d\tau \geq \eta_0.$$

Proof. Proposition 9.2 gives local Seregin compactness for the rescaled sequence on every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$. The local Type I bound passes to the ancient limit as

$$|U(x,t)| \leq M(-t)^{-1/2}, \qquad t < 0.$$

Interior Serrin regularity then gives smoothness of U on $\mathbb{R}^3 \times (-\infty, 0)$ by the argument used in Lemma 14.4. The centered change of variables

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(y\sqrt{-t}, t)$$

therefore gives a smooth bounded solution of the centered equation (10.3). The no-escape part of local admissibility and Lemma 9.6 give positive compact L^3-mass for U away from $t = 0$. On the compact image of that cylinder under $(x,t) = (e^{-\tau/2} y, -e^{-\tau})$, the factor $e^{-\tau}$ in

$$|U|^3 \, dx \, dt = e^{-\tau} |V|^3 \, dy \, d\tau$$

is bounded above and below by positive constants. Hence the positive lower bound for U becomes the displayed local L^3 nonvanishing for V. □

Proposition 9.8 (Local Type I entry into the final assembly)**.** *Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$. Then every terminal point $z_* = (x_*, T)$ satisfying the local pointwise Type I bound (9.1) is regular. More precisely, if such a point were singular, it would generate a raw extracted normalized Seregin ancient limit with compact spacetime nonvanishing; Lemma 14.12 would then give the invariant local nonvanishing used to form the associated generated descendant hull. That raw hull would be exhausted by the Liouville alternatives and the residual class removed by Theorem 17.1.*

Proof. Suppose, toward a contradiction, that $z_* = (x_*, T)$ is a singular point satisfying (9.1). Choose the positive Q_1 velocity concentration sequence supplied by the local concentration theorem. By Corollary 9.5, this sequence satisfies terminal no-escape and is admissible. By Lemma 9.6, the local Type I sequence has local compactness and positive compact L^3-mass away from the terminal slice. By Lemma 9.7, the sequence generates a raw extracted normalized Seregin ancient limit V. By Lemma 14.12, V satisfies the invariant local nonvanishing condition for some data (R_0, η_*). Form the raw generated descendant hull

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*).$$

By construction, $V \in \mathcal{S}$. By the class exhaustion in Proposition 15.11, V lies either in one of the Liouville classes excluded by the stated Liouville theorems or in the remaining class $\mathcal{R}(\mathcal{S})$. The Liouville classes are excluded by the small-amplitude, stationary, uniformly tight, and structure-and-decay Liouville results; the last of these is routed through Theorem 33.1. The residual alternative is removed by the companion residual-closure theorem applied to $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. This contradicts the generated nonvanishing guaranteed by Corollary 14.19. Hence z_* is regular. □

Corollary 9.9 (Exclusion of the local Type I alternative)**.** *The local Type I alternative is excluded for every finite-energy singular point. Hence every finite-energy singular point belongs to the local Type II alternative in the dichotomy above.*

Proof. This is Proposition 9.8 restated in the terminology of the local Type I/local Type II dichotomy. □

Part I dependency diagram.

singular point $z_* = (x_*, T)$
$\downarrow$ Corollary 6.8 (CKN)
positive velocity concentration scales
$\downarrow$ Theorem 9.4 (local) *or* Theorem 7.5 (global)
no-escape away from terminal slice
$\downarrow$ Corollary 9.5 / Corollary 7.6
admissible concentration sequence
$\downarrow$ Lemma 9.7 *or* Theorem 10.1
raw extracted Seregin ancient limit, then raw generated state space and mild strata (Part II)
$\downarrow$ class decomposition Definition 15.7
non-residual classes (excluded here) $\cup$ residual class
$\downarrow$ Theorem 17.1 (companion residual-closure theorem)
contradiction$\Rightarrow$ Theorem 34.1.

The local reduction and the companion residual theorem form one acyclic proof chain; Theorem 17.1 records the exact imported closure node.

Part II. Type I Blow-up Limits and Residual Closure

Overview. An admissible finite-energy Type I singularity for the three-dimensional Navier–Stokes equations is first reduced to a raw extracted nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation, and then to the raw generated state space used in the residual argument. The construction keeps track of pressure gauges, local compactness, stability of the local energy inequality, and the velocity concentration supplied by local Caffarelli–Kohn–Nirenberg theory. It yields a smooth bounded ancient solution for the centered equation on $\mathbb{R}^3 \times \mathbb{R}$, together with a quantitative local L^3 nonvanishing condition.

The possible generated raw Seregin states are organized by the analytic hypotheses used in the class decomposition: small amplitude, stationary L^3, uniform L^3-tightness, and structure or decay assumptions to which cited Liouville theorems apply. The small-amplitude case is excluded by the bounded ancient perturbative Liouville theorem in Lemma 16.4, and the stationary L^3 case by the theorem of Nečas–Růžička–Šverák. The structure and decay class contains only hypotheses for which the cited Liouville theorems force $V \equiv 0$. The uniformly L^3-tight class is excluded in Theorem 26.8. The residual class is closed by the companion theorem Theorem 17.1. Consequently the extraction and companion closure give the Type I conclusion for the raw generated finite-energy and local pointwise Type I extraction classes without an additional assumption.

10. Introduction and main results

10.1. Type I blow-up limits. Let $u : \mathbb{R}^3 \times (0, T) \to \mathbb{R}^3$ solve the incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \Delta u, \qquad \operatorname{div} u = 0. \tag{10.1}$$

The local regularity theory of Caffarelli–Kohn–Nirenberg [3] detects singular points through scale-invariant concentration of velocity and pressure. A terminal point $z_* = (x_*, T)$ is of Type I if

$$\limsup_{t \uparrow T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty. \tag{10.2}$$

The blow-up extraction uses a selected velocity concentration sequence from the local concentration theorem and assumes only that this velocity concentration does not escape into the terminal slice after rescaling. The resulting admissibility condition is defined in Definition 11.4. It replaces a stronger all-scale compactness assumption on scale-invariant energy, pressure, and dissipation quantities by the non-escape condition needed for nontriviality of the ancient limit.

10.2. Normalized Seregin limits. For scales $\lambda \downarrow 0$, the parabolic rescaling about z_* is

$$u^{(\lambda)}(x,t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \qquad p^{(\lambda)}(x,t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

The rescaled time intervals exhaust $(-\infty, 0)$. The Type I bound becomes a uniform bound

$$\|u^{(\lambda)}(t)\|_{L^\infty(\mathbb{R}^3)} \le M(-t)^{-1/2}$$

on compact subintervals of $(-\infty, 0)$, and compactness produces an unrenormalized ancient solution (U, Q). The associated centered variables are

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(y\sqrt{-t}, t).$$

They transform the ancient Navier–Stokes solution into the centered equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla \Pi = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0. \tag{10.3}$$

The selected velocity concentration at the singular point persists through the extraction and gives a compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and a constant $\eta_0 > 0$ such that

$$\iint_K |V|^3 \, dy \, d\tau \geq \eta_0. \tag{10.4}$$

10.3. **Liouville alternatives and the remaining case.** No unconditional Liouville theorem is known for all bounded ancient solutions of (10.3). The reduction therefore uses a finite decomposition based on explicit Liouville theorems. A generated raw Seregin state is compared with the following analytic classes:

$$\mathcal{C}_{\text{small}}, \qquad \mathcal{C}_{\text{stat},L^3}, \qquad \mathcal{C}_{L^3-\text{tight}}, \qquad \mathcal{C}_{\text{sd}}, \qquad \mathcal{R}(\mathcal{S}).$$

They denote, respectively, small amplitude, stationary L^3, uniform L^3-tightness, and structure or decay hypotheses to which the cited Liouville theorems apply (sd), and the complement of these four classes inside a raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. The last set carries no additional analytic hypothesis; it is the remaining case after the stated Liouville alternatives have been removed.

10.4. **Main theorems.**

Theorem 10.1 (Normalized Type I blow-up limit)**.** *Let (u, p, z_*) be an admissible finite-energy Type I singularity. Then there exist scales $\lambda_k \downarrow 0$, pressure gauges $a_k(t)$, an ancient suitable weak solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0)$, and a centered solution (V, Π) on $\mathbb{R}^3 \times \mathbb{R}$, such that, after passing to a subsequence,*

$$u_k \to U \quad \textit{strongly in } L^q_{\text{loc}}(\mathbb{R}^3 \times (-\infty, 0)), \qquad 1 \leq q < \infty,$$

and

$$p_k - a_k(t) \rightharpoonup Q \quad \textit{weakly in } L^{3/2}_{\text{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

The centered velocity V is smooth, bounded, solves (10.3), and is a raw extracted normalized Seregin ancient limit with the compact spacetime L^3 nonvanishing condition (10.4).

For a raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$, the class definitions give, for each $V \in \mathcal{S}$, the exhaustive alternative

$$V \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat},L^3} \cup \mathcal{C}_{L^3-\text{tight}} \cup \mathcal{C}_{\text{sd}} \cup \mathcal{R}(\mathcal{S}).$$

This exhaustive alternative is recorded in Proposition 15.11. It is a direct consequence of the definition of the remaining class, not an independent Liouville theorem.

Theorem 10.2 (Type I exclusion)**.** *No admissible finite-energy Type I singularity exists.*

The argument uses the explicit local class closures and the companion residual-closure theorem. The uniformly L^3-tight class is excluded by Theorem 26.8. The remaining class is excluded by the companion theorem Theorem 17.1. The structure and decay class is restricted to hypotheses to which cited Liouville theorems apply.

The resulting implication is

$$\begin{gathered}
\text{admissible finite-energy Type I singularity} \\
\Downarrow \\
\text{raw extracted nonzero Seregin limit} \\
\Downarrow \\
\text{raw generated Seregin state with invariant nonvanishing} \\
\Downarrow \\
\text{small/stationary/tight/structure and decay/remaining alternative} \\
\Downarrow \\
\text{contradiction by the Liouville-class exclusions and companion residual closure.}
\end{gathered}$$

- Small amplitude: excluded in Section 16.
- Stationary L^3: excluded by [17].
- Uniformly L^3-tight: proved in Theorem 16.8, using Theorem 26.8.
- Structure and decay: routed through Theorem 33.1 and excluded in Theorem 16.9.
- Remaining class: excluded by the companion residual-closure theorem Theorem 17.1.

10.5. **Dependencies and companion closure.** The proof of the extraction theorem uses the local velocity concentration result stated as Proposition 11.7; no other result from the local concentration theory enters the blow-up construction. The extraction first produces a raw extracted limit carrying the local L^3 lower bound (10.4). The subsequent hull construction uses Lemma 14.12 and Lemma 14.16, which provide the invariant nonvanishing and the raw generated structure. Mildness is imposed only in the branch definitions that need it. The uniformly L^3-tight Liouville theorem and the companion residual-closure theorem is invoked only after this raw generated class is formed and the class exhaustion is applied. The statement Theorem 16.8 follows from the endpoint L^3 closure in Theorem 26.8, and Theorem 17.1 is the residual-closure theorem for $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$. The structure and decay class is routed through Theorem 33.1, whose hypotheses are recorded in the structured Liouville section. Thus the final Type I conclusion follows from the admissible extraction in Corollary 7.6, the raw generated closure statements, the local Liouville-class exclusions, and the companion residual-closure theorem. Each input is a proved theorem; none is a new analytic assumption on the incoming singular profile.

10.6. **Two entry routes for the reduction.** There are two routes to the same raw generated Seregin state space; they have different hypotheses and first produce the same type of raw extracted ancient solution.

(1) *Local pointwise Type I.* The Part I local concentration and local endpoint weak-Serrin estimate Theorem 9.4 produce an admissible local Type I concentration sequence at any singular point satisfying (9.1), and Lemma 9.6 feeds the resulting rescalings into the Seregin extraction with positive compact L^3 velocity mass on a fixed negative-time cylinder. This is the entry point used by the final assembly theorem Theorem 34.1.

(2) *Global pointwise Type I.* The Part II extraction estimates, including the pressure decomposition in Lemma 12.2, are written for an admissible Type I solution in the sense of Definition 11.4, which encodes a global pointwise Type I bound on $\mathbb{R}^3$. This streamlines the harmonic-pressure tail estimates, which would otherwise require the local pressure-tail decomposition recorded in Proposition 9.2.

In both cases the conclusion is first a raw extracted normalized Seregin ancient limit satisfying (10.4). The subsequent hull construction uses the invariant normalization (14.1). The local pointwise entry uses Part I to produce an admissible concentration sequence and pressure-tail control, while Part II then runs on the resulting compact cylinders. No analytic hypothesis beyond local pointwise Type I and finite Leray energy is added in this reduction.

Global route	global pointwise Type I	global no-escape $\Rightarrow$ finite-energy Seregin extraction
Local route	local pointwise Type I	AB endpoint weak-Serrin bounds $\Rightarrow$ terminal no-escape $\Rightarrow$ local Seregin extraction

10.7. **Organization.** Section 11 introduces suitable solutions, scale-invariant quantities, the admissibility condition, scaling, pressure gauges, and local concentration. Section 12 proves the compactness theorem for Type I rescalings away from the terminal time and constructs the ancient suitable weak limit. Section 13 proves persistence of concentration and nontriviality.

Section 14 derives the centered equation and defines raw extracted limits, the raw generated state space, and its mild stratum. Section 15 defines the Liouville classes and the remaining class. Section 16 proves the class exclusions and the uniformly tight theorem. Section 17 imports the companion residual-closure theorem and proves the remaining-case criterion Theorem 10.2. Appendix 18 gives the elementary symmetries of the centered equation.

11. Suitable solutions, scale-invariant quantities, and local concentration

For $z_0 = (x_0, t_0)$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

When $z_0 = (0, 0)$, write $Q_r = Q_r(0, 0)$. The spatial average of f over $B_r(x_0)$ is

$$(f)_{B_r(x_0)}(t) = \frac{1}{|B_r|} \int_{B_r(x_0)} f(x,t)\, dx.$$

Throughout this part, $\mathbb{P}$ denotes the Helmholtz–Leray projection onto divergence-free vector fields on $\mathbb{R}^3$.

Definition 11.1 (Terminal singular point)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. A terminal point $z_* = (x_*, T)$ is regular if there are $r > 0$ and $K < \infty$ such that

$$\operatorname*{ess\,sup}_{Q_r(z_*)} |u| \le K.$$

It is singular otherwise.

Definition 11.2 (Suitable weak solution)**.** This definition repeats the Part I notion in the notation used here; the two definitions agree. Let $\Omega \subset \mathbb{R}^3 \times \mathbb{R}$ be open. A pair (u, p) is a suitable weak solution of (10.1) in Ω if

$$u \in L^\infty_t L^2_x(\Omega') \cap L^2_t \dot H^1_x(\Omega'), \qquad p \in L^{3/2}(\Omega')$$

for every compact cylinder $\Omega' \Subset \Omega$, if (u, p) solves (10.1) distributionally, $\operatorname{div} u = 0$, and if the local energy inequality

$$\begin{aligned}
&\int_{\mathbb{R}^3} |u(x,t_2)|^2 \phi(x,t_2)\, dx + 2\int_{t_1}^{t_2}\int_{\mathbb{R}^3} |\nabla u|^2 \phi \, dx\, dt \\
&\le \int_{\mathbb{R}^3} |u(x,t_1)|^2 \phi(x,t_1)\, dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} |u|^2(\partial_t \phi + \Delta\phi)\, dx\, dt \\
&\quad + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} (|u|^2 + 2p) u \cdot \nabla\phi \, dx\, dt
\end{aligned} \tag{11.1}$$

holds for every non-negative $\phi \in C_c^\infty(\Omega)$ and almost every $t_1 < t_2$ for which the displayed terms are defined.

This is the Caffarelli–Kohn–Nirenberg notion of suitability [3]; the global finite-energy background is Leray's Leray–Hopf class [7].

Definition 11.3 (Suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$)**.** A suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, T)$ satisfying

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot H^1(\mathbb{R}^3)), \qquad u \in C_w([0, T); L^2(\mathbb{R}^3)),$$

and the global Leray energy inequality. The pressure is taken in $L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times (0, T))$, modulo functions of time.

Throughout the global extraction and final assembly, the phrase *finite-energy suitable weak solution on* $\mathbb{R}^3 \times (0,T)$ means a suitable Leray–Hopf solution in the sense of Definition 11.3. Purely local statements use the phrase "suitable weak solution in a cylinder" and do not invoke the global energy or whole-space pressure representative unless explicitly stated.

Define

$$A(u; z_0, r) = r^{-1} \operatorname*{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x,t)|^2 \, dx, \tag{11.2}$$

(11.3)
$$C(u; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |u|^3 \, dx \, dt, \qquad D(p; z_0, r) = r^{-2} \iint_{Q_r(z_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} \, dx \, dt,$$

$$E(u; z_0, r) = r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 \, dx \, dt, \tag{11.4}$$

Definition 11.4 (Admissible finite-energy Type I singularity)**.** A triple (u, p, z_*), with $z_* = (x_*, T)$, is an admissible finite-energy Type I singularity if:

- (i) (u, p) is a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$;
- (ii) $z_* = (x_*, T)$ is a singular point;
- (iii) the Type I bound (10.2) holds;
- (iv) there are scales $\lambda_k \downarrow 0$ and a constant $\eta_v > 0$ such that
$$C(u; z_*, \lambda_k) \geq \eta_v \qquad \text{for every } k;$$
- (v) for the corresponding rescaled velocities
$$u_k(x,t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t),$$
the velocity no-escape condition holds:
$$\lim_{\sigma \downarrow 0} \sup_k \int_{-\sigma}^0 \int_{B_1} |u_k(x,t)|^3 \, dx \, dt = 0. \tag{11.5}$$

The sequence $\{\lambda_k\}$ is called an admissible concentration sequence at z_*.

Remark 11.5. Item (iv) is the velocity-only concentration supplied by the local concentration theorem; see Proposition 11.7. Thus, for a singular point, the additional admissibility requirement is the no-escape condition (11.5). It is the sole terminal-slice compactness assumption used to prove nontriviality of the ancient limit. No uniform bound on $A + C + D + E$ over all terminal scales is assumed.

Lemma 11.6 (Uniform kinetic energy implies no escape)**.** *Let* (u, p) *satisfy the Type I bound* (10.2)*, and let* $\lambda_k \downarrow 0$*. Suppose that*

$$\sup_k A(u; z_*, \lambda_k) \leq B < \infty.$$

Then the corresponding rescaled velocities satisfy the no-escape condition (11.5)*.*

Proof. Choose $M < \infty$ and $t_0 < T$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq M(T-t)^{-1/2}, \qquad t_0 < t < T.$$

For k sufficiently large, $T + \lambda_k^2 t > t_0$ whenever $-1 < t < 0$. Hence

$$\|u_k(t)\|_{L^\infty(B_1)} \leq M(-t)^{-1/2}, \qquad -1 < t < 0.$$

By scale invariance of A,

$$\operatorname*{ess\,sup}_{-1<t<0} \int_{B_1} |u_k(x,t)|^2 \, dx = A(u; z_*, \lambda_k) \le B.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^{0} \int_{B_1} |u_k|^3 \, dx \, dt &\le \int_{-\sigma}^{0} \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x,t)|^2 \, dx \, dt \\ &\le MB \int_{-\sigma}^{0} (-t)^{-1/2} \, dt = 2MB\sigma^{1/2}. \end{aligned}$$

Taking the supremum over the tail $k \ge k_0$ and then letting $\sigma \downarrow 0$ gives (11.5) for the tail of the sequence. For each of the finitely many remaining indices $1 \le k < k_0$, the rescaled velocity is defined on $B_1 \times (-\delta_k, 0)$ for some $\delta_k > 0$, and belongs to L^3 there. Therefore absolute continuity of the L^3-integral gives

$$\lim_{\sigma \downarrow 0} \max_{1 \le k < k_0} \int_{-\sigma}^{0} \int_{B_1} |u_k|^3 \, dx \, dt = 0.$$

Combining the finite initial segment with the tail proves the full supremum in (11.5). □

Proposition 11.7 (Local velocity concentration)**.** *Let (u,p) be suitable in a parabolic neighborhood of $z_* = (x_*, T)$. If z_* is singular, then there exist $\lambda_k \downarrow 0$ and $\eta_v > 0$ such that*

$$C(u; z_*, \lambda_k) \ge \eta_v \qquad \text{for every } k.$$

Proof. This is the velocity-only concentration corollary of the local concentration result in Theorem 8.3. That result follows from the velocity ε-regularity criterion: if

$$\limsup_{r \downarrow 0} C(u; z_*, r) = 0,$$

then u is bounded in a backward parabolic cylinder ending at z_*, contradicting singularity of z_*. □

Lemma 11.8 (Scaling)**.** *Let (u,p) be suitable in a neighborhood of (x_*, T). For $\lambda > 0$, set*

$$u^\lambda(x,t) = \lambda u(x_* + \lambda x, T + \lambda^2 t), \qquad p^\lambda(x,t) = \lambda^2 p(x_* + \lambda x, T + \lambda^2 t).$$

Then (u^λ, p^λ) is suitable on the rescaled domain. Moreover A, C, D, E are invariant under this scaling.

Proof. Suitability of the rescaled pair and preservation of pressure gauges follow by applying (11.1) to the test function

$$\phi^\lambda(X,S) = \phi\left(\frac{X - x_*}{\lambda}, \frac{S - T}{\lambda^2}\right)$$

and then changing variables $(X,S) = (x_* + \lambda x, T + \lambda^2 t)$. The differential identities

$$\begin{aligned} \partial_t u^\lambda = \lambda^3 (\partial_t u)(X,S), &\quad (u^\lambda \cdot \nabla) u^\lambda = \lambda^3 ((u \cdot \nabla) u)(X,S), \\ \nabla p^\lambda = \lambda^3 (\nabla p)(X,S), &\qquad \Delta u^\lambda = \lambda^3 (\Delta u)(X,S) \end{aligned}$$

give the rescaled equation and the divergence condition.

For the pressure average,

$$(p^\lambda)_{B_r}(t) = \lambda^2 (p)_{B_{\lambda r}(x_*)}(S).$$

Consequently

$$\iint_{Q_r} |u^\lambda|^3 \, dx \, dt = \lambda^{-2} \iint_{Q_{\lambda r}(z_*)} |u|^3 \, dX \, dS$$

and

$$\iint_{Q_r} |p^\lambda - (p^\lambda)_{B_r}(t)|^{3/2}\,dx\,dt = \lambda^{-2}\iint_{Q_{\lambda r}(z_*)} |p - (p)_{B_{\lambda r}(x_*)}(S)|^{3/2}\,dX\,dS.$$

Multiplication by r^{-2} therefore gives invariance of C and D. The remaining two quantities have the same critical dimensional balance:

$$\int_{B_r} |u^\lambda(x,t)|^2\,dx = \lambda^{-1}\int_{B_{\lambda r}(x_*)} |u(X,S)|^2\,dX,$$
$$\iint_{Q_r} |\nabla u^\lambda|^2\,dx\,dt = \lambda^{-1}\iint_{Q_{\lambda r}(z_*)} |\nabla u|^2\,dX\,dS.$$

Multiplication by r^{-1} on the rescaled cylinder is the same as multiplication by $(\lambda r)^{-1}$ on the original cylinder. Hence A and E are invariant as well. □

Lemma 11.9 (Pressure gauges do not change the equations)**.** *Let (v,q) be suitable in a cylinder Q, and let $\alpha(t) \in L^{3/2}_{\mathrm{loc}}$ depend only on time. Then $(v, q-\alpha(t))$ satisfies the same distributional Navier–Stokes equation, the same divergence condition, and the same local energy inequality.*

Proof. In the distributional equation, replacing q by $q - \alpha(t)$ adds $-\nabla\alpha(t) = 0$. In the local energy inequality the only new term is

$$-2\int \alpha(t)v(x,t)\cdot\nabla\phi(x,t)\,dx\,dt.$$

The product is integrable on the support of ϕ: by suitability, $v \in L^\infty_t L^2_x$ locally, and

$$\left|\int v(\cdot,t)\cdot\nabla\phi(\cdot,t)\,dx\right| \le \|v(t)\|_{L^2(\operatorname{supp}\phi(t))}\|\nabla\phi(t)\|_{L^2}.$$

For almost every t, $\operatorname{div} v(\cdot,t) = 0$ in distributions and $\phi(\cdot,t)$ is compactly supported, so

$$\int v(x,t)\cdot\nabla\phi(x,t)\,dx = 0.$$

Thus the additional term vanishes. □

12. The Type I blow-up sequence and compactness

12.1. **The rescaled sequence.** Choose $M < \infty$ so that, after decreasing $T_0 < T$ if necessary,

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{T-t}}, \qquad T_0 < t < T. \tag{12.1}$$

For $\lambda_k \downarrow 0$, define

$$u_k(x,t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t), \qquad p_k(x,t) = \lambda_k^2 p(x_* + \lambda_k x, T + \lambda_k^2 t). \tag{12.2}$$

Then u_k is defined on $\mathbb{R}^3 \times (-T/\lambda_k^2, 0)$, and these domains exhaust $\mathbb{R}^3 \times (-\infty, 0)$.

12.2. **Inherited Type I bounds.**

Lemma 12.1 (Inherited Type I bound)**.** *For every compact interval $I \Subset (-\infty, 0)$,*

$$\sup_k \|u_k\|_{L^\infty(\mathbb{R}^3\times I)} < \infty.$$

More precisely, for every fixed $t < 0$ and all sufficiently large k,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{-t}}.$$

Proof. Since $T-(T+\lambda_k^2 t)=\lambda_k^2(-t)$, (12.1) gives, for all large k,

$$\|u_k(t)\|_\infty=\lambda_k\|u(T+\lambda_k^2 t)\|_\infty\le \lambda_k\frac{M}{\sqrt{\lambda_k^2(-t)}}=\frac{M}{\sqrt{-t}}.$$

Taking the supremum over $t\in I$ gives the interval statement. □

12.3. **Local energy, pressure, and time-derivative estimates.**

Lemma 12.2 (Pressure gauges and local pressure bounds)**.** *After fixing the whole-space Leray pressure representative below, set*

$$a_k(t)=(p_k)_{B_1}(t)$$

whenever the rescaled solution is defined. Then for every

$$Q=B_R\times[-S,-\sigma]\Subset\mathbb{R}^3\times(-\infty,0)$$

one has

$$\sup_k\|p_k-a_k(t)\|_{L^{3/2}(Q)}<\infty.$$

The estimates below use the global pointwise Type I bound encoded in the admissible Type I assumption Definition 11.4 to control L^2-mass on every annulus uniformly in k; the corresponding local-Type-I version replaces the outer annular control by the Leray finite-energy tail estimate already recorded in Proposition 9.2.

Proof. For large k, the original times $T+\lambda_k^2 t$, $t\in[-S,-\sigma]$, lie in the interval where (12.1) holds. At the point where the whole-space Riesz pressure representation is used, the ambient solution is a suitable Leray–Hopf solution on $\mathbb{R}^3\times(0,T)$ in the sense of Definition 11.3; the pressure representative is therefore chosen, modulo functions of time, as the whole-space Leray pressure

$$p=\mathcal{R}_i\mathcal{R}_j(u_iu_j).$$

On those slices $u(t)\in L^2(\mathbb{R}^3)\cap L^\infty(\mathbb{R}^3)$, hence

$$u_i(t)u_j(t)\in L^1(\mathbb{R}^3)\cap L^{3/2}(\mathbb{R}^3).$$

Use the whole-space Leray pressure representative

$$p^L=\mathcal{R}_i\mathcal{R}_j(u_iu_j),$$

modulo functions of time. The standard convention is

$$\widehat{\mathcal{R}_i f}(\xi)=i\xi_i|\xi|^{-1}\widehat{f}(\xi),$$

so that $\mathcal{R}_i\mathcal{R}_j$ has multiplier $-\xi_i\xi_j|\xi|^{-2}$ and $-\Delta(\mathcal{R}_i\mathcal{R}_j f)=\partial_i\partial_j f$. Thus this is the pressure reconstruction from $-\Delta p=\partial_i\partial_j(u_iu_j)$ and the Calderon–Zygmund theory of Riesz transforms [12, 19]. Since $u(t)\in L^2(\mathbb{R}^3)\cap L^\infty(\mathbb{R}^3)$, interpolation gives $u(t)\in L^3(\mathbb{R}^3)$, hence $u_i(t)u_j(t)\in L^{3/2}(\mathbb{R}^3)$ and $p^L(t)\in L^{3/2}(\mathbb{R}^3)$ for a.e. t.

The given pressure and p^L differ by a function of time. Indeed, applying the Helmholtz projection to the whole-space weak momentum equation gives

$$\partial_t u-\Delta u+\mathbb{P}\operatorname{div}(u\otimes u)=0$$

in solenoidal distributions. The complementary gradient part of $\operatorname{div}(u\otimes u)$ is precisely $-\nabla p^L$ with the convention above. Therefore the same velocity field solves the distributional equation with p^L in place of p. Subtracting the two formulations gives

$$\nabla(p-p^L)=0\quad\text{in }\mathcal{D}'(\mathbb{R}^3)$$

for a.e. time. By the de Rham theorem on $\mathbb{R}^3$, $p-p^L$ is spatially constant for a.e. time. Replacing the pressure by p^L is therefore a time-dependent gauge change and does not affect suitability by Lemma 11.9.

Fix $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R = 1$ on B_{4R}. For $t \in [-S, -\sigma]$, choose the representative so that, in B_{2R},

$$p_k = q_k + h_k, \qquad q_k = \mathcal{R}_i\mathcal{R}_j(\chi_R u_{k,i} u_{k,j}),$$

and $h_k(\cdot, t)$ is harmonic in B_{4R} for a.e. t. This choice is made before $a_k(t) = (p_k)_{B_1}(t)$ is evaluated. The harmonicity holds because $(1-\chi_R)u_{k,i}u_{k,j}$ is supported outside B_{4R}, so the localized Riesz potential generated by this exterior part has zero Laplacian in B_{4R}. Calderon–Zygmund boundedness gives, for a.e. t,

$$\|q_k(t)\|_{L^{3/2}(B_{2R})} \le C\|u_k(t)\|^2_{L^3(B_{8R})} \le C(R, \sigma, M).$$

Integrating over $[-S, -\sigma]$ gives a uniform $L^{3/2}_{x,t}$-bound for q_k. Let $c_{k,R}(t) = h_k(0, t)$, where $h_k(\cdot, t)$ is represented by its smooth harmonic representative. For $x \in B_R$, the kernel estimate

$$|K_{ij}(x-z) - K_{ij}(-z)| \le CR|z|^{-4}, \qquad |z| > 4R,$$

implies

$$|h_k(x,t) - c_{k,R}(t)| \le CR \sum_{m=2}^{\infty} (2^m R)^{-4} \int_{2^m R < |z| < 2^{m+1} R} |u_k(z,t)|^2 \, dz.$$

Using Lemma 12.1,

$$\int_{2^m R<|z|<2^{m+1}R} |u_k(z,t)|^2 \, dz \le CM^2\sigma^{-1}(2^m R)^3,$$

and therefore

$$|h_k(x,t) - c_{k,R}(t)| \le CM^2\sigma^{-1} \sum_{m=2}^{\infty} 2^{-m}, \qquad x \in B_R,$$

which implies

$$\|h_k(t) - c_{k,R}(t)\|_{L^\infty(B_R)} \le C(M, \sigma).$$

Thus

$$\sup_k \|p_k - c_{k,R}(t)\|_{L^{3/2}(B_R \times [-S,-\sigma])} < \infty.$$

If $R < 1$, enlarge R to 1. Otherwise $B_1 \subset B_R$. Jensen's inequality gives, for a.e. t,

$$|a_k(t) - c_{k,R}(t)|^{3/2} \le |B_1|^{-1} \int_{B_1} |p_k - c_{k,R}(t)|^{3/2} \, dx.$$

Integrating in time and adding constants on B_R gives the desired bound for $p_k - a_k(t)$. □

Lemma 12.3 (Compatibility of pressure gauges)**.** *Let $B_1 \subset B_R$, $I \Subset (-\infty, 0)$, and suppose*

$$p_k - c_k(t) \quad \textit{is bounded in } L^{3/2}(B_R \times I)$$

for some time-dependent gauges $c_k(t)$. If $a_k(t) = (p_k)_{B_1}(t)$, then $p_k - a_k(t)$ is bounded in $L^{3/2}(B_{R'} \times I)$ for every $R' \le R$. Changing between such gauges changes the pressure only by a function of time and therefore does not affect the equation, the local energy inequality, or the centered equation.

Proof. For a.e. t, Jensen's inequality gives

$$|a_k(t) - c_k(t)|^{3/2} \leq |B_1|^{-1} \int_{B_1} |p_k - c_k(t)|^{3/2}\, dx.$$

Integrating in t and using $B_{R'} \subset B_R$ yields

$$\|a_k - c_k\|_{L^{3/2}(I)}^{3/2} \leq |B_1|^{-1} \|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}^{3/2}.$$

Thus

$$\|p_k - a_k(t)\|_{L^{3/2}(B_{R'} \times I)} \leq \|p_k - c_k(t)\|_{L^{3/2}(B_{R'} \times I)} + |B_{R'}|^{2/3} \|a_k - c_k\|_{L^{3/2}(I)}$$

and the right-hand side is bounded by

$$C(R')\|p_k - c_k(t)\|_{L^{3/2}(B_R \times I)}.$$

This proves the asserted bound. The last assertion is Lemma 11.9, and centering transforms the time-dependent gauge into another time-dependent gauge. □

Lemma 12.4 (Pressure gauge atlas)**.** *Let (u_n, p_n) be a rescaled sequence of suitable weak solutions satisfying the local compactness bounds on every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$. For every ball B_R and compact time interval $I \Subset (-\infty, 0)$ there are time-dependent functions $a_{n,R}(t)$ such that*

$$p_n - a_{n,R}(t) \quad \textit{is bounded in } L^{3/2}(B_R \times I).$$

On nested balls, any two such gauges for the same n differ by functions of time only. After passing to a diagonal subsequence over R and I, the gauge-fixed pressures converge weakly in $L^{3/2}_{\mathrm{loc}}$ to a pressure Q defined modulo functions of time. These gauge changes do not affect the distributional Navier–Stokes equations or the local energy inequality, and the same compatibility passes to pressure limits.

Proof. The local pressure bounds are those proved in Lemma 12.2 for the global Type I extraction and in Proposition 9.2 for the local Type I rescaling construction. On each fixed $B_R \times I$, choose any of the gauges provided there, for instance a spatial average on a unit ball contained in B_R after enlarging R if necessary. Compatibility on nested balls is Lemma 12.3: subtracting two admissible representatives leaves the same pressure gradient, hence the difference is spatially constant for a.e. time. A countable exhaustion by balls and compact time intervals, together with Banach–Alaoglu in $L^{3/2}$, gives a diagonal subsequence and a compatible local pressure limit. Invariance of the equation and local energy inequality under time-only pressure changes is Lemma 11.9. □

Remark 12.5 (Pressure-gauge atlas)*.* In the sequel, a pressure limit Q means a compatible family of local representatives modulo time-dependent gauges. On overlaps, the gradients agree; hence the representatives differ only by functions of time. This is the pressure atlas of Lemma 12.4 for local suitable limits. When local compactness is run on a fixed compact cylinder K, we use the gauge $a_{k,K}(t) = (p_k)_{B_1}(t)$ supplied by Lemma 12.2, and Lemma 12.3 identifies a single pressure limit on overlaps modulo functions of time.

Lemma 12.6 (Local energy and dissipation)**.** *Let*

$$Q = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0).$$

Then

$$\sup_k \|u_k\|_{L^\infty_t L^2_x(Q)} + \sup_k \|\nabla u_k\|_{L^2(Q)} < \infty.$$

Proof. The $L_t^\infty L_x^2$ estimate follows from

$$\|u_k(t)\|_{L^2(B_R)} \le |B_R|^{1/2} M\sigma^{-1/2}, \qquad t \in [-S, -\sigma].$$

Choose $\phi \in C_c^\infty(B_{2R} \times [-2S, -\sigma/2])$, $0 \le \phi \le 1$, with $\phi = 1$ on Q. Apply the local energy inequality to $(u_k, p_k - a_k(t))$, using Lemma 12.2 on the larger cylinder. For a.e. admissible times and then by monotone approximation of temporal cutoffs,

$$2 \iint |\nabla u_k|^2 \phi \le \iint |u_k|^2 |\partial_t \phi + \Delta\phi| + \iint (|u_k|^2 + 2|p_k - a_k(t)|)|u_k||\nabla\phi|.$$

The first term is bounded because $\partial_t\phi + \Delta\phi$ is bounded and u_k is uniformly bounded in L^∞ on the finite-measure support of ϕ. The cubic velocity term satisfies

$$\iint |u_k|^3 |\nabla\phi| \le \|\nabla\phi\|_\infty |\operatorname{supp}\nabla\phi|\, \|u_k\|^3_{L^\infty(\operatorname{supp}\nabla\phi)} \le C(Q, \phi, M, \sigma).$$

For the pressure term, Hölder's inequality on $\operatorname{supp}\nabla\phi$ gives

$$\iint |p_k - a_k(t)|\, |u_k|\, |\nabla\phi| \le \|p_k - a_k(t)\|_{L^{3/2}} \|u_k \nabla\phi\|_{L^3},$$

and both factors are uniformly bounded by Lemma 12.2 and Lemma 12.1. Since $\phi = 1$ on Q, the dissipation bound follows. ☐

Lemma 12.7 (Time derivative bound)**.** *For every compact cylinder $Q \Subset \mathbb{R}^3 \times (-\infty, 0)$,*

$$\partial_t u_k \quad \textit{is bounded in } L_t^{3/2} W_x^{-1,3/2}(Q).$$

Proof. The equation is

$$\partial_t u_k = \Delta u_k - \operatorname{div}(u_k \otimes u_k) - \nabla(p_k - a_k(t)).$$

Let $Q = B_R \times [-S, -\sigma]$. Throughout the proof we use the embedding $W_0^{1,3}(B_R) \hookrightarrow H_0^1(B_R)$ on the bounded domain B_R; duality then gives $H^{-1}(B_R) \hookrightarrow W^{-1,3/2}(B_R)$, so that any $L_t^2 H_x^{-1}(Q)$-bound becomes an $L_t^{3/2} W_x^{-1,3/2}(Q)$-bound by Hölder in time on the finite interval $[-S, -\sigma]$. The term Δu_k is bounded in $L_t^2 H_x^{-1}(Q)$ by Lemma 12.6; since $W_0^{1,3}(B_R) \subset H_0^1(B_R)$ and the time interval is finite, this is also an $L_t^{3/2} W_x^{-1,3/2}$-bound. Explicitly, for $\zeta \in W_0^{1,3}(B_R)$,

$$|\langle \Delta u_k, \zeta\rangle| = \left| \int_{B_R} \nabla u_k : \nabla\zeta \, dx \right| \le C(R) \|\nabla u_k\|_{L^2(B_R)} \|\zeta\|_{W^{1,3}(B_R)}.$$

The L_t^2-bound on ∇u_k, together with the finite length of $[-S, -\sigma]$, gives the claimed $L_t^{3/2}$-bound.

For the nonlinear term, for $\zeta \in W_0^{1,3}(B_R)$,

$$\left| \int_{B_R} (u_k \otimes u_k) : \nabla\zeta \, dx \right| \le \|u_k \otimes u_k\|_{L^{3/2}(B_R)} \|\nabla\zeta\|_{L^3(B_R)}.$$

Moreover

$$\|u_k \otimes u_k\|_{L^{3/2}(Q)} \le \|u_k\|_{L^\infty(Q)} \|u_k\|_{L^{3/2}(Q)} \le C(Q).$$

Thus $\operatorname{div}(u_k \otimes u_k)$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. For the pressure term,

$$\left| \int_{B_R} (p_k - a_k(t)) \operatorname{div}\zeta \, dx \right| \le \|p_k - a_k(t)\|_{L^{3/2}(B_R)} \|\nabla\zeta\|_{L^3(B_R)}.$$

Lemma 12.2 gives

$$p_k - a_k(t) \quad \text{bounded in } L^{3/2}(Q).$$

Hence $\nabla(p_k - a_k(t))$ is bounded in $L_t^{3/2} W_x^{-1,3/2}(Q)$. Summing the three terms gives the result. ☐

12.4. **Compactness away from the terminal time.**

Proposition 12.8 (Compactness)**.** *After passing to a subsequence,*

$$u_k \to U$$

strongly in $L^q_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0))$ *for every finite* q*, and*

$$p_k - a_k(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times (-\infty, 0)).$$

Proof. Fix $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$. By Lemma 12.6, u_k is bounded in $L^2_t H^1_x(Q_0)$, and by Lemma 12.7, $\partial_t u_k$ is bounded in $L^{3/2}_t W^{-1,3/2}_x(Q_0)$. On bounded balls,

$$H^1 \Subset L^2 \hookrightarrow W^{-1,3/2}.$$

The Aubin–Lions–Simon theorem [13, 16, 18] gives relative compactness in $L^2(Q_0)$. Hence, after passing to a subsequence, $u_k \to U$ strongly in $L^2(Q_0)$ and a.e. on Q_0.

The inherited Type I bound gives a common $L^\infty(Q_0)$ bound, and the limit belongs to the same L^∞ ball by a.e. convergence. For $q \geq 2$,

$$\|u_k - U\|^q_{L^q(Q_0)} \leq (2 \sup_j \|u_j\|_{L^\infty(Q_0)} + 2\|U\|_{L^\infty(Q_0)})^{q-2} \|u_k - U\|^2_{L^2(Q_0)}.$$

For $1 \leq q < 2$, Hölder's inequality on the finite-measure set Q_0 gives

$$\|u_k - U\|_{L^q(Q_0)} \leq |Q_0|^{1/q-1/2} \|u_k - U\|_{L^2(Q_0)} \to 0.$$

Choose an exhaustion $Q^{(m)} \Subset \mathbb{R}^3 \times (-\infty, 0)$ by compact cylinders. The preceding argument gives a subsequence converging on $Q^{(1)}$; extracting successively on $Q^{(m)}$ and taking the diagonal subsequence gives strong local convergence on every compact cylinder.

For pressure, Lemma 12.2 gives boundedness of $p_k - a_k(t)$ in $L^{3/2}(Q^{(m)})$ for every m. Since $L^{3/2}$ is reflexive, each restriction has a weakly convergent subsequence. The same diagonal extraction gives a single limit Q in $L^{3/2}_{\mathrm{loc}}$, because the weak limits agree on overlaps by uniqueness of distributional limits. ∎

12.5. **The ancient suitable weak limit.**

Lemma 12.9 (Limit equation)**.** *The limit* (U, Q) *solves*

$$\partial_t U + (U \cdot \nabla) U + \nabla Q = \Delta U, \qquad \operatorname{div} U = 0$$

in distributions on $\mathbb{R}^3 \times (-\infty, 0)$*.*

Proof. For $\varphi \in C^\infty_c(\mathbb{R}^3 \times (-\infty, 0); \mathbb{R}^3)$, the weak formulation for $(u_k, p_k - a_k(t))$ is

$$\iint [-u_k \cdot \partial_t \varphi - (u_k \otimes u_k) : \nabla\varphi + (p_k - a_k(t)) \operatorname{div}\varphi + \nabla u_k : \nabla\varphi] \, dx\, dt = 0.$$

The linear time term passes by strong L^2_{loc}-convergence of u_k. The pressure term passes by weak $L^{3/2}_{\mathrm{loc}}$-convergence of $p_k - a_k(t)$, since $\operatorname{div}\varphi \in L^3$. The viscous term passes by weak L^2_{loc}-convergence of ∇u_k, which follows from the uniform local dissipation bound. The nonlinear term passes because

$$\|u_k \otimes u_k - U \otimes U\|_{L^1(\operatorname{supp}\varphi)} \leq \|u_k - U\|_{L^2} \big(\|u_k\|_{L^2} + \|U\|_{L^2}\big) \to 0.$$

Testing with gradients of scalar functions gives $\operatorname{div} U = 0$. ∎

Lemma 12.10 (Stability of suitability)**.** *The pair* (U, Q) *is a suitable weak solution on* $\mathbb{R}^3 \times (-\infty, 0)$*.*

Proof. Use the distributional form of the local energy inequality. For every non-negative $\phi \in C_c^\infty(\mathbb{R}^3 \times (-\infty, 0))$,

$$2 \iint |\nabla u_k|^2 \phi \le \iint |u_k|^2 (\partial_t \phi + \Delta \phi) + \iint (|u_k|^2 + 2P_k) u_k \cdot \nabla \phi, \qquad P_k = p_k - a_k(t).$$

This follows from (11.1) by applying (11.1) with temporal cutoffs which increase to 1 on the time projection of $\operatorname{supp} \phi$. The monotone convergence theorem passes the nonnegative dissipation term to the displayed distributional form.

The terms with $|u_k|^2$ converge strongly in L^1 because

$$\| |u_k|^2 - |U|^2 \|_{L^1(\operatorname{supp} \phi)} \le \|u_k - U\|_{L^2} \big(\|u_k\|_{L^2} + \|U\|_{L^2} \big) \to 0.$$

The cubic flux converges strongly since $u_k \to U$ strongly in L^3_{loc}, hence

$$|u_k|^2 u_k \to |U|^2 U \quad \text{in } L^1_{\mathrm{loc}}.$$

For the pressure flux, $u_k \cdot \nabla \phi \to U \cdot \nabla \phi$ strongly in L^3, while $P_k \rightharpoonup Q$ weakly in $L^{3/2}$. Therefore

$$\iint P_k u_k \cdot \nabla \phi \to \iint Q\, U \cdot \nabla \phi.$$

Finally, $\nabla u_k \rightharpoonup \nabla U$ in L^2_{loc}, and

$$w \mapsto \iint |w|^2 \phi$$

is weakly lower semicontinuous because $\phi \ge 0$ and the expression equals $\|\sqrt{\phi}\, w\|_{L^2}^2$. Passing to the limit gives the local energy inequality for (U, Q) in distributional form. The endpoint form in Definition 11.2 is recovered by applying the distributional inequality to standard time cutoffs approximating characteristic functions of (t_1, t_2); weak L^2_{loc}-continuity of local energy-class solutions gives the endpoint traces for a.e. $t_1 < t_2$. ☐

Lemma 12.11 (Inherited ancient Type I bound)**.** *For every compact interval $I = [-S, -\sigma] \Subset (-\infty, 0)$,*

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I)} \le M \sigma^{-1/2}.$$

Proof. Lemma 12.1 gives

$$\|u_k\|_{L^\infty(\mathbb{R}^3 \times I)} \le M \sigma^{-1/2}.$$

Fix $R < \infty$. Since $u_k \to U$ a.e. on $B_R \times I$, the essential supremum of U on $B_R \times I$ cannot exceed $M\sigma^{-1/2}$. If it did, $|u_k|$ would exceed this bound by a positive amount on a positive measure subset for large k. Letting $R \to \infty$ through integers gives the result on $\mathbb{R}^3 \times I$. ☐

Proposition 12.12 (Ancient suitable Type I limit)**.** *The pair (U, Q) is an ancient suitable weak solution on $\mathbb{R}^3 \times (-\infty, 0)$ satisfying the interval Type I bound in Lemma 12.11.*

Proof. Combine Lemmas 12.9 to 12.11. ☐

13. Persistence of concentration and nontriviality

13.1. **Persistence of concentration away from the terminal time.** The persistence of concentration is tracked through the velocity concentration sequence in Definition 11.4, supplied by Proposition 11.7. The no-escape condition says that this velocity mass does not concentrate entirely in the terminal layer $t = 0$ after rescaling.

13.2. **Scale selection.**

Lemma 13.1 (Scale selection from local concentration)**.** *Let $\lambda_k \downarrow 0$ be an admissible concentration sequence in the sense of Definition 11.4. Then there are $\sigma \in (0,1)$, a compact cylinder*

$$K_0 = B_1 \times (-1, -\sigma) \Subset \mathbb{R}^3 \times (-\infty, 0),$$

and $\eta_1 > 0$ such that

$$\iint_{K_0} |u_k|^3 \, dx \, dt \geq \eta_1 \tag{13.1}$$

for all k.

Proof. By scale invariance of C,

$$\int_{-1}^{0} \int_{B_1} |u_k|^3 \, dx \, dt = C(u; z_*, \lambda_k) \geq \eta_v.$$

The no-escape condition (11.5) gives $\sigma \in (0,1)$ such that

$$\sup_k \int_{-\sigma}^{0} \int_{B_1} |u_k|^3 \, dx \, dt \leq \frac{\eta_v}{2}.$$

Therefore

$$\int_{-1}^{-\sigma} \int_{B_1} |u_k|^3 \, dx \, dt \geq \frac{\eta_v}{2}$$

for every k. The assertion follows with $\eta_1 = \eta_v/2$. □

13.3. **Nonzero ancient limits.**

Proposition 13.2 (Nonzero ancient limit)**.** *For the scales chosen in Lemma 13.1, the limit U in Proposition 12.12 is nonzero. More precisely, there are a compact cylinder $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$ and $\eta_1 > 0$ such that*

$$\iint_{K_0} |U|^3 \, dx \, dt \geq \eta_1.$$

Proof. Let $K_0 = B_1 \times (-1, -\sigma)$ and $\eta_1 > 0$ be given by Lemma 13.1. Since $K_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, the compactness theorem gives, after passing to the subsequence used to define U,

$$u_k \to U \qquad \text{strongly in } L^3(K_0).$$

Hence

$$\int_{K_0} |U|^3 \, dx \, dt = \lim_{k \to \infty} \int_{K_0} |u_k|^3 \, dx \, dt \geq \eta_1.$$

□

Thus concentration generated by the singular point propagates through the compactness limit. The velocity concentration and no-escape condition are the hypotheses that prevent the ancient limit from being identically zero.

14. Centered variables and normalized Seregin limits

14.1. The centered self-similar equation. Set

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(y\sqrt{-t}, t),$$

and

$$\Pi(y,\tau) = (-t)Q(y\sqrt{-t}, t),$$

up to addition of a function of τ.

Remark 14.1 (Pressure gauge transformation in centered variables). If Q is replaced by $Q + a(t)$ for a function a of t only, then Π is replaced by $\Pi + e^{-\tau}a(-e^{-\tau})$, which is a function of τ only. Thus the centered equation (10.3) is gauge-invariant: changing the pressure by a function of physical time amounts to changing the centered pressure by a function of centered time.

Lemma 14.2 (Centered pullback identities). *The centered and physical variables are related by*

$$t = -e^{-\tau}, \qquad x = e^{-\tau/2}y, \qquad U(x,t) = e^{\tau/2}V(y,\tau),$$

or equivalently

$$V(y,\tau) = e^{-\tau/2}U(e^{-\tau/2}y, -e^{-\tau}).$$

For every $\tau \in \mathbb{R}$, with $t = -e^{-\tau}$,

$$\int_{\mathbb{R}^3} |U(x,t)|^3\, dx = \int_{\mathbb{R}^3} |V(y,\tau)|^3\, dy.$$

Proof. The first identities are the definitions with $-t = e^{-\tau}$. Since

$$|U(x,t)|^3 = e^{3\tau/2}|V(y,\tau)|^3, \qquad dx = e^{-3\tau/2}\, dy,$$

the critical L^3 identity follows by multiplication and integration. ∎

Lemma 14.3 (Renormalized equation). *The pair (V,Π) solves (10.3).*

Proof. Write $s = -t = e^{-\tau}$, $x = s^{1/2}y$, and

$$U(x,t) = s^{-1/2}V(y,\tau).$$

The identities are first obtained in distributions by testing the equation for U against the inverse change of variables. After Lemma 14.4, the same identities hold classically for the smooth representative. Because $s_t = -1$, $\tau_t = s^{-1}$, and $y_t = (2s)^{-1}y$,

$$\partial_t U = s^{-3/2}\left(\partial_\tau V + \frac{1}{2}V + \frac{1}{2}y\cdot\nabla V\right),$$

$$(U\cdot\nabla_x)U = s^{-3/2}(V\cdot\nabla)V, \qquad \Delta_x U = s^{-3/2}\Delta V, \qquad \nabla_x Q = s^{-3/2}\nabla\Pi.$$

Substitution into the equation for U gives (10.3), and the divergence condition is preserved by the spatial scaling. ∎

14.2. **Boundedness, smoothness, and local energy.**

Lemma 14.4 (Smoothness)**.** *The ancient solution U is smooth on $\mathbb{R}^3 \times (-\infty, 0)$, and (V, Π) is smooth on $\mathbb{R}^3 \times \mathbb{R}$ after fixing a pressure gauge.*

Proof. On each compact cylinder $Q_0 \Subset \mathbb{R}^3 \times (-\infty, 0)$, Lemma 12.11 gives $U \in L^\infty(Q_0)$. Hence $U \in L^s_t L^r_x(Q_0)$ for every finite r, s. Choose $r = s = 10$, so

$$\frac{2}{s} + \frac{3}{r} = \frac{1}{2} < 1.$$

Serrin's interior regularity criterion [11, 19] applies and gives local Hölder regularity of U on a slightly smaller cylinder. Since Q_0 was arbitrary, this removes every possible interior singular point in $\mathbb{R}^3 \times (-\infty, 0)$. The bootstrap starts from

$$\partial_t U - \Delta U + \nabla Q = -\operatorname{div}(U \otimes U), \qquad \operatorname{div} U = 0,$$

with $U \otimes U$ locally bounded after Serrin regularity. Interior Stokes estimates on nested cylinders give $W^{2,1}_q$-regularity for every finite q. Parabolic Sobolev embedding then improves the Hölder regularity of ∇U, and differentiating the equation and iterating gives

$$U \in C^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

The pressure is recovered locally from $-\Delta Q = \partial_i \partial_j (U_i U_j)$ plus a harmonic function and is smooth after fixing a time-dependent gauge. The centered change of variables is a smooth diffeomorphism for $t < 0$, so V and Π are smooth. □

Corollary 14.5 (Classical Type I bound)**.** *For the smooth representative from Lemma 14.4,*

$$\|U(t)\|_{L^\infty(\mathbb{R}^3)} \le \frac{M}{\sqrt{-t}}, \qquad t < 0.$$

Proof. Fix $t_0 < 0$. Choose $\delta_n \downarrow 0$ with

$$I_n = [t_0 - \delta_n, t_0 + \delta_n] \Subset (-\infty, 0).$$

Lemma 12.11 gives

$$\|U\|_{L^\infty(\mathbb{R}^3 \times I_n)} \le M(-t_0 - \delta_n)^{-1/2}.$$

Since the representative is continuous, the essential spacetime bound holds pointwise. Letting $n \to \infty$ gives the estimate at t_0. □

Lemma 14.6 (Renormalized boundedness and nontriviality)**.** *The centered field satisfies*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \le M.$$

Moreover (10.4) *holds for some compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and $\eta_0 > 0$.*

Proof. The L^∞ bound follows from Corollary 14.5:

$$|V(y, \tau)| = \sqrt{-t}\,|U(y\sqrt{-t}, t)| \le M.$$

For nontriviality, Proposition 13.2 gives a positive local L^3 lower bound on a compact cylinder in (x, t). Under $(x, t) = (e^{-\tau/2} y, -e^{-\tau})$,

$$dx\,dt = e^{-5\tau/2}\,dy\,d\tau, \qquad U = e^{\tau/2} V,$$

and hence

$$|U|^3\,dx\,dt = e^{-\tau}|V|^3\,dy\,d\tau.$$

On the compact image cylinder the factor $e^{-\tau}$ is bounded above and below by positive constants, so the positive lower bound becomes (10.4), with a possibly different η_0. □

Lemma 14.7 (Local energy bounds in centered variables)**.** *Let V be obtained from an admissible finite-energy Type I singularity. For every $R < \infty$ and compact interval $I \subset \mathbb{R}$,*

$$\int_I \int_{B_R} |V|^2 \, dy \, d\tau + \int_I \int_{B_R} |\nabla V|^2 \, dy \, d\tau < \infty.$$

After subtracting a function of τ, the associated pressure satisfies

$$\Pi - b(\tau) \in L^{3/2}(B_R \times I).$$

Proof. The unrenormalized limit (U, Q) is suitable and has local energy, dissipation, and pressure bounds on compact negative-time cylinders by Lemmas 12.2 and 12.6 and Proposition 12.12; the bounds for the limit follow from the weak lower semicontinuity and weak compactness already used in Lemma 12.10. On each compact (y, τ)-cylinder, the map

$$(y, \tau) \mapsto (x, t) = (e^{-\tau/2} y, -e^{-\tau})$$

has Jacobian $dx \, dt = e^{-5\tau/2} \, dy \, d\tau$, with

$$V = e^{-\tau/2} U, \qquad \nabla_y V = e^{-\tau} \nabla_x U, \qquad \Pi = e^{-\tau} Q.$$

If $I = [\tau_1, \tau_2]$, then $e^{-\tau}$ is bounded above and below by positive constants on I. Therefore

$$\int_I \int_{B_R} |V|^2 \, dy \, d\tau \le C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |U|^2 \, dx \, dt,$$

and similarly

$$\int_I \int_{B_R} |\nabla_y V|^2 \, dy \, d\tau \le C(I) \int_{-e^{-\tau_1}}^{-e^{-\tau_2}} \int_{B_{C(I)R}} |\nabla_x U|^2 \, dx \, dt.$$

For the pressure, subtract a spatial-average gauge $a(t)$ for Q on the corresponding x-ball and set $b(\tau) = e^{-\tau} a(-e^{-\tau})$. Then $\Pi - b = e^{-\tau}(Q - a)$, and the same bounded-weight change of variables gives $\Pi - b \in L^{3/2}(B_R \times I)$. □

Remark 14.8. Lemma 14.7 is local. It does not give uniform L^3-tightness in y, nor a global $L^\infty_\tau L^3_y$ bound. This is the reason for retaining the remaining case.

14.3. **Normalized Seregin ancient limits.**

Definition 14.9 (Admissible Type I extraction)**.** An admissible Type I extraction is one of the following two blow-up procedures:

(i) the finite-energy Type I extraction in Definition 11.4 and Theorem 10.1;
(ii) the local pointwise Type I extraction in Definition 9.1 and Lemma 9.7.

In both cases the conclusion is a smooth bounded solution of the centered equation (10.3) with compact local L^3 nonvanishing.

Definition 14.10 (Raw extracted normalized Seregin ancient limit)**.** A smooth bounded solution (V, Π) of (10.3) is a *raw extracted normalized Seregin ancient limit* at z_* if it is produced by an admissible finite-energy Type I extraction in the sense of Definition 14.9, or from an admissible local pointwise Type I concentration sequence at z_* in the sense of Definition 9.1 and Lemma 9.7, and satisfies the compact spacetime L^3 nonvanishing normalization (10.4).

Corollary 14.11 (Raw extracted limits are nonzero)**.** *Every raw extracted normalized Seregin ancient limit satisfies the compact spacetime nonvanishing condition* (10.4)*. In particular, it is not identically zero.*

Proof. This is part of the definition of a raw extracted normalized Seregin ancient limit. For the finite-energy extraction the condition is produced by Lemmas 13.1 and 14.6 and Proposition 13.2; for the local pointwise Type I concentration sequence it is produced by Lemmas 9.6 and 9.7. □

Lemma 14.12 (Time-translation invariant nonvanishing)**.** *Let (V,Π) be a raw extracted normalized Seregin ancient limit satisfying (10.4) on the compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ with constant $\eta_{\mathrm{raw}} = \eta_0 > 0$. Then there exist $R_0 > 0$ and $\eta_* > 0$, depending only on K and η_{raw}, such that*

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \geq \eta_*. \tag{14.1}$$

Proof. Choose $R_0 > 0$ and an interval $[a,b] \subset \mathbb{R}$ with $K \subset B_{R_0} \times [a,b]$. Then

$$\eta_{\mathrm{raw}} \leq \iint_K |V|^3\, dy\, d\tau \leq \int_a^b \int_{B_{R_0}} |V(y,\tau)|^3\, dy\, d\tau \leq (b-a) \sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy.$$

Take $\eta_* = \eta_{\mathrm{raw}}/(b-a)$. □

Definition 14.13 (Raw generated Seregin state space)**.** For a fixed triple (u,p,z_*), fix invariant normalization data (R_0,η_*) obtained from a raw extracted profile by Lemma 14.12. Let

$$\mathcal{S}_{\mathrm{raw}}(u,p,z_*;R_0,\eta_*)$$

be the collection of raw extracted normalized Seregin ancient limits produced by admissible finite-energy Type I or admissible local pointwise Type I extractions at z_* which satisfy the invariant nonvanishing condition

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \geq \eta_*.$$

If different raw extractions yield different invariant normalization pairs, the construction below is carried out separately for each pair.

The *raw generated descendant hull*

$$\mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$$

is, by definition, the smallest class containing $\mathcal{S}_{\mathrm{raw}}(u,p,z_*;R_0,\eta_*)$ and closed under logarithmic time translations and locally smooth limits that retain:

- (i) the inherited centered L^∞ bound;
- (ii) smoothness and the centered local-pressure ancient formulation;
- (iii) the invariant local velocity nonvanishing condition (14.1) with the fixed parameters (R_0,η_*).

The class is closed under locally smooth limits that still satisfy the invariant (R_0,η_*)-nonvanishing condition; no claim is made that this condition is lower semicontinuous under arbitrary uncentered locally smooth limits. An element of $\mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$ is called a *generated raw Seregin state*. When the invariant normalization data are clear from the extracted profile under discussion, we write $\mathcal{S}_{\mathrm{raw}}(u,p,z_*)$ for $\mathcal{S}_{\mathrm{raw}}(u,p,z_*;R_0,\eta_*)$ and $\mathcal{S}_{\text{raw-gen}}(u,p,z_*)$ for $\mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$. For notational economy, $\mathcal{S}(u,p,z_*)$ denotes this raw generated class.

Definition 14.14 (Mild stratum of the generated state space)**.** The mild stratum of the raw generated state space is

$$\mathcal{S}_{\mathrm{mild}}(u,p,z_*;R_0,\eta_*) \subset \mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$$

consisting of those generated states V satisfying the finite-interval projected centered mild formulation, where

$$L = \Delta - \frac{1}{2} y\cdot\nabla - \frac{1}{2}, \qquad S(s) = e^{sL}.$$

For every $\sigma < \tau$,

$$V(\tau) = S(\tau - \sigma)V(\sigma) - \int_{\sigma}^{\tau} S(\tau - s)\mathbb{P}\nabla \cdot (V \otimes V)(s)\, ds$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

Remark 14.15 (Raw generated descendant hulls). No assertion is made here that an arbitrary locally smooth limit of translates is itself an extracted limit from the original physical sequence without a separate diagonal generation argument. Instead, the raw generated descendant hull is included in the definition of the class to which the companion residual-closure theorem is applied. This is the class on which the residual set $\mathcal{R}(\mathcal{S})$ is formed in the final assembly. The projected mild formulation is imposed only on the mild stratum, not on the whole raw hull.

Lemma 14.16 (Raw generated closure preserves normalized ancient structure)**.** *Every element of $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ is a smooth bounded centered ancient local-pressure solution, satisfies the inherited centered L^∞ bound, and retains the time-translation invariant local L^3 nonvanishing normalization* (14.1) *with the fixed* (R_0, η_*).

Proof. Raw extracted elements have these raw properties by Theorem 10.1 and Lemma 9.7 together with Lemma 14.12, which converts the raw compact cylinder mass lower bound into the invariant condition (14.1). Logarithmic time translations preserve the autonomous centered equation, the centered L^∞ bound, smoothness, and the local-pressure formulation; they also preserve the supremum on the left of (14.1), since the supremum is over all $\tau \in \mathbb{R}$. For a locally smooth limit of such translates, smoothness and the equation pass by local convergence; the L^∞ bound passes by pointwise convergence. The raw generated hull is closed only under limits retaining the invariant nonvanishing condition with the fixed (R_0, η_*), so (14.1) remains part of the structure. □

Lemma 14.17 (Mild stratum stability)**.** *The class $\mathcal{S}_{\text{mild}}(u, p, z_*; R_0, \eta_*)$ is invariant under logarithmic time translations. Moreover, locally smooth limits of elements of $\mathcal{S}_{\text{mild}}$ with a common L^∞ bound remain in $\mathcal{S}_{\text{mild}}$, provided the limiting state belongs to the raw generated hull and retains the fixed invariant nonvanishing normalization.*

Proof. Time translation preserves the finite-interval projected mild identity because the centered equation and the centered Stokes semigroup are autonomous. Consider a locally smooth sequence $V_j \in \mathcal{S}_{\text{mild}}$ with a common L^∞ bound, converging locally smoothly to V. For fixed $\sigma < \tau$, the linear term $S(\tau - \sigma)V_j(\sigma)$ converges to $S(\tau - \sigma)V(\sigma)$ in distributions and weak-* L^∞_{loc}. The Duhamel terms pass to the limit against compactly supported test fields by the Oseen-kernel estimate in Lemma 16.1, the common L^∞ bound, and local smooth convergence of $V_j \otimes V_j$. Indeed, after testing against a compactly supported solenoidal field, the adjoint Oseen kernel belongs to L^1_y, with time-singularity integrable on (σ, τ); truncating this L^1_y kernel outside a large ball and using the common L^∞ bound controls the spatial tails uniformly, while local smooth convergence handles the truncated part. Hence the projected mild formula holds for V on $[\sigma, \tau]$. The convergence against compactly supported tests upgrades to weak-* L^∞ convergence against arbitrary $L^1(\mathbb{R}^3)$ tests by approximating the L^1 test function by compactly supported functions and using the common L^∞ bound on the sequence and on the limit. The remaining raw generated and nonvanishing requirements are part of the stated limiting hypotheses. □

Remark 14.18. The raw compact spacetime L^3 nonvanishing condition is part of the normalization, not an additional hypothesis. It is inherited from the selected velocity concentration and the no-escape condition through Lemma 13.1 and Proposition 13.2 for the finite-energy extraction, and, for the local pointwise Type I extraction, through Lemmas 9.6 and 9.7. The raw generated descendant hull uses the time-translation invariant consequence (14.1); this is the nonzero normalization used below for generated raw states.

Corollary 14.19 (Generated raw Seregin states are nonzero)**.** *Every generated raw Seregin state in* $\mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$ *satisfies the time-translation invariant local* L^3 *nonvanishing condition* (14.1)*; in particular it is not identically zero.*

Proof. For raw finite-energy extractions this is Lemmas 13.1 and 14.6 and Proposition 13.2 together with Lemma 14.12. For the raw local pointwise Type I concentration sequence it is Lemmas 9.6 and 9.7 together with Lemma 14.12. Generated closure elements retain (14.1) by Lemma 14.16. □

Proof of Theorem 10.1. Choose an admissible concentration sequence $\lambda_k \downarrow 0$ from Definition 11.4, and form the rescaled sequence (12.2). The compact ancient limit is Proposition 12.12; the strong and weak convergence are Proposition 12.8. The no-escape argument in Lemma 13.1, followed by strong L^3-compactness away from $t=0$, gives nontriviality through Proposition 13.2. The pointwise Type I bound is Corollary 14.5, and the centered equation, smoothness, boundedness, and raw compact L^3 nonvanishing are Lemmas 14.3, 14.4 and 14.6. □

15. Liouville classes and the remaining class

Fix a raw generated descendant hull

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$$

in the sense of Definition 14.13. Complements below are taken relative to this raw hull. Write

$$\mathcal{S}_{\text{mild}} = \mathcal{S}_{\text{mild}}(u,p,z_*;R_0,\eta_*)$$

for its mild stratum. Structural conditions involving the unrenormalized ancient field refer to

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

Definition 15.1 (Small-amplitude class)**.** Let $\varepsilon_{\text{small}} > 0$ be the small-data Liouville constant for bounded mild ancient solutions of (10.3). We say $V \in \mathcal{C}_{\text{small}}$ if

$$V \in \mathcal{S}_{\text{mild}} \qquad \text{and} \qquad \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \varepsilon_{\text{small}}.$$

Definition 15.2 (Stationary L^3 class)**.** We say $V \in \mathcal{C}_{\text{stat},L^3}$ if

$$V(y,\tau) = W(y) \quad \text{and} \quad W \in L^3(\mathbb{R}^3).$$

Definition 15.3 (Uniformly L^3-tight class)**.** We say $V \in \mathcal{C}_{L^3-\text{tight}}$ if

$$V \in \mathcal{S}_{\text{mild}}$$

and

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0. \tag{15.1}$$

Every $V \in \mathcal{S}$ is bounded. Theorem 26.8 shows that, for mild states, the tail condition (15.1) gives the endpoint sequence-L^3 control needed after physical pullback; no separate $L^\infty_\tau L^3_y$ hypothesis is required for the tight-class exclusion.

Definition 15.4 (Structure and decay class)**.** A generated raw Seregin state $V \in \mathcal{S}_{\text{raw-gen}}$ belongs to $\mathcal{C}_{\text{sd}}$ if $V \in \mathcal{S}_{\text{mild}}$ and its physical ancient representative U satisfies at least one of the following analytic hypotheses:

(i) U is axisymmetric without swirl;

(ii) U is axisymmetric and satisfies the pointwise scale-invariant bound

$$|U(x,t)| \leq \frac{C}{r}$$

for some $C < \infty$, where r is the cylindrical radius;

(iii) U is axisymmetric and

$$rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0))$$

for some $1 \leq p < \infty$;

(iv) U is axisymmetric, periodic in the axial variable, and

$$rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0));$$

(v) the physical ancient representative U satisfies the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2.

Remark 15.5. Only hypotheses for which a cited Liouville theorem gives $U \equiv 0$ are included in $\mathcal{C}_{\mathrm{sd}}$. Broader qualitative descriptions are not included unless they imply one of the listed hypotheses. In the anchored weighted-vorticity alternative, the Biot–Savart normalization is used at the anchor times to identify the Gallay–Wayne Cauchy problem with the anchored physical flow. After the anchored argument gives $\nabla \times U \equiv 0$, boundedness gives a spatially constant velocity field, and the Type I bound removes the constant. Membership in $\mathcal{C}_{\mathrm{sd}}$ is checked profile by profile. No closedness of the structural class under all generated limits is used in the final contradiction; whenever a generated limit satisfies one of the listed conditions and lies in the mild stratum, the corresponding Liouville theorem is applied directly to that profile.

Lemma 15.6 (Structural quantities under centering)**.** *The centered change of variables preserves axisymmetry and absence of swirl. If (r, z, t) and (ρ, Z, τ) are related by*

$$\rho = \frac{r}{\sqrt{-t}}, \qquad Z = \frac{z}{\sqrt{-t}}, \qquad \tau = -\log(-t),$$

then the swirl variables satisfy

$$\rho V_\theta(\rho, Z, \tau) = rU_\theta(r, z, t).$$

Proof. The map $x \mapsto y = x/\sqrt{-t}$ is radial in the cylindrical variables and does not mix the e_r, e_θ, e_z frame. Since $V(y,\tau) = \sqrt{-t}\,U(x,t)$, the θ-components obey $V_\theta = \sqrt{-t}\,U_\theta$, and the displayed identities follow. ☐

Definition 15.7 (Remaining class)**.** The remaining class is

$$\mathcal{R}(\mathcal{S}) = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*) \setminus \left(\mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}}\right).$$

Remark 15.8. The first four classes are not intended to be disjoint. The remaining class is the complement of their union, so overlaps among the Liouville classes cause no ambiguity. The remaining class is a set-theoretic complement, not a new structural hypothesis: it carries no analytic content beyond "not in any of the four named classes." The companion residual-closure theorem Theorem 17.1 is the sole input used to remove this complement in the final contradiction.

Remark 15.9 (Non-mild states are residual unless otherwise classified)*.* The ambient state space is the raw generated Seregin hull. The explicit Liouville classes include exactly the hypotheses needed to invoke their corresponding Liouville theorems. In particular, a generated state for which the projected mild formulation has not been established is not placed in the small, tight, or structure/decay Liouville classes merely because it satisfies the corresponding pointwise or tail condition. Such a state lies in the remaining class unless it belongs to the stationary L^3 branch.

Remark 15.10 (Single-profile versus collection-level tightness)*.* The exclusion of a single generated raw Seregin state in $\mathcal{C}_{L^3-\mathrm{tight}}$ uses only the single-profile tail condition together with mildness and boundedness, via Theorem 26.8. The stronger collection-level tightness modulus appearing in Definition 23.6 is used only in the auxiliary compactness and minimal-element material of Part III; no hidden common modulus is imposed on individual generated raw Seregin states.

Proposition 15.11 (Exhaustion by defined classes)**.** *Let*

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be a raw generated descendant hull. Then every $V \in \mathcal{S}$ belongs either to one of

$$\mathcal{C}_{\text{small}}, \qquad \mathcal{C}_{\text{stat},L^3}, \qquad \mathcal{C}_{L^3-\text{tight}}, \qquad \mathcal{C}_{\text{sd}},$$

or to

$$\mathcal{R}(\mathcal{S}) = \mathcal{S} \setminus \left(\mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat},L^3} \cup \mathcal{C}_{L^3-\text{tight}} \cup \mathcal{C}_{\text{sd}}\right).$$

Proof. Let

$$\mathcal{U} = \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat},L^3} \cup \mathcal{C}_{L^3-\text{tight}} \cup \mathcal{C}_{\text{sd}}.$$

For $V \in \mathcal{S}$, either $V \in \mathcal{U}$, in which case it lies in at least one Liouville class, or $V \in \mathcal{S} \setminus \mathcal{U} = \mathcal{R}(\mathcal{S})$. $\square$

16. Liouville hypotheses and class exclusions

Lemma 16.1 (Centered Stokes kernel estimates)**.** *Let*

$$L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}, \qquad S(s) = e^{sL}.$$

There is a universal constant C_0 such that, for all $s > 0$,

$$\|S(s)f\|_{L^\infty} \le e^{-s/2} \|f\|_{L^\infty}$$

and

$$\|S(s)\mathbb{P}\nabla \cdot F\|_{L^\infty} \le C_0 e^{-s/2} (1 - e^{-s})^{-1/2} \|F\|_{L^\infty}.$$

Proof. The first estimate follows from the explicit Ornstein–Uhlenbeck kernel:

$$S(s)f(y) = e^{-s/2} \int_{\mathbb{R}^3} G_{1-e^{-s}}(e^{-s/2} y - z) f(z)\, dz,$$

where G_a is the heat kernel with variance a. The kernel has total mass one. For the projected-divergence estimate, write the corresponding kernel as the Oseen projected-gradient kernel after the same change of variables. More explicitly, the semigroup for $\Delta - \frac{1}{2} y \cdot \nabla$ is

$$(e^{s(\Delta - \frac{1}{2} y \cdot \nabla)} g)(y) = (e^{(1-e^{-s})\Delta} g)(e^{-s/2} y).$$

Applying this to the distribution $g = \mathbb{P}\nabla \cdot F$ and using the projected-gradient Oseen kernel estimate

$$\|e^{\theta\Delta} \mathbb{P}\nabla \cdot F\|_\infty \le C\theta^{-1/2} \|F\|_\infty, \qquad \theta > 0,$$

gives the displayed estimate with $\theta = 1 - e^{-s}$, together with the extra factor $e^{-s/2}$ from the zeroth-order term in L. The projected-gradient Oseen estimate follows from the L^1-bound

$$\|\nabla O(\cdot, \theta)\|_{L^1} \le C\theta^{-1/2}$$

for the gradient of the Oseen kernel $O(\cdot, \theta)$. This is the bounded-data Stokes estimate used by Kato [14]; it does not require boundedness of the Leray projector itself on L^∞. $\square$

Remark 16.2 (Oseen interpretation in L^∞)*.* All occurrences of $S(t)\mathbb{P}\nabla \cdot F$ in L^∞ are understood through the projected heat-divergence, or Oseen, kernel representation. They do not use boundedness of the Leray projector as a Calderon–Zygmund operator on L^∞.

Lemma 16.3 (Mild representation for generated mild states)**.** *Every $V \in \mathcal{S}_{\mathrm{mild}}(u, p, z_*; R_0, \eta_*)$ satisfies, for all $\Theta > 0$,*

$$V(\tau) = S(\Theta)V(\tau - \Theta) - \int_0^{\Theta} S(\Theta - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - \Theta + \sigma)\, d\sigma.$$

Proof. This is the finite-interval projected mild formulation in Definition 14.14, applied with $\sigma = \tau - \Theta$. The centered Stokes estimates in Lemma 16.1 ensure that the Duhamel term is well-defined in weak-* L^∞. □

Lemma 16.4 (Small bounded ancient Liouville theorem)**.** *There is $\varepsilon_{\mathrm{small}} > 0$ such that every bounded mild ancient solution of* (10.3) *with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \le \varepsilon_{\mathrm{small}}$$

is identically zero.

Proof. Write the projected centered equation as

$$\partial_\tau V = LV - \mathbb{P}\nabla \cdot (V \otimes V), \qquad L = \Delta - \frac{1}{2} y \cdot \nabla - \frac{1}{2}.$$

Let $S(s) = e^{sL}$. By Lemma 16.1, the nonlinear kernel is integrable on $(0, \infty)$; set

$$A = C_0 \int_0^\infty e^{-s/2}(1 - e^{-s})^{-1/2}\, ds < \infty.$$

For fixed τ and $T > 0$, the mild identity on $[\tau - T, \tau]$ gives

$$V(\tau) = S(T)V(\tau - T) - \int_0^T S(T - \sigma)\mathbb{P}\nabla \cdot (V \otimes V)(\tau - T + \sigma)\, d\sigma.$$

This Duhamel identity is valid for every $T > 0$ by Lemma 16.3. If $M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$, then

$$\|V(\tau)\|_\infty \le e^{-T/2} M + AM^2.$$

Letting $T \to \infty$ and taking the supremum in τ gives

$$M \le AM^2.$$

If $0 < M < 1/A$, this is impossible. Choose $\varepsilon_{\mathrm{small}} = 1/(2A)$; then $M \le \varepsilon_{\mathrm{small}}$ implies $M = 0$. □

Lemma 16.5 (Stationary centered profiles)**.** *If $V(y, \tau) = W(y)$ is a smooth solution of* (10.3)*, then after subtracting a function of τ from the pressure, W solves*

$$(W \cdot \nabla)W + \nabla P = \Delta W - \frac{1}{2}(W + y \cdot \nabla W), \qquad \operatorname{div} W = 0.$$

Proof. Since V is independent of τ, $\partial_\tau V = 0$. Set

$$F(y) = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W.$$

The centered equation gives $\nabla_y \Pi(\cdot, \tau) = F$ for a.e. τ. Fix one such time τ_0 and put $P(y) = \Pi(y, \tau_0)$. Then $\nabla_y(\Pi(\cdot, \tau) - P) = 0$ for a.e. τ, so $\Pi(y, \tau) - P(y) = b(\tau)$ is spatially constant. Subtracting this gauge from Π leaves the equation unchanged and yields the displayed stationary self-similar equation for W. □

Lemma 16.6 (Spatially constant normalized profiles vanish)**.** *Let U be the physical ancient representative of a generated raw Seregin state V in the mild stratum. If U is spatially constant in the physical Type I frame, then $U \equiv 0$ and hence $V \equiv 0$.*

Proof. Mildness excludes parasitic spatially constant solutions: if $U(x,t) = c(t)$, the mild formula has zero nonlinear term and the heat semigroup fixes constants, so $c(t) = c(s)$ for all $s < t < 0$. Thus $U(x,t) \equiv c$. The centered profile is then

$$V(y,\tau) = e^{-\tau/2}c.$$

Since generated raw Seregin states are bounded for all $\tau \in \mathbb{R}$, letting $\tau \to -\infty$ forces $c = 0$. $\square$

Theorem 16.7 (Small and stationary class exclusions)**.** *Let V be a generated raw Seregin state. Then*

$$V \notin \mathcal{C}_{\mathrm{small}}, \qquad V \notin \mathcal{C}_{\mathrm{stat},L^3}.$$

Proof. If $V \in \mathcal{C}_{\mathrm{small}}$, then by definition $V \in \mathcal{S}_{\mathrm{mild}}$, and Lemmas 16.3 and 16.4 give $V \equiv 0$, contradicting the invariant nonvanishing normalization of Corollary 14.19.

If $V \in \mathcal{C}_{\mathrm{stat},L^3}$, then $V(y,\tau) = W(y)$ with $W \in L^3(\mathbb{R}^3)$. By Lemma 16.5, W satisfies the stationary self-similar profile equation after a pressure gauge change. The stationary Leray theorem Theorem 22.4 gives $W \equiv 0$. Again the invariant nonvanishing normalization of Corollary 14.19 gives the contradiction. $\square$

Theorem 16.8 (Uniformly tight L^3 closure)**.** *No generated raw Seregin state in a raw generated descendant hull $\mathcal{S}_{\text{raw-gen}}(u,p,z_*;R_0,\eta_*)$ belongs to $\mathcal{C}_{L^3-\mathrm{tight}}$.*

Proof. Let V be a generated raw Seregin state in $\mathcal{C}_{L^3-\mathrm{tight}}$. By definition $V \in \mathcal{S}_{\mathrm{mild}}$ and satisfies the uniform tail condition. Thus Theorem 26.8 gives $V \equiv 0$, contradicting Corollary 14.19. $\square$

Theorem 16.9 (Structure and decay exclusions from Liouville theorems)**.** *No nonzero generated raw Seregin state belongs to $\mathcal{C}_{\mathrm{sd}}$.*

Proof. Let $V \in \mathcal{C}_{\mathrm{sd}}$. By definition of the structure/decay class, $V \in \mathcal{S}_{\mathrm{mild}}$, and its physical representative

$$U(x,t) = (-t)^{-1/2}V(x/\sqrt{-t}, -\log(-t))$$

satisfies one of the listed structural hypotheses. The inherited centered L^∞ bound gives the physical Type I estimate

$$|U(x,t)| \le \|V\|_\infty(-t)^{-1/2}.$$

By Lemma 26.4, for every $s < t < 0$ the standard projected mild formula holds on the interval $[s,t]$: choose $T = t$ and apply the finite-terminal restriction lemma. Thus U is a smooth mild Type I ancient solution in the sense of Definition 29.1. Therefore Theorem 33.1 gives $U \equiv 0$, and hence $V \equiv 0$, contradicting the invariant nonvanishing normalization of Corollary 14.19. $\square$

Proposition 16.10 (Exclusion of the Liouville classes)**.** *No nonzero generated raw Seregin state can belong to*

$$\mathcal{C}_{\mathrm{small}} \cup \mathcal{C}_{\mathrm{stat},L^3} \cup \mathcal{C}_{L^3-\mathrm{tight}} \cup \mathcal{C}_{\mathrm{sd}}.$$

Proof. If $V \in \mathcal{C}_{\mathrm{small}}$, then by definition $V \in \mathcal{S}_{\mathrm{mild}}$, and the small mild Liouville theorem applies. If $V \in \mathcal{C}_{\mathrm{stat},L^3}$, then the stationary self-similar profile equation and the stationary Leray theorem apply. These two branches are excluded in Theorem 16.7. If $V \in \mathcal{C}_{L^3-\mathrm{tight}}$, then by definition $V \in \mathcal{S}_{\mathrm{mild}}$, so Theorem 16.8 applies. If $V \in \mathcal{C}_{\mathrm{sd}}$, then by definition $V \in \mathcal{S}_{\mathrm{mild}}$ and the physical representative satisfies the relevant Type I mild structural hypothesis, so Theorem 16.9 applies. $\square$

17. Residual closure imported from the companion paper

The preceding sections prove that every generated raw Seregin state in a raw generated descendant hull associated with an admissible finite-energy Type I extraction or an admissible local pointwise Type I concentration sequence is either in one of the four non-residual Liouville classes or in the complement $\mathcal{R}(\mathcal{S})$. The four non-residual classes ($\mathcal{C}_{\mathrm{small}}$, $\mathcal{C}_{\mathrm{stat},L^3}$, $\mathcal{C}_{L^3-\mathrm{tight}}$, $\mathcal{C}_{\mathrm{sd}}$) are excluded in Proposition 16.10. The companion residual paper proves the literal closure of the remaining raw set. No result in the preceding sections depends on that later theorem; it is used only in Theorem 10.2 and in the final assembly Theorem 34.1.

Theorem 17.1 (Residual closure for raw generated Seregin state spaces)**.** *Fix a finite-energy suitable weak solution (u,p), a terminal point $z_* = (x_*, T)$ satisfying the local pointwise Type I bound, and an admissible local Type I concentration sequence at z_*. Fix the incoming raw extracted profile and its single ancestry and invariant-normalization ledger, and form the corresponding raw generated descendant hull*

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*),$$

one has

$$\mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big) = \varnothing.$$

The identical local conclusion holds when the incoming profile is produced by an admissible finite-energy Type I extraction in Theorem 10.2.

The residual complement is formed after removing only states satisfying the full hypotheses of the explicit Liouville branches. The companion proof acts locally on a hypothetical incoming residual profile and on realized descendants carrying the same ancestry and normalization ledger. It assumes no projected mildness for the raw hull: mildness is used only on the setup mild stratum or after the companion paper's local inheritance theorem applies.

Proof. Fix the incoming profile, the raw hull, and its normalization ledger. If $V \in \mathcal{R}(\mathcal{S})$ existed, the exact setup-to-residual interface in the companion paper routes this same profile either to its closed affine/parasitic row or to its retained residual row [1, Proposition 1.5]. Its Type I residual closure theorem closes the retained row by local estimates on the current profile and its realized descendants [1, Theorem 12.1]. The two rows are exhaustive, so neither can contain the hypothetical incoming profile. Hence no such V exists and $\mathcal{R}(\mathcal{S}) = \varnothing$. This is a contrapositive argument from a hypothetical singular profile, not a residual existence assumption and not a global ancient-solution estimate. ☐

Remark 17.2 (Companion-theorem import)*.* Theorem 17.1 is an imported theorem, not an additional analytic assumption. Its producer is the Type I residual-closure theorem in the companion paper, whose conclusion is literal emptiness of the same raw generated residual set. The combined proof is therefore unconditional in the stated finite-energy local pointwise Type I setting.

Remark 17.3 (Interface with the residual paper)*.* The companion residual paper proves

$$\mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big) = \varnothing$$

for the exact raw generated descendant hull defined in Definition 14.13. The final assembly uses this same hull and no enlarged or quotient replacement.

Corollary 17.4 (Closure of the remaining raw class)**.** *For every raw generated descendant hull $\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ associated with an admissible finite-energy Type I extraction or an admissible local pointwise Type I concentration sequence,*

$$\mathcal{R}(\mathcal{S}) = \varnothing.$$

Proof. This is Theorem 17.1. □

Proposition 17.5 (Pointwise contrapositive formulation). *Fix a generated raw Seregin hull and its ancestry and normalization ledger. If one supposes that $V \in \mathcal{R}(\mathcal{S})$, the companion paper's local routing closes the unique branch selected for V, giving a contradiction. Since V is arbitrary, this pointwise contradiction is equivalent to $\mathcal{R}(\mathcal{S}) = \varnothing$.*

Proof. This is the elementwise form of the proof of Theorem 17.1. Universal generalization over an arbitrary hypothetical $V \in \mathcal{R}(\mathcal{S})$ is precisely emptiness of that set; no existence premise is introduced. □

Proof of Theorem 10.2. Suppose that (u, p, z_*) is an admissible finite-energy Type I singularity. By Theorem 10.1, there exists a raw extracted normalized nonzero Seregin ancient limit V. By Lemma 14.12, V satisfies the invariant local nonvanishing condition. Let

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be the raw generated descendant hull associated with z_*, the admissible extraction scales, and the fixed invariant normalization component. By construction, $V \in \mathcal{S}$.

By Proposition 15.11, either V belongs to one of the Liouville classes or $V \in \mathcal{R}(\mathcal{S})$. The first alternative is impossible by Proposition 16.10. The second is impossible by Theorem 17.1, which states literally that $\mathcal{R}(\mathcal{S}) = \varnothing$. Hence the hypothetical admissible finite-energy Type I singularity cannot exist. □

The preceding theorem uses the companion residual-closure theorem for the raw generated hull associated with the finite-energy extraction. The compactness construction, pressure normalization, compact-to-invariant nonvanishing, raw-hull stability, and class exhaustion are established above. The structure and decay class is restricted to the assumptions in Definition 15.4 and is routed through Theorem 33.1; every other structured behavior remains in $\mathcal{R}(\mathcal{S})$ and is closed by Theorem 17.1. No whole-space critical-norm estimate on the original solution is used in this local closure.

18. Elementary symmetries of the centered equation

Lemma 18.1 (Logarithmic-time translation). *If V solves (10.3), then*

$$V_{\tau_0}(y, \tau) = V(y, \tau + \tau_0)$$

also solves (10.3). If

$$U^\lambda(x, t) = \lambda U(\lambda x, \lambda^2 t),$$

then the renormalized field associated with U^λ is

$$V^\lambda(y, \tau) = V(y, \tau - 2\log\lambda).$$

Proof. The first statement follows because (10.3) is autonomous in τ. For the second, put $s = -t = e^{-\tau}$. Then

$$\sqrt{s}\, U^\lambda(y\sqrt{s}, -s) = \lambda\sqrt{s}\, U(\lambda y\sqrt{s}, -\lambda^2 s) = V(y, -\log(\lambda^2 s)) = V(y, \tau - 2\log\lambda).$$

□

Proposition 18.2 (Dilation and translation are not symmetries). *The centered equation (10.3) is autonomous in τ, but it is not invariant under*

$$V(y, \tau) \mapsto \alpha V(\alpha y, \alpha^2 \tau)$$

unless $\alpha = 1$. It is also not invariant under constant translations

$$V(y, \tau) \mapsto V(y - a, \tau)$$

unless $a = 0$.

Proof. For scaling, set $Y = \alpha y$, $T = \alpha^2 \tau$, and $\widetilde{V}(y, \tau) = \alpha V(Y, T)$, $\widetilde{\Pi}(y, \tau) = \alpha^2 \Pi(Y, T)$. The non-drift part scales with factor α^3, while

$$-\frac{1}{2}(\widetilde{V} + y \cdot \nabla \widetilde{V}) = -\frac{\alpha}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

Using the equation for V, the remaining term in the equation for $\widetilde{V}$ is

$$\frac{\alpha - \alpha^3}{2}(V + Y \cdot \nabla_Y V)(Y, T).$$

If $\alpha \neq 1$, this forces $V + Y \cdot \nabla V = 0$. Along each ray,

$$\frac{d}{dr}\big(rV(r\omega, T)\big) = 0.$$

Boundedness and smoothness at the origin force the constant to be zero, hence $V \equiv 0$. Thus a nonzero centered solution has no such scaling symmetry.

For translations, all translation-invariant terms transform by composition, but the drift satisfies

$$y \cdot \nabla V(y - a, \tau) = (y - a) \cdot \nabla V(y - a, \tau) + a \cdot \nabla V(y - a, \tau).$$

The extra term is absent from the original equation. Thus spatial translation is not a symmetry of the fixed centered equation unless $a = 0$, apart from exceptional fixed solutions satisfying $a \cdot \nabla V \equiv 0$. $\square$

Part III. Uniformly Tight Ancient Dynamics

Overview. Bounded ancient solutions of the three-dimensional Navier–Stokes equations are considered in backward self-similar variables under uniform L^3-tightness. The compactness theory shows that bounded mild ancient solutions are smooth, locally compact under time translation, and that the common $L^\infty_\tau L^3_y$ and tightness bounds used in compactness statements are stable under locally smooth limits. A minimal-element reduction is then obtained for closed normalized uniformly tight collections: any nonempty such collection contains a nonzero solution of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ size.

For the time-translation hull of a bounded ancient solution, compact invariant hulls and invariant probability measures are constructed by Krylov–Bogolyubov averaging. These measures satisfy a stationary statistical form of the self-similar Navier–Stokes equation. Passing to the barycenter produces an explicit quadratic covariance term, isolating the obstruction to obtaining a single stationary self-similar profile. This compact-hull and invariant-measure material is auxiliary only; it is not used in the final Type I contradiction.

Finally, the uniformly L^3-tight class is closed directly from its defining hypotheses. The tail condition, together with the boundedness of the ancient solution, gives the sequence-L^3 bound needed after the physical pullback. Since this pullback preserves the critical L^3-norm on each time slice, the endpoint ancient Liouville theorem of Albritton–Barker then forces the pullback, and hence the centered solution, to vanish. Therefore the uniformly L^3-tight class of the Type I ancient-limit reduction is excluded without any compact-hull Lyapunov hypothesis.

19. Introduction and main results

19.1. **The uniformly tight ancient class.** Consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The centered velocity V and pressure P solve

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \qquad \operatorname{div} V = 0 \tag{19.1}$$

on $\mathbb{R}^3 \times \mathbb{R}$. In Theorem 10.1, admissible Type I singularities are reduced to nonzero normalized ancient solutions of (19.1). The uniformly L^3-tight class is the class in which

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0. \tag{19.2}$$

The compactness statements also record the common bound

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \tag{19.3}$$

but the endpoint exclusion of the tight class uses only boundedness and the tail condition (19.2).

This condition on the centered limit is distinct from the original-solution condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3)).$$

For finite-energy suitable weak solutions, the original-solution assumption $u \in L^\infty_t L^3_x$ near a putative singular time is incompatible with singularity by the theorem of Escauriaza–Seregin–Šverák [4]. Uniform L^3-tightness is imposed on centered ancient solutions after blow-up.

19.2. **Minimal elements and compact hulls.**

Remark 19.1 (Auxiliary diagnostic material). The compact-hull, minimal-element, and invariant-measure material in this part is auxiliary diagnostic material. It is not used in the proof of Theorem 34.1; the final Type I contradiction uses only the uniform L^3-tight Liouville theorem Theorem 26.8 and its interface corollary Corollary 27.3. It is recorded here because the covariance obstruction it identifies is the precise reason the residual class cannot be removed by a stationary statistical reduction alone, and because the same constructions are reused in the residual-branch paper.

Remark 19.2 (Paper I/Paper III interface and normalization alignment). Raw extractions in Part II produce the local lower bound (10.4). By Lemma 14.12, every raw extracted profile also satisfies the invariant local normalization used in (23.1). The raw generated class $\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$ is closed under time translations and locally smooth limits only within this invariantly normalized class, so the closed normalized collections of Definition 23.1 include every raw generated descendant hull as a special case (with (R_0, η_0) given by the raw extraction parameters of Lemma 14.12).

The first reduction is variational:

$$\text{nonempty tight normalized class} \quad \Longrightarrow \quad L^\infty\text{-minimal ancient element.}$$

The normalization is a fixed local velocity nonvanishing condition. It prevents minimizing sequences from losing all mass by time translation. Once a minimal element V_* is selected, its time-translation hull is

$$\mathcal{H}(V_*) = \overline{\{\Theta_s V_* : \ s \in \mathbb{R}\}}^{C^\infty_{\text{loc}}}, \qquad (\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The hull is compact because bounded mild ancient solutions are locally smooth with estimates depending only on their common L^∞ bound.

19.3. **Invariant measures and covariance.** Every compact invariant hull K carries invariant probability measures. Indeed, Krylov–Bogolyubov averages

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds$$

have weak subsequential limits. Such invariant measures satisfy a stationary statistical form of (19.1). If

$$\overline{W}(y) = \int_K W(y, 0)\, d\mu(W),$$

then the averaged nonlinearity decomposes as

$$\int_K W_i W_j\, d\mu = \overline{W}_i\, \overline{W}_j + C_{ij}.$$

Thus the barycenter solves the stationary self-similar equation only up to the explicit covariance term C. This invariant-measure construction does not prove that a recurrent compact hull contains a single stationary profile.

19.4. **Endpoint L^3 closure of the tight class.** The compact-hull and invariant-measure constructions identify the covariance obstruction in a purely local topology, but they are not needed to close the uniformly L^3-tight class. The tail condition already gives the sequence-L^3 information required by the endpoint ancient Liouville theorem, because the centered solution is bounded. Under the physical change of variables

$$u(x, t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t < 0,$$

the L^3-norm is invariant:

$$\|u(t)\|_{L^3_x} = \|V(-\log(-t))\|_{L^3_y}.$$

Consequently every tail-tight centered ancient solution produces, after restriction to any finite terminal time and translation of that terminal time to 0, a bounded physical mild ancient solution whose terminal datum lies in $L^3(\mathbb{R}^3)$ and whose L^3-norm is bounded along a sequence $t_k \downarrow -\infty$.

The endpoint theorem of Albritton–Barker [2] then gives $u \equiv 0$, and the change of variables gives $V \equiv 0$.

The endpoint closure uses only the boundedness and tail tightness already present in the ancient solution class and introduces no additional estimate, recurrence assumption, or Lyapunov functional. The compactness and invariant-measure results establish the local smooth compactness theory, the existence of minimal elements and compact trajectory hulls, and the stationary statistical identity with its covariance term. The endpoint L^3 exclusion of the uniformly tight class is then obtained by the physical pullback and the Albritton–Barker ancient Liouville theorem.

Thus the uniformly L^3-tight class is closed under local smooth limits by its defining hypotheses. The compact-hull construction describes recurrent dynamics and the covariance obstruction, but no compact-recurrence or Lyapunov assumption enters the tight-class exclusion in Theorem 26.8.

19.5. **Main results.** The first main theorem is the analytic compactness and closure statement.

Theorem 19.3 (Compactness and closure of uniformly tight ancient solutions)**.** *Let $\{V_n\}$ be a sequence of bounded mild ancient solutions of (19.1). Assume*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V.

If in addition there are $L < \infty$ and a modulus $\omega : [1, \infty) \to [0, \infty)$, $\omega(R) \to 0$, such that

$$\sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y, \tau)|^3 \, dy \le \omega(R) \qquad (R \ge 1),$$

then

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L, \qquad \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y, \tau)|^3 \, dy \le \omega(R),$$

and in particular V is uniformly L^3-tight.

Compatible pressure representatives also converge locally weakly modulo time-dependent gauges under the additional $L^\infty_\tau L^3_y$ assumption; see Lemma 21.3.

The second main theorem is the minimal-element reduction.

Theorem 19.4 (Minimal element in a normalized uniformly tight class)**.** *Let A be a nonempty collection of bounded mild ancient solutions satisfying a common L^∞ bound, a common $L^\infty_\tau L^3_y$ bound, a common uniform L^3-tightness modulus, and a fixed local nonvanishing normalization*

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 \, dy \ge \eta_0 > 0.$$

Assume A is invariant under time translation and closed under locally smooth limits which retain the normalization. Then A contains an element V_* minimizing

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}$$

among all $V \in A$. In particular $V_* \not\equiv 0$.

The third main theorem is the endpoint L^3 tight-class Liouville criterion.

Theorem 19.5 (Uniformly tight Liouville criterion)**.** *Let A be a collection of bounded mild ancient solutions of* (19.1)*. Assume there is a modulus $\omega : [1,\infty) \to [0,\infty)$, $\omega(R) \to 0$, such that*

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy \le \omega(R) \qquad \text{for every } V \in A,\ R \ge 1,$$

and assume that A has a fixed local nonvanishing normalization:

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \ge \eta_0 > 0 \qquad \text{for every } V \in A.$$

Then

$$A = \varnothing.$$

The proof of the last theorem is summarized by the implications

$$\begin{gathered}
\text{uniformly } L^3\text{-tight locally nonzero class } A \\
\Downarrow \quad \text{boundedness plus tail tightness} \\
\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for a sequence } \tau_k \to -\infty \\
\Downarrow \quad \text{physical pullback and critical } L^3\text{-invariance} \\
\text{mild ancient Navier–Stokes solution } u \\
\text{with } \sup_k \|u(t_k)\|_{L^3(\mathbb{R}^3)} < \infty \quad \text{for some } t_k \downarrow -\infty \\
\Downarrow \quad \text{Albritton–Barker endpoint theorem} \\
u = 0 \\
\Downarrow \\
\text{contradiction with local velocity nonvanishing in } A.
\end{gathered}$$

19.6. **Relation to the Type I reduction.** Theorem 10.1 first reduces admissible Type I singularities to raw extracted Seregin ancient limits. The subsequent normalization step fixes the time-translation invariant lower bound, and the generated descendant hull is then organized into small, stationary L^3, uniformly tight, structure and decay, and residual classes. The tight-class exclusion requires no additional compact-dynamical hypothesis: the uniformly L^3-tight class contains no nonzero generated raw ancient state in the mild stratum.

19.7. **Organization.** Section 20 fixes the centered equation, mild formulation, time translation, and local smooth topology. Section 21 proves smoothing, pressure reconstruction, stability, and compactness. Section 22 proves closure of uniform L^3-tightness and recalls stationary L^3 rigidity. Section 23 proves the minimal-element reduction. Section 24 constructs compact trajectory hulls and minimal invariant subsets. Section 25 constructs invariant measures and identifies the covariance obstruction. Section 26 proves the physical pullback identities, applies the endpoint L^3 ancient Liouville theorem, and states the consequence used in the Type I reduction.

20. Centered ancient solutions and the local smooth topology

20.1. **The centered equation.** Throughout this part V denotes a centered ancient velocity and P its pressure. An element of a compact hull is denoted by W, and a stationary profile by $w(y)$. Define

$$LV = \Delta V - \frac{1}{2} y \cdot \nabla V - \frac{1}{2} V.$$

After applying the Helmholtz–Leray projection $\mathbb{P}$, the centered equation becomes

$$\partial_\tau V = LV - \mathbb{P}\,\mathrm{div}(V \otimes V). \tag{20.1}$$

20.2. **The Ornstein–Uhlenbeck–Stokes semigroup.** Let $S(a) = e^{aL}$, $a > 0$. It is given by

$$(S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\Big(-\frac{|z-e^{-a/2}y|^2}{4(1-e^{-a})}\Big)}{[4\pi(1-e^{-a})]^{3/2}} f(z)\,dz. \tag{20.2}$$

The factor $e^{-a/2}$ represents the damping caused by the term $-V/2$.

Lemma 20.1 (Centered Stokes kernel estimate)**.** *For $a > 0$ and $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,*

$$\|(S(a)\mathbb{P}\,\mathrm{div})^*\phi\|_{L^1(\mathbb{R}^3)} \le Ce^{-a/2}(1-e^{-a})^{-1/2}\|\phi\|_{L^1(\mathbb{R}^3)}. \tag{20.3}$$

Consequently the Duhamel term in (20.4) is well-defined by duality for bounded V.

Proof. Set $\lambda = e^{-a/2}$ and $\theta = 1 - e^{-a}$. From (20.2),

$$S(a)f(y) = e^{-a/2}(e^{\theta\Delta}f)(\lambda y).$$

Thus

$$S(a)^*\phi(z) = e^{-a/2}\lambda^{-3}(e^{\theta\Delta}\psi)(z), \qquad \psi(x) = \phi(x/\lambda),$$

and $\|\psi\|_{L^1} = \lambda^3\|\phi\|_{L^1}$. The Leray projection is a homogeneous Fourier multiplier of degree zero, so it commutes with this dilation. The classical heat–Stokes kernel estimate

$$\|(e^{\theta\Delta}\mathbb{P}\,\mathrm{div})^*\phi\|_{L^1} \le C\theta^{-1/2}\|\phi\|_{L^1}, \qquad \theta > 0,$$

follows from the Oseen kernel bound; see [21, 22, 19]. Substituting $\theta = 1 - e^{-a}$ and using

$$e^{-a/2}\lambda^{-3}\|\psi\|_{L^1} = e^{-a/2}\|\phi\|_{L^1}$$

gives (20.3). Since $(1-e^{-a})^{-1/2}$ is integrable at $a = 0$, the duality pairing in (20.4) is integrable on finite time intervals for $V \in L^\infty$. ◻

20.3. **Riesz transform convention.** The convention is that R_i has Fourier symbol $i\xi_i/|\xi|$. Thus R_iR_j has symbol $-\xi_i\xi_j/|\xi|^2$, and

$$P = R_iR_j(F_{ij}) \iff -\Delta P = \partial_i\partial_j F_{ij}.$$

Pressures are determined only modulo functions of time unless a whole-space Leray gauge is explicitly chosen.

20.4. **Bounded mild ancient solutions.**

Definition 20.2 (Bounded mild ancient solution)**.** A vector field $V : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}^3$ is a bounded mild ancient solution of (19.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \qquad \mathrm{div}\, V = 0$$

in distributions, and for every $\sigma < \tau$,

$$V(\tau) = S(\tau-\sigma)V(\sigma) - \int_\sigma^\tau S(\tau-s)\mathbb{P}\,\mathrm{div}(V\otimes V)(s)\,ds \tag{20.4}$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

The Duhamel term in (20.4) is interpreted by duality. For $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,

$$\langle S(\tau-s)\mathbb{P}\,\mathrm{div}(V\otimes V)(s), \phi\rangle = \langle V\otimes V(s), (S(\tau-s)\mathbb{P}\,\mathrm{div})^*\phi\rangle\,,$$

and Lemma 20.1 makes the scalar integrand absolutely integrable on $s \in (\sigma,\tau)$.

20.5. **Time translation.** For $s \in \mathbb{R}$ define

$$(\Theta_s V)(y,\tau) = V(y,\tau+s).$$

The equation, the mild formulation, the L^∞ norm, and the $L^\infty_\tau L^3_y$ and tail bounds used below are invariant under Θ_s. Indeed, replacing (σ,τ) by $(\sigma+s,\tau+s)$ in (20.4) and changing variables in the time integral gives the mild formula for $\Theta_s V$.

20.6. **The locally smooth topology.**

Definition 20.3 (Local smooth topology)**.** The locally smooth topology on $C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R}; \mathbb{R}^3)$ is generated by the seminorms

$$\|V\|_{m,N} = \max_{|\alpha|+j\le m} \sup_{\substack{|y|\le N \\ |\tau|\le N}} |\partial_y^\alpha \partial_\tau^j V(y,\tau)|, \qquad m,N \in \mathbb{N}.$$

One compatible metric is

$$d(U,V) = \sum_{m,N=1}^{\infty} 2^{-(m+N)} \frac{\|U-V\|_{m,N}}{1+\|U-V\|_{m,N}}.$$

Thus $V_n \to V$ locally smoothly if and only if the convergence is in $C^m(K)$ for every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and every $m \ge 0$.

21. Smoothing, pressure reconstruction, and compactness

21.1. **Local smoothing.**

Lemma 21.1 (Smoothing of bounded mild solutions)**.** *Every bounded mild ancient solution is smooth on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies the projected equation (20.1) classically. Moreover, for each compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and each integer $m \ge 0$,*

$$\|V\|_{C^m(K)} \le C\big(m, K, \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}\big).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. Fix a compact time interval $J \Subset \mathbb{R}$ containing the time projection of K, and set $t = -e^{-\tau}$. On the corresponding compact interval $t(J) \Subset (-\infty, 0)$, define

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map $(x,t) \mapsto (x/\sqrt{-t}, -\log(-t))$ is a smooth diffeomorphism between $\mathbb{R}^3 \times t(J)$ and $\mathbb{R}^3 \times J$, with all derivatives bounded on compact subcylinders.

The centered mild formula (20.4) transforms, under this parabolic change of variables, into the usual projected Navier–Stokes mild formula on every compact subinterval of $t(J)$:

$$U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(U \otimes U)(r)\, dr, \qquad s < t.$$

This transformation uses only the projected mild equation: the Leray projection is homogeneous of degree zero and commutes with the parabolic dilation, while the Ornstein–Uhlenbeck kernel in (20.2) becomes the heat kernel with time parameter $t - s$. No pressure formulation is used at this stage. The L^∞ norm of U on $\mathbb{R}^3 \times t(J)$ is bounded by a constant depending only on J and $\|V\|_{L^\infty}$.

The standard local smoothing theorem for bounded projected mild Navier–Stokes solutions [14, 22] gives first local derivative bounds away from the initial time in any mild representation. In the present ancient setting one may choose the initial time strictly before the compact subcylinder under consideration. Iterating the differentiated Duhamel formula and then applying the interior parabolic bootstrap for the Stokes system [24, 19] gives uniform bounds for all derivatives of U on compact subcylinders of $\mathbb{R}^3 \times t(J)$. The mild formula may then be differentiated in time, so U solves the projected Navier–Stokes equation classically. Returning to (y, τ) gives the asserted estimates for V and the classical projected equation (20.1). The argument is uniform for families with a common L^∞ bound. □

21.2. **Pressure reconstruction.**

Lemma 21.2 (Local pressure reconstruction)**.** *Let V be a smooth bounded solution of the projected centered equation (20.1). Then for every ball B_R and compact interval $I \Subset \mathbb{R}$ there exists*

$$P_R \in C^\infty(B_R \times I),$$

unique up to an additive function of time, such that

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P_R = \Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \operatorname{div} V = 0,$$

on $B_R \times I$.

Proof. Set

$$F = \Delta V - \frac{1}{2}(V + y\cdot\nabla V) - \partial_\tau V - (V\cdot\nabla)V.$$

Since $(V\cdot\nabla)V = \operatorname{div}(V\otimes V)$ and V satisfies (20.1), self-adjointness of the Leray projection gives

$$\int_I \int_{B_R} F\cdot\Phi \, dy\, d\tau = 0$$

for every $\Phi \in C_c^\infty(B_R \times I; \mathbb{R}^3)$ satisfying $\operatorname{div}_y \Phi = 0$. The local de Rham theorem, applied in the spatial variables with τ as a parameter, gives a scalar distribution P_R on $B_R \times I$, unique up to a distribution depending only on time, such that $\nabla_y P_R = F$. Since F is smooth, all spatial derivatives of P_R of positive order are smooth. Fixing, for instance, the spatial mean of P_R over B_R to be zero for each τ selects a representative which is smooth on $B_R \times I$. This proves the local pressure formulation. □

Lemma 21.3 (Leray pressure and local gauges)**.** *Let V be a bounded mild ancient solution satisfying*

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then the whole-space Leray pressure

$$P = R_i R_j (V_i V_j)$$

belongs to $L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3))$. Moreover, if $V_n \to V$ locally smoothly, the V_n have a common $L^\infty_\tau L^3_y$ bound, and

$$P_n = R_i R_j (V_{n,i} V_{n,j}),$$

then on every compact cylinder $B_R \times [\tau_1, \tau_2]$ there are bounded measurable gauges $a_n(\tau)$ and $a(\tau)$ such that, after passing to a subsequence,

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

Equivalently, after replacing the limiting pressure by an equivalent representative modulo a function of time, the weak limit may be denoted by P.

Proof. The first assertion follows from the Calderon–Zygmund boundedness of Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ [12]:

$$\|P(\tau)\|_{L^{3/2}} \le C\|V_i(\tau)V_j(\tau)\|_{L^{3/2}} \le C\|V(\tau)\|_{L^3}^2.$$

For the local stability statement, choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R}. Decompose on each time slice

$$P_n = R_iR_j(\chi V_{n,i}V_{n,j}) + R_iR_j((1-\chi)V_{n,i}V_{n,j}).$$

The local term converges strongly in $L^q(B_{2R} \times [\tau_1,\tau_2])$ for every finite q. This is because $V_n \to V$ smoothly on $\overline{B}_{4R} \times [\tau_1,\tau_2]$, so $\chi V_{n,i}V_{n,j} \to \chi V_iV_j$ strongly in every finite L^q, and the Riesz transforms are Calderon–Zygmund operators on L^q, $1 < q < \infty$.

It remains to control the exterior part modulo constants. Let K_{ij} be the convolution kernel of R_iR_j. For $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x-z) - K_{ij}(-z)| \le C_R|z|^{-4}.$$

Define the exterior gauges

$$b_n(\tau) = \int_{\mathbb{R}^3} K_{ij}(-z)(1-\chi(z))V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz$$

and $b(\tau)$ by the same formula with V in place of V_n. The cutoff removes the singularity at the origin, and the integral is absolutely convergent at infinity because, on the dyadic annulus $A_m = \{2^mR < |z| < 2^{m+1}R\}$,

$$\int_{A_m} |z|^{-3}|V_n(z,\tau)|^2\,dz \le C(2^mR)^{-3}\|V_n(\tau)\|_{L^3(A_m)}^2|A_m|^{1/3} \le CL^2(2^mR)^{-2}.$$

The same dyadic estimate shows that b_n and b are bounded uniformly on $[\tau_1,\tau_2]$. The maps $\tau \mapsto b_n(\tau)$ and $\tau \mapsto b(\tau)$ are measurable, because they are limits of measurable truncated integrals. After subtracting $b_n(\tau)$, the exterior contribution on B_R is

$$E_n(x,\tau) = \int_{\mathbb{R}^3} [K_{ij}(x-z) - K_{ij}(-z)](1-\chi(z))V_{n,i}(z,\tau)V_{n,j}(z,\tau)\,dz.$$

The same dyadic estimate, now using $|z|^{-4}$, gives

$$\|E_n(\cdot,\tau)\|_{L^\infty(B_R)} \le C(R)L^2$$

uniformly in n and τ. Splitting the exterior integral into $\{4R < |z| < R'\}$ and $\{|z| > R'\}$, the first part converges by local smooth convergence, while the second part is bounded uniformly by CL^2/R'. Letting first $n \to \infty$ and then $R' \to \infty$ identifies the local limit of E_n with the corresponding exterior contribution for V. Therefore, with $a_n = b_n$ and $a = b$, the decompositions of $P_n - a_n$ and $P - a$ have local parts converging strongly and exterior parts converging distributionally with a uniform local $L^{3/2}$ bound. Hence

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1,\tau_2]).$$

□

21.3. **Stability under locally smooth limits.**

Lemma 21.4 (Closure of the mild formulation)**.** *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

If $V_n \to V$ locally smoothly, then V is a bounded mild ancient solution of (19.1).

Proof. The common L^∞ bound passes to V by pointwise convergence on compact subsets. Divergence freeness passes distributionally. Fix $\sigma < \tau$ and test (20.4) for V_n against $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$. The endpoint terms are

$$\langle V_n(\tau), \phi\rangle, \qquad \langle V_n(\sigma), S(\tau-\sigma)^*\phi\rangle.$$

The first converges by local smooth convergence. For the second, note that $S(\tau-\sigma)^*\phi \in L^1(\mathbb{R}^3)$. Approximate this L^1 function by a compactly supported smooth function and use the common L^∞ bound to control the L^1 approximation error uniformly in n.

For the Duhamel term, write

$$\langle S(\tau-s)\mathbb{P}\operatorname{div}(V_n\otimes V_n)(s), \phi\rangle = \langle V_n(s)\otimes V_n(s), (S(\tau-s)\mathbb{P}\operatorname{div})^*\phi\rangle\,.$$

For fixed s, the adjoint test belongs to L^1; pairing convergence again follows by compact approximation in L^1, local smooth convergence, and the common L^∞ bound on $V_n\otimes V_n$. The membership in L^1 follows from Lemma 20.1: the operator $(S(\tau-s)\mathbb{P}\operatorname{div})^*$ acts on the compactly supported smooth test field ϕ by integration against the (smooth) adjoint OU–Stokes kernel, which has finite L^1 norm in the spatial variable for every $\tau - s > 0$. The approximation error is uniform in n for this fixed s. Moreover Lemma 20.1 gives the integrable domination

$$C\sup_n \|V_n\|_{L^\infty}^2 e^{-(\tau-s)/2}(1-e^{-(\tau-s)})^{-1/2}\|\phi\|_{L^1}, \qquad s\in(\sigma,\tau).$$

Dominated convergence in s passes the Duhamel integral to the limit. Thus V satisfies (20.4). □

21.4. **Compactness of bounded families.**

Proposition 21.5 (Compactness of bounded ancient families)**.** *Let $\{V_n\}$ be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V.

Proof. By Lemma 21.1, the sequence is bounded in every C^m-norm on every compact cylinder. Arzela–Ascoli on a countable exhaustion $B_N\times[-N,N]$, followed by a diagonal extraction over (m,N), gives a subsequence converging locally smoothly to a smooth limit. Lemma 21.4 shows that the limit is again a bounded mild ancient solution. □

22. Uniform L^3-tightness and stationary L^3 rigidity

22.1. **Uniform L^3-tightness.**

Definition 22.1 (Uniform L^3-tightness)**.** A bounded ancient solution V is uniformly L^3-tight if

$$\lim_{R\to\infty}\sup_{\tau\in\mathbb{R}}\int_{|y|>R}|V(y,\tau)|^3\,dy = 0.$$

For the compactness statements in this section we sometimes assume, in addition, a common $L^\infty_\tau L^3_y$ bound. The endpoint Liouville argument in Section 26 does not take this bound as a separate hypothesis; it obtains the required sequence estimate from boundedness and the tail condition above.

22.2. **Closedness under locally smooth convergence.**

Lemma 22.2 (Closure of uniform L^3-tightness)**.** *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \sup_{\tau\in\mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \le L < \infty$$

and assume that they have a common tightness modulus: for some $\omega : [1,\infty) \to [0,\infty)$ with $\omega(R) \to 0$,

$$\sup_n \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V_n(y,\tau)|^3 \, dy \le \omega(R), \qquad R \ge 1.$$

If $V_n \to V$ locally smoothly, then

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le \omega(R), \qquad R \ge 1.$$

In particular V is uniformly L^3-tight.

Proof. Fix τ. On B_R, local smooth convergence gives

$$\int_{B_R} |V(y,\tau)|^3 \, dy = \lim_{n\to\infty} \int_{B_R} |V_n(y,\tau)|^3 \, dy \le L^3.$$

Here L bounds the L^3 *norm* (so L^3 bounds the integral of the cube), while $\omega(R)$ below bounds the integral on the tail directly, not its cube. Letting $R \to \infty$ and using monotone convergence gives $\|V(\tau)\|_{L^3} \le L$. Since τ was arbitrary, the same bound is uniform in time.

For the tail bound, fix $1 \le R < R'$. On the bounded annulus $\{R < |y| < R'\}$, dominated convergence gives

$$\int_{R<|y|<R'} |V(y,\tau)|^3 \, dy = \lim_{n\to\infty} \int_{R<|y|<R'} |V_n(y,\tau)|^3 \, dy \le \omega(R).$$

Letting $R' \to \infty$ and using monotone convergence gives

$$\int_{|y|>R} |V(y,\tau)|^3 \, dy \le \omega(R).$$

The estimate is independent of τ, so taking the supremum in time gives the asserted tail bound. Since $\omega(R) \to 0$, V is uniformly L^3-tight. ☐

Proof of Theorem 19.3. The compactness assertion is Proposition 21.5. The $L^\infty_\tau L^3_y$ and tightness closure assertions follow from Lemma 22.2 with the displayed constants L and ω. ☐

22.3. **Stationary time-translate limits.** The stationary self-similar profile equation used below is

$$(w\cdot\nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y\cdot\nabla w), \qquad \operatorname{div} w = 0. \tag{22.1}$$

Theorem 22.3 (Stationary limits under uniform L^3-tightness)**.** *Let V be uniformly L^3-tight. Let $\tau_n \in \mathbb{R}$, and suppose*

$$\Theta_{\tau_n} V \to W$$

locally smoothly. If W is stationary, then $W \equiv 0$.

Proof. The sequence $\Theta_{\tau_n} V$ has the same L^3 bound and tightness modulus as V. By Lemma 22.2, the limit W belongs to $L^\infty_\tau L^3_y$. If W is stationary, write $W(y,\tau) = w(y)$. Then $w \in L^3(\mathbb{R}^3)$.

By Lemma 21.4, W is a bounded mild ancient solution. The smoothness from Lemma 21.1 and the local pressure lemma Lemma 21.2 give smooth local pressures p_N on B_N at time 0. Since $\partial_\tau W = 0$, each p_N satisfies

$$(w \cdot \nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w) \quad \text{on } B_N.$$

On overlaps the gradients of the local pressures agree, so the pressures differ only by constants. Adjusting those constants and passing to the compatible union gives a scalar distribution p on $\mathbb{R}^3$ for which (22.1) holds in distributions. The stationary L^3 rigidity theorem Theorem 22.4 applies and gives $w = 0$. □

22.4. **Stationary self-similar rigidity.**

Theorem 22.4 (Stationary self-similar rigidity)**.** *Let $w \in C^\infty(\mathbb{R}^3;\mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ solve (22.1) in distributions for some scalar distribution p, with $\operatorname{div} w = 0$. Then $w \equiv 0$.*

Proof. We use the theorem of Nečas, Růžička, and Šverák [17], applied to the stationary Leray self-similar system. The smooth distributional formulation stated here is a special case of their weak $L^3(\mathbb{R}^3)$ theorem. □

23. Normalized collections and minimal ancient solutions

23.1. **Closed normalized collections.**

Definition 23.1 (Closed normalized collection)**.** A collection A of bounded mild ancient solutions is called a closed normalized collection if:

(i) there is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M \qquad \text{for all } V \in A;$$

(ii) there are $R_0 > 0$ and $\eta_0 > 0$ such that

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \ge \eta_0 \qquad \text{for all } V \in A; \tag{23.1}$$

(iii) A is invariant under time translation:

$$V \in A,\ s \in \mathbb{R} \quad \Longrightarrow \quad \Theta_s V \in A;$$

(iv) if $V_n \in A$, $V_n \to V$ locally smoothly, and V satisfies the same (R_0,η_0)-nonvanishing condition (23.1), then $V \in A$.

23.2. **Time-centering.**

Lemma 23.2 (Time-centering)**.** *Let A be a closed normalized collection. If $V \in A$, then for every $\delta > 0$ there exists $s \in \mathbb{R}$ such that*

$$\int_{B_{R_0}} |\Theta_s V(y,0)|^3\, dy \ge \eta_0 - \delta.$$

Proof. By (23.1), the supremum in time of the local L^3 mass is at least η_0. Choose s within δ of this supremum. Since $(\Theta_s V)(y,0) = V(y,s)$, the displayed estimate follows. Time-translation invariance keeps $\Theta_s V$ in A. □

23.3. **Stability of local nonvanishing.**

Lemma 23.3 (Stability of local nonvanishing). *If $V_n \to V$ locally smoothly and*

$$\int_{B_{R_0}} |V_n(y,0)|^3 \, dy \geq \eta_0 - o(1),$$

then

$$\int_{B_{R_0}} |V(y,0)|^3 \, dy \geq \eta_0.$$

Proof. Local smooth convergence implies uniform convergence on $\overline{B}_{R_0} \times \{0\}$, hence convergence in $L^3(B_{R_0})$. The map $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$, and the claim follows by passing to the limit. □

23.4. **Lower semicontinuity of size.**

Lemma 23.4 (Lower semicontinuity of L^∞ size). *If $V_n \to V$ locally smoothly and*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} < \infty,$$

then

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \leq \liminf_{n\to\infty} \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Proof. Let L_0 be the liminf on the right. If $|V(y_0,\tau_0)| > L_0 + \varepsilon$, choose a subsequence with $\|V_n\|_{L^\infty} \to L_0$. Local uniform convergence at (y_0,τ_0) gives $|V_n(y_0,\tau_0)| > L_0 + \varepsilon/2$ for all large n, contradicting $\|V_n\|_{L^\infty} \to L_0$. Hence $|V| \leq L_0$ pointwise. Since the smooth representative has pointwise supremum equal to its essential L^∞ norm, the result follows. □

23.5. **Existence of a minimal element.**

Theorem 23.5 (Existence of a minimal ancient solution). *Let A be nonempty and closed normalized. Set*

$$m = \inf_{V\in A} \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Then $0 < m < \infty$, and there exists $V_* \in A$ such that

$$\|V_*\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} = m.$$

In particular $V_* \not\equiv 0$.

Proof. The common L^∞ bound gives $m < \infty$. The local nonvanishing gives a positive lower bound:

$$\eta_0 \leq \sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3 \, dy \leq |B_{R_0}| \|V\|^3_{L^\infty(\mathbb{R}^3\times\mathbb{R})},$$

so

$$m \geq (\eta_0/|B_{R_0}|)^{1/3} > 0.$$

Choose $V_n \in A$ with

$$m \leq \|V_n\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \leq m + \frac{1}{n}.$$

By Lemma 23.2, after replacing V_n by a time translate, which stays in A and preserves the L^∞ norm, we may assume

$$\int_{B_{R_0}} |V_n(y,0)|^3 \, dy \geq \eta_0 - \frac{1}{n}.$$

The common L^∞ bound and Proposition 21.5 give a locally smooth subsequential limit V_*, which is a bounded mild ancient solution. By Lemma 23.3,

$$\int_{B_{R_0}} |V_*(y,0)|^3\,dy \geq \eta_0,$$

so V_* retains the fixed (R_0,η_0)-normalization. Closedness of A then gives $V_* \in A$. Finally, Lemma 23.4 gives

$$\|V_*\|_{L^\infty} \leq \liminf_{n\to\infty} \|V_n\|_{L^\infty} = m.$$

Since m is the infimum over A and $V_* \in A$, equality holds. The positive lower bound on m rules out $V_* \equiv 0$. □

23.6. **The uniformly tight normalized class.**

Definition 23.6 (Uniformly tight normalized collection)**.** Fix $M, L, R_0, \eta_0 > 0$ and a modulus

$$\omega : [1,\infty) \to [0,\infty), \qquad \omega(R) \to 0.$$

A collection A of bounded mild ancient solutions is called a uniformly tight normalized collection with data $(M, L, \omega, R_0, \eta_0)$ if:

(1) for every $V \in A$,
$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \leq M;$$
(2) for every $V \in A$,
$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L;$$
(3) for every $V \in A$ and every $R \geq 1$,
$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\,dy \leq \omega(R);$$
(4) for every $V \in A$,
$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\,dy \geq \eta_0;$$
(5) A is invariant under time translation;
(6) A is closed under locally smooth limits retaining item (4).

Proposition 23.7 (Uniformly tight normalized collections are closed normalized)**.** *Every uniformly tight normalized collection is a closed normalized collection.*

Proof. The L^∞ bound, local nonvanishing, time-translation invariance, and closedness clause in Definition 23.1 are part of Definition 23.6. It remains to verify that the stated L^3 bound and fixed tightness modulus are preserved under locally smooth limits. This follows from Lemma 22.2. □

Proof of Theorem 19.4. By Proposition 23.7, the collection is closed normalized. Theorem 23.5 gives the asserted nonzero minimizer. □

24. Compact trajectory hulls

The material in Sections 24 and 25 is auxiliary: it is included to clarify why a purely recurrent compactness argument does not by itself produce a stationary profile, since the barycenter carries a covariance defect. None of these results is used in the proof of Theorem 26.8 or in the final assembly Theorem 34.1.

On compact hulls we use the metric induced by the seminorms $\|\cdot\|_{C^m(B_N\times[-N,N])}$, $m, N \in \mathbb{N}$; this gives the locally smooth topology a complete separable metric structure on bounded sets, used by the probability-measure arguments in Section 25.

24.1. **Trajectory hulls.**

Definition 24.1 (Trajectory hull)**.** For a bounded mild ancient solution V, its trajectory hull is

$$\mathcal{H}(V) = \overline{\{\Theta_s V :\ s \in \mathbb{R}\}}^{C^\infty_{\mathrm{loc}}}.$$

24.2. **Compactness of bounded hulls.**

Theorem 24.2 (Compact trajectory hull)**.** *If V is a bounded mild ancient solution, then $\mathcal{H}(V)$ is compact in the locally smooth topology and invariant under every time translation Θ_s. Every $W \in \mathcal{H}(V)$ is a bounded mild ancient solution with*

$$\|W\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}.$$

Proof. Every translate $\Theta_s V$ is a bounded mild ancient solution with the same L^∞ bound as V. Hence every sequence of translates has a locally smoothly convergent subsequence by Proposition 21.5. If $(W_n) \subset \mathcal{H}(V)$, choose translates $\Theta_{s_n} V$ with $d(W_n, \Theta_{s_n} V) < 1/n$. A subsequence of $\Theta_{s_n} V$ converges locally smoothly, and the corresponding subsequence of W_n has the same limit. Thus $\mathcal{H}(V)$ is sequentially compact. Since the locally smooth topology is metrizable, it is compact.

The L^∞ estimate and the mild equation pass to elements of the hull by local smooth convergence and Lemma 21.4. For invariance, note that Θ_t maps the orbit of V onto itself. If $W_n \to W$ locally smoothly, then $\Theta_t W_n \to \Theta_t W$ locally smoothly because time translation only shifts the compact time interval on which the seminorms are evaluated. Hence $\Theta_t \mathcal{H}(V) = \mathcal{H}(V)$. ◻

Corollary 24.3 (Uniform tightness on hulls)**.** *If V is uniformly L^3-tight with bound L and modulus ω, then every $W \in \mathcal{H}(V)$ satisfies*

$$\sup_{\tau\in\mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \le L$$

and

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |W(y,\tau)|^3\,dy \le \omega(R), \qquad R \ge 1.$$

The same estimates hold uniformly on every compact invariant subset $K \subset \mathcal{H}(V)$.

Proof. Every translate of V has the same L^3 bound and tightness modulus. Applying Lemma 22.2 to any sequence of translates converging locally smoothly to $W \in \mathcal{H}(V)$ gives the displayed estimates for W. Taking the supremum over $W \in K \subset \mathcal{H}(V)$ gives the final assertion. ◻

24.3. **Compact invariant sets.**

Definition 24.4 (Compact invariant set)**.** A nonempty compact subset K of the locally smooth space of bounded mild ancient solutions is invariant if

$$\Theta_t K = K \qquad \text{for every } t \in \mathbb{R}.$$

It is minimal if it contains no proper nonempty compact invariant subset.

Lemma 24.5 (Continuity of the translation action)**.** *Let K be a compact invariant set. For each $t \in \mathbb{R}$, $\Theta_t : K \to K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. For fixed t, convergence $W_n \to W$ locally smoothly implies

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R\times[-T,T])} = \|W_n - W\|_{C^m(B_R\times[-T+t,T+t])} \to 0.$$

Thus Θ_t is continuous, and its inverse is Θ_{-t}. If $t_n \to t$ and $W_n \to W$, then on any fixed compact cylinder the difference

$$\Theta_{t_n} W_n - \Theta_t W$$

is bounded by a term involving $W_n - W$ on a slightly larger compact cylinder and a term involving the uniform continuity of finitely many derivatives of W on that larger cylinder. Both terms tend to zero, proving joint continuity. □

24.4. **Minimal compact invariant subsets.**

Proposition 24.6 (Minimal compact invariant subsets)**.** *Let K be a nonempty compact invariant set. Then K contains a nonempty compact minimal invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K, ordered by reverse inclusion. For a chain $\{K_\alpha\}$, the intersection $\bigcap_\alpha K_\alpha$ is nonempty by compactness and the finite intersection property, since the chain is totally ordered by ordinary inclusion. The intersection is compact and invariant: invariance follows from $\Theta_t K_\alpha = K_\alpha$ for all α and the fact that Θ_t is a homeomorphism. Thus every chain has an upper bound in the reverse-inclusion order. Zorn's lemma gives a maximal element for that order, equivalently a minimal compact invariant subset under ordinary inclusion. □

24.5. **The zero trajectory.**

Lemma 24.7 (The zero trajectory)**.** *The singleton $\{0\}$ is compact and invariant. If a minimal compact invariant set contains the zero trajectory, then it equals $\{0\}$.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for all t. Therefore $\{0\}$ is a nonempty compact invariant subset of any invariant set containing zero. Minimality forces equality. □

25. Invariant measures and the covariance obstruction

25.1. **Krylov–Bogolyubov averages.** The following is the standard Krylov–Bogolyubov averaging argument [23, 20].

Theorem 25.1 (Krylov–Bogolyubov averages)**.** *Let K be a compact invariant set and $W_0 \in K$. For $T > 0$, define*

$$\mu_T = \frac{1}{T}\int_0^T \delta_{\Theta_s W_0}\, ds.$$

Then every sequence $T_n \to \infty$ has a subsequence such that

$$\mu_{T_n} \rightharpoonup \mu$$

weakly as Borel probability measures on K, and every such limit μ is invariant:

$$(\Theta_t)_\# \mu = \mu \qquad \text{for all } t \in \mathbb{R}.$$

If K is minimal, then every invariant Borel probability measure on K has support equal to K.

Proof. The orbit map $s \mapsto \Theta_s W_0$ is continuous by Lemma 24.5; thus μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$. Since K is compact and metrizable, the space of Borel probability measures on K is weakly sequentially compact [20]. Hence μ_{T_n} has a weakly convergent subsequence.

For $F \in C(K)$ and $t > 0$,

$$\int_K F(\Theta_t W)\,d\mu_T(W) - \int_K F(W)\,d\mu_T(W) = \frac{1}{T}\int_T^{T+t} F(\Theta_s W_0)\,ds - \frac{1}{T}\int_0^t F(\Theta_s W_0)\,ds.$$

The absolute value is at most $2t\|F\|_{C(K)}/T$. Letting $T = T_{n_k} \to \infty$ gives invariance for $t > 0$; negative times follow by applying the positive-time result to $F \circ \Theta_t$.

If K is minimal and μ is invariant, then $\operatorname{supp}\mu$ is nonempty, closed, and invariant. Indeed, if U is a neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is a neighborhood of W, and invariance shows $\mu(U) = \mu(\Theta_{-t}U) > 0$ whenever $W \in \operatorname{supp}\mu$. Hence $\Theta_t(\operatorname{supp}\mu) \subset \operatorname{supp}\mu$. Applying the same argument with $-t$ gives the reverse inclusion, so $\operatorname{supp}\mu$ is invariant. It is nonempty because $\mu(K) = 1$, and it is compact because it is closed in compact K. Minimality forces $\operatorname{supp}\mu = K$. □

25.2. **Support on minimal hulls.**

Corollary 25.2 (Nonzero support)**.** *Let K be a minimal compact invariant set with $K \neq \{0\}$. Then zero is not in K, and every invariant probability measure on K has support equal to K.*

Proof. If zero were in K, Lemma 24.7 would give $K = \{0\}$, contrary to the assumption. The support conclusion is the minimal-support part of Theorem 25.1. □

25.3. **Stationary statistical identity.**

Theorem 25.3 (Stationary statistical form)**.** *Let K be a compact invariant set and μ an invariant Borel probability measure on K. Then for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$,*

$$\begin{aligned}0 = \int_K \Bigg[&\int_{\mathbb{R}^3} W(y,0)\cdot\left(\Delta\phi + \phi + \frac{1}{2}y\cdot\nabla\phi\right)dy \\ &+ \int_{\mathbb{R}^3} W_i(y,0)W_j(y,0)\partial_j\phi_i(y)\,dy\Bigg]\,d\mu(W).\end{aligned} \tag{25.1}$$

Proof. Define

$$F(W) = \int_{\mathbb{R}^3} W(y,0)\cdot\phi(y)\,dy.$$

Local smooth convergence makes F continuous on K: if $W_n \to W$, then $W_n(\cdot,0) \to W(\cdot,0)$ uniformly on the compact support of ϕ. Invariance gives

$$\int_K \frac{F(\Theta_h W) - F(W)}{h}\,d\mu(W) = 0 \qquad (0 < |h| \le 1).$$

Choose R with $\operatorname{supp}\phi \subset B_R$. On $\overline{B}_R \times [-1,1]$, compactness of K in the locally smooth topology gives

$$M_\phi = \sup_{W\in K}\|\partial_\tau W\|_{L^\infty(\overline{B}_R\times[-1,1])} < \infty.$$

For $0 < |h| \le 1$,

$$\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3}\left(\frac{1}{h}\int_0^h \partial_\tau W(y,s)\,ds\right)\cdot\phi(y)\,dy,$$

and the absolute value is bounded by $|B_R|\|\phi\|_{L^\infty}M_\phi$, uniformly in W and h. For each fixed W, smoothness gives uniform convergence of the inner average to $\partial_\tau W(y,0)$ on $\operatorname{supp}\phi$. Dominated convergence on K yields

$$\int_K\int_{\mathbb{R}^3}\partial_\tau W(y,0)\cdot\phi(y)\,dy\,d\mu(W) = 0.$$

For each W, the smooth local equation (19.1) at time zero, tested against solenoidal ϕ, gives

$$\int_{\mathbb{R}^3} \partial_\tau W \cdot \phi \, dy = \int_{\mathbb{R}^3} W \cdot \Delta\phi \, dy + \int_{\mathbb{R}^3} W \cdot \phi \, dy + \frac{1}{2}\int_{\mathbb{R}^3} W \cdot (y \cdot \nabla\phi) \, dy + \int_{\mathbb{R}^3} W_i W_j \partial_j \phi_i \, dy.$$

The pressure term vanishes because $\operatorname{div}\phi = 0$; the drift identity uses

$$\int y_j \partial_j W_i \, \phi_i \, dy = -3 \int W \cdot \phi \, dy - \int W \cdot (y \cdot \nabla\phi) \, dy.$$

The right-hand side is a continuous bounded function of $W \in K$. Averaging over K gives (25.1). □

25.4. **Barycenter and covariance.**

Theorem 25.4 (Barycenter and covariance)**.** *Let K and μ be as in Theorem 25.3. Define*

$$\overline{W}(y) = \int_K W(y,0) \, d\mu(W)$$

and

$$C_{ij}(y) = \int_K W_i(y,0) W_j(y,0) \, d\mu(W) - \overline{W}_i(y)\overline{W}_j(y).$$

Then $\overline{W}$ and C are C^∞_{loc}, $\overline{W}$ is solenoidal, and for every solenoidal $\phi \in C^\infty_c(\mathbb{R}^3;\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \left(\Delta\phi + \phi + \frac{1}{2} y \cdot \nabla\phi \right) dy + \int_{\mathbb{R}^3} \overline{W}_i \overline{W}_j \partial_j \phi_i \, dy + \int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i \, dy = 0. \tag{25.2}$$

In particular, $\overline{W}$ is a weak stationary self-similar solution against solenoidal test fields if and only if

$$\int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i \, dy = 0$$

for every solenoidal test field ϕ.

Proof. Compactness of K in the locally smooth topology gives, for every compact ball and every derivative order, a uniform bound on $\partial_y^\alpha W(y,0)$ for $W \in K$. Hence one may differentiate $\overline{W}$ under the integral sign. More explicitly, for every multiindex α and every y in a fixed compact ball,

$$\partial_y^\alpha \overline{W}(y) = \int_K \partial_y^\alpha W(y,0) \, d\mu(W),$$

because the difference quotients are dominated by the next derivative seminorm, uniformly on K. The same argument applies to the second moment

$$M_{ij}(y) = \int_K W_i(y,0) W_j(y,0) \, d\mu(W),$$

using Leibniz' rule and the uniform bounds on derivatives of W. This gives C^∞_{loc} regularity of $\overline{W}$, M, and $C = M - \overline{W} \otimes \overline{W}$. Since every W is divergence free, Fubini against scalar test functions gives $\operatorname{div}\overline{W} = 0$: for $\eta \in C^\infty_c(\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \nabla\eta \, dy = \int_K \int_{\mathbb{R}^3} W(y,0) \cdot \nabla\eta(y) \, dy \, d\mu(W) = 0.$$

For each compactly supported test field ϕ, the maps

$$W \mapsto \int_{\mathbb{R}^3} W(y,0) \cdot \phi(y) \, dy, \qquad W \mapsto \int_{\mathbb{R}^3} W_i(y,0) W_j(y,0) \partial_j \phi_i(y) \, dy$$

are continuous and bounded on K, again by compactness in C^∞_{loc}. Thus all integrals below are finite, and Fubini's theorem applies.

Applying Theorem 25.3 and passing the linear term through the K-integral gives the first integral in (25.2). Passing the nonlinear term through the integral gives

$$\int_K W_i W_j \, d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Substitution yields (25.2). Comparing this identity with the pressure-free stationary weak formulation for $\overline{W}$ gives the stated equivalence with vanishing covariance contribution. □

26. Endpoint L^3 closure of the uniformly tight class

26.1. **Physical pullback.** The pullback below may be unbounded as $t \uparrow 0$; it is bounded only on every truncated ancient interval $(-\infty, T]$ with $T < 0$ by the factor $(-T)^{-1/2}$ below. For this reason the Albritton–Barker endpoint Liouville theorem is applied after translating such a truncated interval to terminal time 0, not directly to the unbounded pullback on $(-\infty, 0)$.

Let V be a smooth centered ancient velocity and let P be a compatible local pressure. For $t < 0$ define

$$u(x,t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \tag{26.1}$$

and

$$p(x,t) = (-t)^{-1} P\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right). \tag{26.2}$$

Equivalently,

$$V(y,\tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}), \qquad P(y,\tau) = e^{-\tau} p(e^{-\tau/2} y, -e^{-\tau}).$$

Lemma 26.1 (Pullback solves physical Navier–Stokes)**.** *Let V be a bounded mild ancient solution of (19.1). Then the pullback u in (26.1) is a smooth divergence-free solution of*

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \Delta u, \qquad \operatorname{div} u = 0, \tag{26.3}$$

on $\mathbb{R}^3 \times (-\infty, 0)$, with pressure p locally given by (26.2). Moreover, for every $T < 0$,

$$\|u\|_{L^\infty(\mathbb{R}^3 \times (-\infty, T])} \le (-T)^{-1/2} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. By Lemma 21.1, V is smooth. By Lemma 21.2, on every compact parabolic cylinder there is a smooth pressure representative P, unique up to a function of τ, for which (19.1) holds classically. Adding a function of τ to P only adds a function of t to p, and hence does not affect ∇p.

Set

$$\lambda(t) = (-t)^{-1/2}, \qquad y = \lambda(t) x, \qquad \tau = -\log(-t).$$

Then $\dot{\tau} = \lambda^2$ and

$$\partial_t y = \frac{1}{2}\lambda^2 y, \qquad \dot{\lambda} = \frac{1}{2}\lambda^3.$$

Using $u = \lambda V(y,\tau)$, one obtains

$$\partial_t u = \lambda^3 \left(\partial_\tau V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V\right),$$

while

$$(u \cdot \nabla_x) u = \lambda^3 (V \cdot \nabla_y) V, \qquad \Delta_x u = \lambda^3 \Delta_y V, \qquad \nabla_x p = \lambda^3 \nabla_y P.$$

Substituting these identities into (19.1) gives (26.3). The divergence identity follows similarly:

$$\operatorname{div}_x u = \lambda^2 \operatorname{div}_y V = 0.$$

Finally, if $t \le T < 0$, then $(-t)^{-1/2} \le (-T)^{-1/2}$, and the stated L^∞ bound follows directly from (26.1). ☐

26.2. **Critical L^3 invariance.**

Lemma 26.2 (Invariance of the critical norm)**.** *Let u be the physical pullback of V. Then, for every $t < 0$,*

$$\|u(t)\|_{L^3(\mathbb{R}^3)} = \|V(-\log(-t))\|_{L^3(\mathbb{R}^3)}. \tag{26.4}$$

Proof. With $y = x/\sqrt{-t}$, (26.1) gives

$$|u(x,t)|^3 = (-t)^{-3/2}|V(y, -\log(-t))|^3, \qquad dx = (-t)^{3/2}dy.$$

Multiplying these two identities and integrating over $\mathbb{R}^3$ gives (26.4). ☐

Lemma 26.3 (Tail tightness gives an endpoint sequence)**.** *Let V be a bounded function on $\mathbb{R}^3 \times \mathbb{R}$ satisfying*

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0.$$

Then, for every sequence $\tau_k \in \mathbb{R}$,

$$\sup_k \|V(\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. Let $M = \|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})}$. Choose $R \ge 1$ such that

$$\sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le 1.$$

Then, for every k,

$$\int_{\mathbb{R}^3} |V(y,\tau_k)|^3 \, dy \le M^3|B_R| + \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy \le M^3|B_R| + 1.$$

This gives the claimed endpoint sequence bound. ☐

26.3. **Mildness after finite terminal-time restriction.**

Lemma 26.4 (Physical pullbacks are mild ancient solutions on finite intervals)**.** *Let V be a bounded mild ancient solution of (19.1), and let u be its physical pullback. For every $T < 0$, the restriction of u to $\mathbb{R}^3 \times (-\infty, T]$ is a bounded mild ancient solution of the physical Navier–Stokes equations after translating the terminal time T to 0.*

Proof. Let $s < t \le T < 0$, and set

$$\sigma = -\log(-s), \qquad \tau = -\log(-t).$$

Then $\sigma < \tau$. Apply the centered mild formula for V on $[\sigma, \tau]$:

$$V(\tau) = S(\tau-\sigma)V(\sigma) - \int_\sigma^\tau S(\tau-\alpha)\mathbb{P}\nabla\cdot(V\otimes V)(\alpha)\,d\alpha.$$

Put

$$a = \tau - \sigma, \qquad \lambda = e^{-a/2} = \sqrt{\frac{-t}{-s}}, \qquad \theta = 1 - e^{-a} = \frac{t-s}{-s}.$$

For the linear term, use the explicit Ornstein–Uhlenbeck semigroup formula (20.2). Writing $x = \sqrt{-t}\, y$ and changing variables $z = \zeta/\sqrt{-s}$, we obtain

$$\begin{aligned}(-t)^{-1/2}(S(a)V(\sigma))(y) &= (-t)^{-1/2}e^{-a/2}\int_{\mathbb{R}^3}\frac{\exp\Big(-\frac{|e^{-a/2}y-z|^2}{4(1-e^{-a})}\Big)}{[4\pi(1-e^{-a})]^{3/2}}V(z,\sigma)\,dz\\ &= \int_{\mathbb{R}^3}\frac{\exp\Big(-\frac{|x-\zeta|^2}{4(t-s)}\Big)}{[4\pi(t-s)]^{3/2}}u(\zeta,s)\,d\zeta\\ &= (e^{(t-s)\Delta}u(s))(x).\end{aligned}$$

For the Duhamel term, write $r = -e^{-\alpha}$ for $\alpha \in [\sigma, \tau]$. Then $dr = (-r)\,d\alpha$,

$$V(\xi,\alpha)\otimes V(\xi,\alpha) = (-r)\,(u\otimes u)(\sqrt{-r}\,\xi, r),$$

and the homogeneity of $\mathbb{P}\nabla\cdot$ together with the same scaling calculation gives

$$\begin{aligned}&(-t)^{-1/2}\big(S(\tau-\alpha)\mathbb{P}\nabla\cdot(V\otimes V)(\alpha)\big)(y)\\ &\quad= (-r)\,\big(e^{(t-r)\Delta}\mathbb{P}\nabla\cdot(u\otimes u)(r)\big)(x).\end{aligned}$$

Indeed, if $F_\alpha(\xi) = V(\xi,\alpha)\otimes V(\xi,\alpha)$ and $H_r(x) = u(x,r)\otimes u(x,r)$, then

$$F_\alpha(\xi) = (-r)H_r(\sqrt{-r}\,\xi),$$

so

$$\mathbb{P}\nabla_\xi\cdot F_\alpha = (-r)^{3/2}\big(\mathbb{P}\nabla_x\cdot H_r\big)(\sqrt{-r}\,\xi),$$

and using $S(b)f(\eta) = e^{-b/2}(e^{(1-e^{-b})\Delta}f)(e^{-b/2}\eta)$ with $b = \tau-\alpha$ yields the displayed identity because $e^{-b/2}\sqrt{-r} = \sqrt{-t}$ and $(-r)(1-e^{-b}) = t-r$. Therefore, after the change of variables $r = -e^{-\alpha}$,

$$\begin{aligned}&(-t)^{-1/2}\int_\sigma^\tau S(\tau-\alpha)\mathbb{P}\nabla\cdot(V\otimes V)(\alpha)\,d\alpha\\ &\quad= \int_s^t e^{(t-r)\Delta}\mathbb{P}\nabla\cdot(u\otimes u)(r)\,dr.\end{aligned}$$

Under the parabolic change of variables

$$u(x,t) = (-t)^{-1/2}V(x/\sqrt{-t}, -\log(-t)),$$

the centered mild formula therefore becomes

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-r)\Delta}\mathbb{P}\operatorname{div}(u\otimes u)(r)\,dr$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$. Since $t \le T < 0$, the physical pullback is bounded on $\mathbb{R}^3\times(-\infty,T]$, and the Oseen kernel estimate makes the Duhamel term finite on every finite interval. Hence the restriction of u to $(-\infty,T]$, after translating T to 0, is a bounded mild ancient solution. □

26.4. Endpoint L^3 ancient Liouville theorem.

Theorem 26.5 (Albritton–Barker endpoint ancient Liouville theorem)**.** *Let u be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3\times(-\infty,0]$. Suppose that there is a sequence $s_k \downarrow -\infty$ such that*

$$\sup_k \|u(s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

and suppose $u(0)\in L^3(\mathbb{R}^3)$. Then $u\equiv 0$. The statement is written after translating the terminal time to 0.

Proof. We use the L^3-sequence formulation of Albritton–Barker's ancient Liouville theorem [2, Theorem 1.2], after translating the terminal time to 0. $\square$

Remark 26.6 (Quantitative Lorentz/Besov form). Albritton–Barker also give a quantitative endpoint statement [2, Theorem 4.1] with additional quantitative endpoint hypotheses. Those hypotheses are not used in this paper. The only imported statement needed below is the L^3-sequence Liouville theorem in Theorem 26.5.

Remark 26.7 (Notation match with Albritton–Barker). In the tight-class application below, the terminal datum is $u_T(0) = u(T)$. Tail tightness and the critical L^3 identity give $u(T) \in L^3(\mathbb{R}^3)$, so the L^3 terminal hypothesis of Theorem 26.5 is automatic. The backward sequence bound is obtained in L^3, which is the form used below.

Theorem 26.8 (Tail-tight centered ancient solutions vanish). *Let V be a bounded mild ancient solution of* (19.1)*. Assume*

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3\, dy = 0.$$

Then $V \equiv 0$.

Proof. Let u be the physical pullback of V. Fix $T < 0$. By Lemma 26.4, after translating the terminal time T to 0, the restriction of u to $(-\infty, T]$ is a bounded mild ancient solution. Moreover $u(T) \in L^3(\mathbb{R}^3)$: indeed, if $\tau_T = -\log(-T)$, then $\|u(T)\|_{L^3} = \|V(\tau_T)\|_{L^3}$, and the latter is finite by boundedness on B_R and tail tightness outside B_R. Choose $t_k = -e^k$. For all sufficiently large k, $t_k < T$, and $\tau_k = -\log(-t_k) = -k$. By Lemma 26.3,

$$\sup_k \|V(-k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Hence (26.4) gives

$$\|u(t_k)\|_{L^3(\mathbb{R}^3)} = \|V(-k)\|_{L^3(\mathbb{R}^3)}$$

uniformly along the same sequence, after discarding the finitely many indices for which $t_k \geq T$. Set $u_T(x,\sigma) = u(x, \sigma + T)$ for $\sigma \leq 0$ and $s_k = t_k - T$. Then $s_k \downarrow -\infty$ and

$$\|u_T(s_k)\|_{L^3(\mathbb{R}^3)} = \|u(t_k)\|_{L^3(\mathbb{R}^3)}$$

is uniformly bounded. Applying Theorem 26.5 to u_T gives $u_T \equiv 0$ on $\mathbb{R}^3 \times (-\infty, 0]$, equivalently $u = 0$ on $\mathbb{R}^3 \times (-\infty, T]$. Since $T < 0$ was arbitrary, $u = 0$ on $\mathbb{R}^3 \times (-\infty, 0)$. The inverse change of variables then gives

$$V(y,\tau) = e^{-\tau/2} u(e^{-\tau/2} y, -e^{-\tau}) = 0$$

for every (y,τ). $\square$

27. Tight-Class Liouville Criterion and Type I Reduction

27.1. **Normalized tight collections.**

Theorem 27.1 (No nonempty normalized uniformly tight collection). *No uniformly tight normalized collection in the sense of Definition 23.6 is nonempty.*

Proof. Suppose A is such a collection and choose $V \in A$. By the tightness modulus in item (3) of Definition 23.6, V satisfies the tail condition in Theorem 26.8. Hence Theorem 26.8 gives $V \equiv 0$. This contradicts the local velocity normalization in the same definition:

$$\sup_{\tau\in\mathbb{R}} \int_{B_{R_0}} |V(y,\tau)|^3\, dy \geq \eta_0 > 0.$$

Hence A is empty. $\square$

Proof of Theorem 19.5. If A were nonempty, choose $V \in A$. The common tail modulus gives the tail condition in Theorem 26.8, so $V \equiv 0$. This contradicts the fixed local nonvanishing normalization in Theorem 19.5. $\square$

27.2. **Use in the Type I reduction.**

Definition 27.2 (Tight-class Liouville consequence)**.** The tight-class Liouville consequence is the assertion that no generated raw Seregin state satisfying the invariant nonvanishing normalization of Corollary 14.19 belongs to $\mathcal{C}_{L^3-\mathrm{tight}}$.

Corollary 27.3 (Tight-class consequence for the Type I reduction)**.** *Definition 27.2 holds.*

Proof. Suppose, contrary to Definition 27.2, that a generated raw Seregin state V belongs to $\mathcal{C}_{L^3-\mathrm{tight}}$. By definition of $\mathcal{C}_{L^3-\mathrm{tight}}$, $V \in \mathcal{S}_{\mathrm{mild}}$ and satisfies the tail condition

$$\lim_{R\to\infty} \sup_{\tau\in\mathbb{R}} \int_{|y|>R} |V(y,\tau)|^3 \, dy = 0.$$

Theorem 26.8 applies and gives $V \equiv 0$, contradicting Corollary 14.19. Hence no generated raw Seregin state can belong to $\mathcal{C}_{L^3-\mathrm{tight}}$. $\square$

The tight-class consequence used in the Type I reduction is proved independently of the companion residual closure. The proof relies only on boundedness, the tail-tightness condition of the uniformly L^3-tight class, the local-in-time physical pullback, and the endpoint ancient Liouville theorem of Albritton–Barker.

Part IV. Structured Ancient Solution Exclusions

Overview. The final part concerns bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. Explicit PDE hypotheses are stated under which such solutions are spatially constant; in the Type I class, these constants vanish. The perturbative weighted-vorticity anchored Gallay–Wayne perturbative exclusion used by the Type I reduction is also proved.

For the unforced equations, the cited axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L^\infty_t L^p_x$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ. When one of these hypotheses holds for a nonzero Type I blow-up limit, the cited theorems reduce the solution to a spatially constant state in the physical Type I frame, and that constant is then forced to vanish, contradicting the nontriviality of the limit.

The perturbative part uses Gallay–Wayne's forward small-data theorem only through anchored forward rescalings launched from arbitrarily early physical times. The anchored small-data estimate forces the vorticity at any fixed physical time to vanish, and the section concludes with a summary of the Liouville hypotheses and cited PDE results.

28. Introduction

28.1. **Ancient solutions from Type I blow-up.** The Type I blow-up argument in Theorem 10.1 produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$\partial_t U + (U\cdot\nabla)U + \nabla P = \Delta U, \qquad \nabla\cdot U = 0, \qquad t<0, \tag{28.1}$$

satisfying the Type I ancient bound

$$\sup_{t<0} \sqrt{-t}\,\|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty. \tag{28.2}$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (28.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \quad\Longrightarrow\quad U\equiv 0 \quad\Longrightarrow\quad \text{contradiction with nonzero blow-up limit.}$$

The hypotheses treated in this part are precisely those to which a cited Liouville theorem or the perturbative argument in Section 32 applies:

$$\text{axisymmetric Liouville hypotheses,} \qquad \text{anchored Gallay–Wayne perturbative vorticity.}$$

No conclusion is asserted for decaying or symmetry-restricted ancient solutions outside the stated hypotheses. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

For the Type I application in Theorem 10.1, the exclusions used are precisely the cases in which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$. Ancient solutions outside these hypotheses are treated separately.

28.2. **Axisymmetric Liouville hypotheses.** Several unforced axisymmetric cases fall under cited Liouville theorems. Koch–Nadirashvili–Seregin–Šverák treat the no-swirl case and the pointwise scale-invariant condition $|u| \le C/r$. The swirl-bearing finite-L^p condition

$$\Gamma = ru_\theta \in L^\infty_t L^p_x, \qquad 1\le p<\infty,$$

is treated by Lei–Zhang–Zhao. The z-periodic case with bounded Γ is treated by Lei–Ren–Zhang.

For Type I ancient limits, a constant conclusion after the relevant Galilean reduction must be converted into zero in the blow-up frame. The proof time-shifts away from the terminal time $t = 0$, applies the bounded ancient Liouville theorem, patches the resulting constants on backward intervals, and uses (28.2): if $U(x,t) \equiv c$, then

$$\sup_{t<0} \sqrt{-t}\,\|U(t)\|_\infty = \sup_{t<0} \sqrt{-t}\,|c|$$

is finite only when $c = 0$.

28.3. **Anchored Gallay–Wayne hypotheses.** The perturbative branch is stated directly for the physical Type I ancient representative U. From anchor times $a_n \to -\infty$ and scales λ_n, one launches the Gallay–Wayne forward rescaled vorticity equation from the anchored physical vorticity data

$$w_n^0(\xi) = \lambda_n^2 (\nabla \times U)(\lambda_n \xi, a_n).$$

If these anchored initial data remain in a fixed perturbative $L^2_\sigma(m)$-ball, with the Biot–Savart normalization at each anchor, then Gallay–Wayne's forward decay estimate applies on every anchored forward interval. Evaluating those anchored forward solutions at any fixed physical time $T < 0$ and sending $n \to \infty$ forces $\nabla \times U(\cdot, T) = 0$, hence $U \equiv 0$. This avoids treating the backward Type I centered vorticity as a single Gallay–Wayne forward orbit.

28.4. **Main results.** The first result states the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 28.1 (Axisymmetric Liouville consequences for Type I ancient solutions)**.** *Let U be a smooth mild Type I ancient solution of* (28.1) *in the sense of Definition 29.1. If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

(i) *axisymmetric without swirl;*
(ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x,t)| \le \frac{C}{r};$$

(iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty;$$

(iv) *axisymmetric, periodic in the physical axial variable z, and with*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 28.2 (Anchored Gallay–Wayne perturbative exclusion)**.** *Let U be a smooth mild Type I ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$. If U satisfies the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2, then*

$$U \equiv 0.$$

The anchored condition is stated directly in terms of the physical ancient representative U. Gallay–Wayne's theorem is used only on forward Cauchy problems launched from the anchor times a_n.

Theorem 28.3 (Consequences for Type I blow-up limits)**.** *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in Theorem 28.1; nor can U satisfy the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2.*

Only the explicit hypotheses in the theorem are used.

28.5. **Relation to the preceding parts.** Theorem 10.1 reduces Type I singularities to nonzero ancient limits satisfying (28.2). Theorem 26.8 treats uniformly L^3-tight ancient solutions. The axisymmetric and perturbative weighted-vorticity exclusions used by the Type I argument are proved in Sections 31 and 32. Cited Liouville theorems rule out the axisymmetric cases, and the anchored Gallay–Wayne argument handles the perturbative weighted-vorticity case. Theorem 33.1 does not depend on Theorem 26.8.

28.6. **Summary of hypotheses.**

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $\lvert u\rvert \le C/r$	$U \equiv 0$	KNSS plus $r \to \infty$ constant removal
$\Gamma \in L^\infty_t L^p_x$, $1 \le p < \infty$	$U \equiv 0$	KNSS/LZZ split plus Type I constant removal
z-periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
anchored Gallay–Wayne perturbative vorticity	$U \equiv 0$	anchored Gallay–Wayne decay plus physical uniqueness comparison

28.7. **Organization.** Section 29 introduces the Type I ancient setting, self-similar variables, the invariance facts, and the time-shift lemma. Section 30 fixes axisymmetric notation and the cited Liouville theorems. Section 31 proves the unforced axisymmetric Liouville consequences. Section 32 proves the anchored Gallay–Wayne perturbative exclusion, and Section 33 summarizes the consequences.

29. Ancient Type I limits and self-similar variables

29.1. Type I ancient solutions.

Definition 29.1 (Type I ancient solution)**.** A smooth mild solution U of (28.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if (28.2) holds. Mild is understood in the standard velocity formulation [14, 19]: for every $-\infty < s < t < 0$,

$$U(t) = e^{(t-s)\Delta}U(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (U(\sigma) \otimes U(\sigma))\, d\sigma \tag{29.1}$$

in distributions, where $\mathbb{P}$ is the Leray projector.

29.2. Nonzero Type I blow-up limits.

Definition 29.2 (Nonzero Type I blow-up limit)**.** A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 29.3 (Nontriviality of blow-up limits)**.** *A nonzero Type I blow-up limit is not identically zero.*

Proof. The definition includes the condition $U \not\equiv 0$. □

29.3. **Backward self-similar variables.** For $t < 0$, define

$$V(y,\tau) = \sqrt{-t}\,U(x,t), \qquad Q(y,\tau) = (-t)P(x,t), \qquad y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t). \tag{29.2}$$

Equivalently,

$$x = e^{-\tau/2}y, \qquad t = -e^{-\tau}.$$

Lemma 29.4 (Self-similar equation)**.** *Let (U,P) be a smooth solution of (28.1), and define (V,Q) by (29.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$\partial_\tau V + (V\cdot\nabla)V + \nabla Q = \Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \nabla\cdot V = 0. \tag{29.3}$$

Moreover (28.2) is equivalent to

$$\sup_{\tau\in\mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t\tau = \frac{1}{-t} = \frac{1}{a^2}, \qquad \partial_t y = \frac{y}{2a^2}, \qquad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3}\left(\partial_\tau V + \frac{1}{2}y\cdot\nabla V + \frac{1}{2}V\right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2}\nabla_y V, \qquad \Delta_x U = a^{-3}\Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3}\nabla_y Q.$$

Thus

$$(U\cdot\nabla_x)U = a^{-3}(V\cdot\nabla_y)V.$$

Multiplying (28.1) by a^3 yields (29.3); the divergence equation follows from $\nabla_x\cdot U = a^{-2}\nabla_y\cdot V$. The spatial map $y \mapsto x = e^{-\tau/2}y$ is a bijection for each τ, and

$$\|V(\tau)\|_\infty = \sqrt{-t}\,\|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau \in \mathbb{R}$, gives the last assertion. ☐

29.4. **Preservation of axisymmetry and swirl variables.** For axisymmetric flows we write

$$x = (r\cos\theta, r\sin\theta, z), \qquad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho\cos\theta, \rho\sin\theta, Z), \qquad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r,z,t) = rU_\theta(r,z,t), \qquad \Gamma_V(\rho,Z,\tau) = \rho V_\theta(\rho,Z,\tau).$$

Lemma 29.5 (Preservation of axisymmetric structure)**.** *Axisymmetry and absence of swirl are preserved under (29.2). If U is axisymmetric, then*

$$\Gamma_V(\rho,Z,\tau) = \Gamma_U(r,z,t), \qquad (r,z) = \sqrt{-t}\,(\rho,Z).$$

Fixed physical z-periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ-dependent period in the Z-coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho, Z, \tau) = \sqrt{-t}\, U_\theta(r, z, t), \qquad r = \sqrt{-t}\rho, \quad z = \sqrt{-t}Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho\sqrt{-t}\, U_\theta = rU_\theta = \Gamma_U.$$

If U has physical axial period L, then V has period $L/\sqrt{-t} = Le^{\tau/2}$ in Z, not a fixed period. □

29.5. **Time shifts away from the singular time.**

Lemma 29.6 (Time shifts away from the singular time)**.** *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x, s) = U(x, T + s), \qquad \Pi_T(x, s) = P(x, T + s), \qquad s < 0,$$

is a smooth bounded mild ancient solution of (28.1) *on $\mathbb{R}^3 \times (-\infty, 0)$. Moreover the axisymmetric, no-swirl, L^p-swirl, bounded-Γ, and physical z-periodic properties are preserved.*

Proof. Time translation preserves the differential equation because $\partial_s W_T(x, s) = \partial_t U(x, T + s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (29.1) to U between the physical times $T + s < T + t < 0$:

$$U(T + t) = e^{(t-s)\Delta}U(T + s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta}\mathbb{P}\nabla \cdot (U(\eta) \otimes U(\eta))\, d\eta.$$

With $\eta = T + \sigma$, this becomes the mild formula for W_T on the interval (s, t). For $s < 0$, $T + s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T + s)\|_\infty \le \frac{1}{\sqrt{-T}} \sup_{\eta<0} \sqrt{-\eta}\, \|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r, z, s) = \Gamma_U(r, z, T + s),$$

so $L^\infty_s L^p_x$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. □

29.6. **PDE hypotheses.** The arguments are stated directly in terms of hypotheses on ancient solutions. The axisymmetric hypotheses are:

(i) axisymmetry and $U_\theta \equiv 0$;
(ii) axisymmetry and the pointwise bound $|U(x, t)| \le C/r$ for $r > 0$;
(iii) axisymmetry and

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big), \qquad 1 \le p < \infty;$$

(iv) axisymmetry, periodicity in the physical axial variable z, and $\Gamma \in L^\infty$.

The additional hypothesis used later is the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2.

30. Axisymmetric notation and cited Liouville theorems

30.1. **Cylindrical variables and the swirl scalar.** For an axisymmetric solution u, write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \qquad \Gamma = r u_\theta, \qquad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r\cos\theta, r\sin\theta, z)$, independent of θ:

$$\|\Gamma(t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r,z,t)|^p r\, dr\, dz, \qquad 1 \le p < \infty.$$

30.2. **Constant solutions in the Type I frame.**

Definition 30.1 (Reduction to a spatial constant in the Type I frame)**.** The cited Liouville theorems are used below only through the consequence that, after passing to the finite-terminal shift in the physical Type I frame, the solution is spatially constant. The Type I bound then forces that constant to be zero.

Lemma 30.2 (Spatially constant mild solutions)**.** *Let W be a smooth mild solution of (28.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If $W(x,t) = b(t)$ for all x and t, then b is constant in time.*

Proof. Insert $W(x,t) = b(t)$ into (29.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. □

Lemma 30.3 (Patching constants on backward intervals)**.** *Let $U : \mathbb{R}^3 \times (-\infty, 0) \to \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x,t) = c_T \qquad (x \in \mathbb{R}^3,\ t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x,t) = c \qquad (x \in \mathbb{R}^3,\ t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. □

Lemma 30.4 (Constant Type I ancient solutions vanish)**.** *Let U be a Type I ancient solution. If $U(x,t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t}\,|c| \le \sup_{s<0} \sqrt{-s}\,\|U(s)\|_\infty < \infty \qquad (t < 0).$$

Letting $t \to -\infty$ gives $c = 0$. □

30.3. **Cited Liouville theorems.** The following Liouville theorems are used in the stated forms.

Remark 30.5 (Match with the cited papers)*.* The axisymmetric theorems below are imported from the cited references and are recorded with the hypotheses used here. In the Type I application they are applied to the finite-terminal time shifts

$$W_T(x,s) = U(x, T+s), \qquad s < 0, \quad T < 0,$$

which are bounded ancient mild solutions by Lemma 29.6. The resulting constants are then patched in T and removed by the Type I bound. The notation matches the cited papers modulo the centered-versus-physical translation recorded in Lemma 29.5.

Theorem 30.6 (KNSS no-swirl Liouville theorem)**.** *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is reduced, in the form used here, to a spatially constant axial drift in the physical Type I frame.*

Proof. Imported theorem. We use the axisymmetric no-swirl Liouville theorem of Koch, Nadirashvili, Seregin, and Šverák [15, Theorem 5.2].

Required hypotheses. The theorem is stated for bounded ancient mild solutions of the unforced Navier–Stokes equations in the whole space which are axisymmetric and have no swirl. No drift-only or circulation-only hypothesis is being substituted for the no-swirl assumption in this branch.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 29.6. The no-swirl branch of $\mathcal{C}_{\mathrm{sd}}$ requires axisymmetry together with $u_\theta \equiv 0$, and this structural hypothesis is preserved under the time shift. Therefore the cited theorem applies to W_T.

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives that W_T is a spatially constant axial drift in the frame used here. The resulting constants patch in T by Lemma 30.3, and Lemma 30.4 then forces the patched constant to vanish. □

Theorem 30.7 (KNSS pointwise scale-invariant condition)**.** *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x,t)| \leq \frac{C}{r} \qquad (r > 0,\ t < 0)$$

is constant up to a Galilean transformation. Only the spatially constant conclusion in the physical Type I frame is used below; if the source theorem is stated modulo Galilean transformations, this branch is invoked only after reducing to that constant representative. Zero is obtained only after the constant-removal step.

Proof. Imported theorem. We use the pointwise scale-invariant axisymmetric Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [15, Theorem 5.3].

Required hypotheses. The source theorem applies to bounded ancient mild axisymmetric whole-space solutions satisfying the full velocity condition

$$r|u(x,t)| \leq C.$$

Equivalently, away from the axis this is $|u(x,t)| \leq C/r$. This branch therefore uses the full velocity hypothesis. Drift-only assumptions or circulation-only assumptions are not included in this class.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 29.6. The pointwise branch of $\mathcal{C}_{\mathrm{sd}}$ requires the full velocity bound $r|u| \leq C$, not merely a bound on the drift $u^r e_r + u^z e_z$ or on the circulation $\Gamma = r u_\theta$, and this structural hypothesis is preserved under the time shift. Therefore the cited theorem applies to W_T.

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives a spatially constant solution in the frame used here. The resulting constants patch in T by Lemma 30.3. In the Type I ancient frame, the patched constant is eliminated by Lemma 30.4; in the pointwise $1/r$ branch it is also excluded directly by letting $r \to \infty$ in the pointwise bound, as recorded in Remark 31.3. □

Theorem 30.8 (Lei–Zhang–Zhao)**.** *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = ru_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty,0)\big), \qquad 1 \le p < \infty,$$

then the no-swirl case $u_\theta \equiv 0$ is handled by Theorem 30.6, while in the nontrivial-swirl case the cited theorem yields that u is a constant vector field.

Proof. Imported theorem. We use the ancient axisymmetric circulation Liouville theorem of Lei–Zhang–Zhao [27, Theorem 1.3].

Required hypotheses. The theorem assumes a bounded ancient mild axisymmetric solution in $\mathbb{R}^3 \times (-\infty,0)$, nontrivial swirl, and finite critical circulation integrability

$$\Gamma = ru_\theta \in L^\infty_t L^p_x(\mathbb{R}^3 \times (-\infty,0)), \qquad 1 \le p < \infty.$$

The no-swirl case $u_\theta \equiv 0$ is handled separately by Theorem 30.6; this theorem is invoked for the finite L^p-circulation branch.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a smooth bounded mild ancient axisymmetric solution by Lemma 29.6. The finite-L^p circulation hypothesis is preserved under the time shift.

If $(W_T)_\theta \equiv 0$, the no-swirl branch Theorem 30.6 applies. If $(W_T)_\theta \not\equiv 0$, the Lei–Zhang–Zhao theorem applies and gives that W_T is a constant vector field.

Conclusion in the Type I frame. In either case W_T is spatially constant. The constants patch in T by Lemma 30.3, and Lemma 30.4 eliminates the patched constant in the Type I ancient frame. □

Theorem 30.9 (Lei–Ren–Zhang)**.** *Let u be a bounded ancient mild axisymmetric solution, periodic in z, with*

$$\Gamma = ru_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty,0)\big).$$

Then u is a constant axial vector field.

Proof. Imported theorem. We use the periodic axisymmetric ancient Liouville theorem of Lei–Ren–Zhang [26, Theorem 1.1].

Required hypotheses. The theorem is stated on the quotient $\mathbb{R}^2 \times \mathbb{T}^1$ for bounded mild ancient axisymmetric solutions with bounded swirl circulation $\Gamma = ru_\theta$. The periodic branch in this paper is deliberately no larger than that source theorem: the physical representative is assumed z-periodic and satisfies the same bounded Γ condition.

Verification for our profile. Let U be the Type I ancient physical representative. For each $T < 0$, the shifted solution

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a smooth bounded mild ancient solution by Lemma 29.6. In the periodic branch the periodicity is imposed on the physical axial variable z, not on the centered field, and the branch also requires bounded circulation $\Gamma = ru_\theta$. These structural hypotheses are preserved under the time shift. Therefore the cited theorem applies to W_T after descending to $\mathbb{R}^2 \times \mathbb{T}^1$ and normalizing the period by a Navier–Stokes scaling, which does not change boundedness, mildness, axisymmetry, or boundedness of Γ.

Conclusion in the Type I frame. For each $T < 0$, the cited theorem gives a spatially constant solution in the frame used here. The resulting constants patch in T by Lemma 30.3, and Lemma 30.4 then eliminates the patched constant in the Type I ancient frame. □

31. Unforced axisymmetric Liouville consequences

31.1. No swirl.

Proposition 31.1 (No-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x,s) = U(x, T+s)$. By Lemma 29.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 30.6 gives that W_T is a spatially constant axial drift, say $W_T(x,s) = b_T(s)e_z$. By Lemma 30.2, $b_T(s)$ is constant in s. Thus $U(x,t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Lemma 30.3 gives $U(x,t) = c$ for all $t < 0$. Finally, Lemma 30.4 gives $c = 0$. □

31.2. Pointwise scale-invariant control.

Proposition 31.2 (Pointwise $1/r$ Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*

$$|U(x,t)| \le \frac{C}{r} \qquad (r > 0,\ t < 0).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 29.6, the shifted solution

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a bounded ancient mild axisymmetric solution and satisfies the same pointwise bound

$$|W_T(x,s)| \le \frac{C}{r}.$$

By Theorem 30.7, W_T is spatially constant in the frame used here. Write $W_T \equiv c_T$. The pointwise bound gives

$$|c_T| \le \frac{C}{r} \qquad \text{for every } r > 0,$$

hence $c_T = 0$ by letting $r \to \infty$. Equivalently, one may patch the constants in T and apply Lemma 30.4. Thus $W_T \equiv 0$, so $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ was arbitrary, $U \equiv 0$. □

Remark 31.3 (Pointwise bound rules out a constant directly). In the pointwise scale-invariant case, even before invoking the Type I constant-removal lemma, the pointwise bound $|c| \le C/r$ for all $r > 0$ forces $c = 0$ by letting $r \to \infty$ when the Liouville theorem returns a constant. The Type I argument used in the proof above is consistent with this stronger conclusion.

31.3. Finite L^p control of Γ.

Proposition 31.4 (Finite-L^p swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \le p < \infty$,*

$$\Gamma = rU_\theta \in L^\infty_t L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 29.6,

$$W_T(x,s) = U(x, T+s), \qquad s < 0,$$

is a bounded ancient mild axisymmetric solution. The finite-L^p circulation hypothesis is preserved by time shift:

$$\Gamma_{W_T} \in L^\infty_s L^p_x\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

If $(W_T)_\theta \equiv 0$, then the no-swirl Liouville branch gives that W_T is spatially constant. If $(W_T)_\theta \not\equiv 0$, then the Lei–Zhang–Zhao nontrivial-swirl theorem applies and again gives that W_T is spatially constant. In either case, $W_T \equiv c_T$. Patch the constants in T using Lemma 30.3, and then apply Lemma 30.4. ☐

31.4. **Periodic bounded swirl.** The periodic hypothesis below is always imposed on the physical ancient representative U, not on the centered profile V. Under the self-similar transform a fixed physical z-period becomes a τ-dependent period in Z, so it is not a fixed-period condition on V (Lemma 29.5).

Proposition 31.5 (Periodic bounded-swirl Type I vanishing)**.** *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z, and satisfies*

$$\Gamma = rU_\theta \in L^\infty\big(\mathbb{R}^3 \times (-\infty, 0)\big).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Lemma 29.6, the shifted solution W_T is bounded, ancient, mild, axisymmetric, physically z-periodic, and has bounded Γ. Theorem 30.9 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Lemma 30.4. ☐

31.5. **Proof of the axisymmetric Liouville theorem.**

Proof of Theorem 28.1. The four cases are precisely the hypotheses of Propositions 31.1, 31.2, 31.4 and 31.5. Each proposition gives $U \equiv 0$. This contradicts Lemma 29.3 when U is a nonzero Type I blow-up limit. ☐

32. Anchored Gallay–Wayne perturbative exclusion

32.1. **The forward rescaled vorticity equation.** The Gallay–Wayne variable used here is the forward self-similar vorticity time θ; its generator Λ below has the opposite drift convention from the centered velocity generator used in Parts II–III. The argument is formulated directly in physical variables: from arbitrarily early physical times $a < 0$ we launch genuine forward Gallay–Wayne Cauchy problems and compare them with the corresponding anchored forward rescalings of the same ancient solution. No claim is made that the backward Type I centered vorticity is itself a single Gallay–Wayne forward orbit.

Let $w = \nabla \times v$. The Gallay–Wayne forward rescaled vorticity equation is

$$\partial_\theta w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \qquad \nabla \cdot w = 0, \tag{32.1}$$

where

$$\Lambda w = \Delta w + \frac{1}{2}\xi \cdot \nabla w + w, \tag{32.2}$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \geq 0$, set

$$L^2_\sigma(m) = \left\{ f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty \right\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 \, d\xi.$$

32.2. **Gallay–Wayne small-data estimate.**

Theorem 32.1 (Gallay–Wayne small-data estimate)**.** *Let $0 < \mu \le 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $r_m > 0$ and $K_m \ge 1$ such that if*

$$w_0 \in L^2_\sigma(m), \qquad \|w_0\|_m \le r_m,$$

then (32.1) has a unique global mild solution $w(\theta) = \Phi_\theta w_0$ in $C([0,\infty); L^2_\sigma(m))$ and

$$\|w(\theta)\|_m \le K_m e^{-\mu\theta} \|w_0\|_m, \qquad \theta \ge 0.$$

In particular, if $m > \frac{7}{2}$, the choice $\mu = 1$ is admissible.

Proof. Imported theorem. We use Gallay–Wayne [25, Theorem 2.3] for the rescaled three-dimensional vorticity equation.

Required hypotheses. Their theorem is stated in the weighted divergence-free space $L^2_\sigma(m)$. It assumes $0 < \mu \le 1$, $m > 2\mu + 1/2$, and small initial data $\|w_0\|_m \le r_0(m, \mu)$. In this formulation no additional moment, orthogonality, spectral-projection, or codimension condition is used beyond membership in the stated small ball of $L^2_\sigma(m)$.

Variable convention. The equation (32.1) is the same forward self-similar vorticity equation after the notational change $\xi \leftrightarrow y$; the sign of the confining drift is the forward Gallay–Wayne convention used in this weighted-vorticity subsection. □

32.3. **Anchored Gallay–Wayne condition.**

Definition 32.2 (Anchored Gallay–Wayne perturbative vorticity condition)**.** Let U be a smooth mild Type I ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$, and set

$$\Omega = \nabla \times U.$$

Fix $m > \frac{7}{2}$. Let $r_m > 0$ and $K_m \ge 1$ be the constants in Theorem 32.1 with $\mu = 1$. Fix $0 < \rho_m < r_m$.

We say that U satisfies the anchored Gallay–Wayne perturbative vorticity condition if there exist anchor times

$$a_n \to -\infty$$

and scales

$$\lambda_n > 0$$

such that

$$\lambda_n^2 - a_n \to +\infty,$$

and, for every n, the anchored initial vorticity

$$w_n^0(\xi) := \lambda_n^2 \Omega(\lambda_n \xi, a_n)$$

satisfies

$$w_n^0 \in L^2_\sigma(m), \qquad \|w_n^0\|_m \le \rho_m.$$

Moreover, the corresponding anchored initial velocity is the Biot–Savart representative:

$$\lambda_n U(\lambda_n \xi, a_n) = \mathrm{BS}[w_n^0](\xi),$$

where

$$\mathrm{BS}[w](\xi) = -\frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{(\xi - \eta) \times w(\eta)}{|\xi - \eta|^3} \, d\eta.$$

32.4. **Anchored forward Gallay–Wayne rescalings.**

Lemma 32.3 (Local weighted-vorticity persistence for anchored smooth data)**.** *Let $m > \frac{7}{2}$, let $a \in \mathbb{R}$, and let $\lambda > 0$. Let*

$$v^0 \in C_b^\infty(\mathbb{R}^3) \cap C_0(\mathbb{R}^3), \qquad \nabla_\xi \cdot v^0 = 0,$$

and suppose

$$w^0 = \nabla_\xi \times v^0 \in L^2_\sigma(m).$$

Assume moreover that v^0 is the decaying Biot–Savart representative of w^0:

$$v^0 = \mathrm{BS}[w^0].$$

Define the physical initial velocity at time a by

$$u_0(x) = \lambda^{-1} v^0(x/\lambda).$$

Let u be the bounded mild Navier–Stokes solution with initial data u_0 on a sufficiently short interval $[a, a+\delta]$. Let

$$\omega = \nabla_x \times u, \qquad t(\theta) = a + \lambda^2(e^\theta - 1),$$

and define the anchored forward self-similar vorticity and velocity by

$$w(\xi,\theta) := \lambda^2 e^\theta \omega(\lambda e^{\theta/2}\xi, t(\theta)).$$

$$v(\xi,\theta) := \lambda e^{\theta/2} u(\lambda e^{\theta/2}\xi, t(\theta)).$$

Then, for some $\theta_0 > 0$,

$$w \in C([0,\theta_0]; L^2_\sigma(m))$$
$$w(\cdot,0) = w^0,$$

and w solves the Gallay–Wayne rescaled vorticity equation

$$\partial_\theta w = \Lambda w - (v\cdot\nabla_\xi)w + (w\cdot\nabla_\xi)v, \qquad v = \mathrm{BS}[w],$$

with initial data w^0.

Proof. At time a,

$$u_0(x) = \lambda^{-1} v^0(x/\lambda),$$

so

$$\omega_0(x) = \nabla_x \times u_0(x) = \lambda^{-2}(\nabla_\xi \times v^0)(x/\lambda) = \lambda^{-2} w^0(x/\lambda).$$

Therefore

$$w(\xi,0) = \lambda^2 \omega_0(\lambda\xi) = w^0(\xi),$$

and

$$v(\xi,0) = \lambda u_0(\lambda\xi) = v^0(\xi).$$

The Biot–Savart normalization at $\theta = 0$ removes the possible bounded harmonic velocity mode. Since $v^0 = \mathrm{BS}[w^0] \in C_0$, the scaled physical initial velocity

$$u_0(x) = \lambda^{-1} v^0(x/\lambda)$$

belongs to $C_0(\mathbb{R}^3)$. The standard local mild theory in $C_0 \cap C_b^\infty$ gives a short-time solution $u(t) \in C_0$, and therefore the rescaled velocity $v(\theta)$ also belongs to C_0 for $0 \le \theta \le \theta_0$.

For each such θ, the difference

$$h(\theta) := v(\theta) - \mathrm{BS}[w(\theta)]$$

is divergence-free and curl-free. Hence $h(\theta)$ is spatially constant. But both $v(\theta)$ and the Biot–Savart representative $\mathrm{BS}[w(\theta)]$ are in the decaying gauge, so $h(\theta) \in C_0(\mathbb{R}^3)$. A spatially constant C_0 function is zero. Thus

$$v = \mathrm{BS}[w]$$

on $[0, \theta_0]$.

For $C_0 \cap C_b^\infty$ divergence-free data, the bounded mild solution is smooth on a short time interval and has bounded spatial derivatives there. Its vorticity solves the physical vorticity equation classically. The anchored forward self-similar change of variables above transforms that physical vorticity equation into

$$\partial_\theta w = \Lambda w - (v \cdot \nabla_\xi) w + (w \cdot \nabla_\xi) v.$$

A standard weighted energy estimate with weight $(1 + |\xi|)^{2m}$, using the boundedness of v and $\nabla_\xi v$ on a sufficiently short interval, gives

$$\sup_{0 \le \theta \le \theta_0} \|w(\theta)\|_m < \infty$$

for some $\theta_0 > 0$, and continuity in $L^2_\sigma(m)$. Hence the transformed vorticity belongs to the Gallay–Wayne uniqueness class on $[0, \theta_0]$. $\square$

Lemma 32.4 (Bounded physical pullback of anchored Gallay–Wayne data)**.** *Let $m > \frac{7}{2}$, let $w^0 \in L^2_\sigma(m)$ satisfy the Gallay–Wayne small-data hypothesis. Assume, in addition, that*

$$v^0 = \mathrm{BS}[w^0] \in C_b^\infty(\mathbb{R}^3) \cap C_0(\mathbb{R}^3), \qquad \nabla_\xi \cdot v^0 = 0, \qquad \nabla_\xi \times v^0 = w^0$$

in distributions. Let

$$w^{GW}(\theta) = \Phi_\theta w^0, \qquad v^{GW}(\theta) = \mathrm{BS}[w^{GW}(\theta)]$$

be the Gallay–Wayne mild solution. Fix $a < T < 0$ and $\lambda > 0$, and define

$$A(t) = \lambda^2 + t - a, \qquad \theta(t) = \log \frac{A(t)}{\lambda^2},$$

$$\widetilde{U}(x,t) = A(t)^{-1/2} v^{GW}\left(\frac{x}{\sqrt{A(t)}}, \theta(t) \right), \qquad a \le t \le T.$$

Then $\widetilde{U}$ is a bounded mild Navier–Stokes solution on every finite interval $[a, b] \subset [a, T]$. In particular,

$$\sup_{a \le t \le b} \|\widetilde{U}(t)\|_{L^\infty} < \infty.$$

Proof. The inverse Gallay–Wayne change of variables sends the rescaled vorticity equation to the physical vorticity equation and sends $v^{GW} = \mathrm{BS}[w^{GW}]$ to a physical velocity field $\widetilde{U}$. It remains to justify that this velocity is bounded and mild on each compact physical interval.

At $\theta = 0$, the assumptions give

$$v^0 = \mathrm{BS}[w^0] \in C_b^\infty \cap C_0, \qquad \nabla_\xi \times v^0 = w^0 \in L^2_\sigma(m).$$

Define the physical initial velocity

$$u_0(x) = \lambda^{-1} v^0(x/\lambda).$$

The standard $C_0 \cap C_b^\infty$-data mild theory gives a bounded mild physical solution with initial data u_0 on a short interval starting at $t = a$. By Lemma 32.3, its anchored transformed vorticity lies in

$$C([0, \theta_0]; L^2_\sigma(m))$$

for some $\theta_0 > 0$ and solves the Gallay–Wayne equation with initial data w^0. By uniqueness in the Gallay–Wayne class, it agrees with w^{GW} on this initial interval. Therefore

$$\sup_{0 \le \theta \le \theta_0} \|v^{GW}(\theta)\|_{L^\infty} < \infty.$$

Next fix any finite rescaled time interval $[0, \Theta]$. Since $m > \frac{3}{2}$, Cauchy–Schwarz against the weight gives

$$\|w^{GW}(\theta)\|_{L^1} \le C_m \|w^{GW}(\theta)\|_m, \qquad 0 \le \theta \le \Theta,$$

so Theorem 32.1 yields

$$\sup_{0\le\theta\le\Theta} \|w^{GW}(\theta)\|_{L^1} < \infty.$$

For every $0 < \theta_0 < \Theta$, standard parabolic smoothing for the mild solution of (32.1) gives

$$\sup_{\theta_0\le\theta\le\Theta} \|w^{GW}(\theta)\|_{L^p} < \infty$$

for any chosen $p > 3$. Splitting the Biot–Savart kernel into the regions $|\xi - \eta| \le 1$ and $|\xi - \eta| > 1$ gives the standard estimate

$$\| \mathrm{BS}[w]\|_{L^\infty} \le C_p\big(\|w\|_{L^1} + \|w\|_{L^p}\big), \qquad p > 3,$$

because the far part is controlled by $\|w\|_{L^1}$ and the near part is controlled by Hölder since $|x|^{-2}\mathbf{1}_{\{|x|\le 1\}} \in L^{p/(p-1)}(\mathbb{R}^3)$ for $p > 3$. Hence

$$\sup_{\theta_0\le\theta\le\Theta} \|v^{GW}(\theta)\|_{L^\infty} < \infty.$$

Together with the initial short-time bounded mild solution, this gives

$$\sup_{0\le\theta\le\Theta} \|v^{GW}(\theta)\|_{L^\infty} < \infty.$$

Now let $[a, b] \subset [a, T]$, and set

$$\Theta = \log \frac{\lambda^2 + b - a}{\lambda^2}.$$

Since $A(t) \ge \lambda^2 > 0$ on $[a, b]$, the inverse transform gives

$$\|\widetilde{U}(t)\|_{L^\infty} = A(t)^{-1/2}\|v^{GW}(\theta(t))\|_{L^\infty} \le \lambda^{-1} \sup_{0\le\theta\le\Theta} \|v^{GW}(\theta)\|_{L^\infty}$$

for $a \le t \le b$. Therefore

$$\sup_{a\le t\le b} \|\widetilde{U}(t)\|_{L^\infty} < \infty.$$

Finally, the Duhamel formula for the Gallay–Wayne mild solution transforms under the inverse forward self-similar map into the standard physical Navier–Stokes mild formula on $[a, b]$. Therefore $\widetilde{U}$ is a bounded mild Navier–Stokes solution on every compact subinterval of $[a, T]$. □

Lemma 32.5 (Physical uniqueness comparison for anchored Gallay–Wayne flows)**.** *Let $m > \frac{7}{2}$. Let U be a smooth mild Navier–Stokes solution on $\mathbb{R}^3 \times [a, T]$, with $a < T < 0$. Let $w^0 \in L^2_\sigma(m)$ satisfy the Gallay–Wayne small-data hypothesis, and let*

$$w^{GW}(\theta) = \Phi_\theta w^0, \qquad v^{GW}(\theta) = \mathrm{BS}[w^{GW}(\theta)]$$

be the Gallay–Wayne mild solution.

Fix $\lambda > 0$, set

$$A(t) = \lambda^2 + t - a, \qquad \theta(t) = \log \frac{A(t)}{\lambda^2},$$

and define the physical pullback of the Gallay–Wayne solution by

$$\widetilde{U}(x, t) = A(t)^{-1/2} v^{GW}\left(\frac{x}{\sqrt{A(t)}}, \theta(t)\right), \qquad a \le t \le T.$$

Assume the anchored initial velocity and vorticity agree with U:

$$\lambda U(\lambda\xi, a) = \mathrm{BS}[w^0](\xi),$$

and

$$w^0(\xi) = \lambda^2 (\nabla_x \times U)(\lambda\xi, a).$$

Then

$$\widetilde{U}(x,t) = U(x,t) \qquad \text{for } a \le t \le T.$$

Proof. The initial agreement gives

$$v^0(\xi) := \mathrm{BS}[w^0](\xi) = \lambda U(\lambda\xi, a).$$

Because $\mathrm{BS}[w^0]$ is the decaying Biot–Savart representative, $v^0 \in C_0(\mathbb{R}^3)$. Since U is smooth and bounded on the finite terminal interval under consideration, the anchored representative also lies in C_b^∞. Hence

$$v^0 \in C_b^\infty(\mathbb{R}^3) \cap C_0(\mathbb{R}^3).$$

Moreover,

$$\nabla_\xi \cdot v^0(\xi) = \lambda^2 (\nabla_x \cdot U)(\lambda\xi, a) = 0,$$

since U is divergence-free. The anchored vorticity agreement gives

$$\nabla_\xi \times v^0 = w^0.$$

Hence the hypotheses of Lemma 32.4 are satisfied. Therefore the inverse Gallay–Wayne pullback $\widetilde{U}$ is a bounded mild Navier–Stokes solution on every compact subinterval of $[a,T]$. At $t = a$, $A(a) = \lambda^2$, so

$$\widetilde{U}(x,a) = \lambda^{-1} v^{GW}\Big(\frac{x}{\lambda}, 0\Big) = \lambda^{-1}\,\mathrm{BS}[w^0]\Big(\frac{x}{\lambda}\Big).$$

By the anchored Biot–Savart normalization, this equals $U(x,a)$.

Let $Z = U - \widetilde{U}$. On any compact subinterval $[a,b] \subset [a,T]$, the two mild solutions satisfy

$$Z(t) = -\int_a^t e^{(t-s)\Delta}\mathbb{P}\nabla \cdot \big(Z \otimes U + \widetilde{U} \otimes Z\big)(s)\,ds.$$

Using the Oseen estimate,

$$\|e^{(t-s)\Delta}\mathbb{P}\nabla \cdot F\|_\infty \le C(t-s)^{-1/2}\|F\|_\infty,$$

we get

$$\sup_{a\le t\le b}\|Z(t)\|_\infty \le C(M_U + M_{\widetilde{U}})(b-a)^{1/2}\sup_{a\le t\le b}\|Z(t)\|_\infty,$$

where

$$M_U = \sup_{a\le t\le b}\|U(t)\|_\infty, \qquad M_{\widetilde{U}} = \sup_{a\le t\le b}\|\widetilde{U}(t)\|_\infty.$$

Choosing $b - a$ small enough forces $Z \equiv 0$ on $[a,b]$. Iterating over finitely many subintervals gives $Z \equiv 0$ on $[a,T]$. □

Lemma 32.6 (Anchored transforms satisfy the Gallay–Wayne equation)**.** *Let U be a smooth mild Type I ancient solution satisfying Definition 32.2. For each anchor-scale pair (a_n, λ_n), define*

$$t_n(\theta) = a_n + \lambda_n^2(e^\theta - 1), \qquad x = \lambda_n e^{\theta/2}\xi,$$

and

$$w_n(\xi,\theta) = \lambda_n^2 e^\theta \Omega(\lambda_n e^{\theta/2}\xi, t_n(\theta)),$$
$$v_n(\xi,\theta) = \lambda_n e^{\theta/2} U(\lambda_n e^{\theta/2}\xi, t_n(\theta)).$$

Then w_n is the restriction, on the interval

$$0 \le \theta < \log\frac{\lambda_n^2 - a_n}{\lambda_n^2},$$

of the Gallay–Wayne mild solution with initial data w_n^0. In particular, for every $\theta \geq 0$ with $t_n(\theta) < 0$,

$$\|w_n(\theta)\|_m \leq K_m e^{-\theta}\|w_n^0\|_m \leq K_m \rho_m.$$

Proof. The change of variables is exactly the forward self-similar change of variables used by Gallay–Wayne, translated to the initial time a_n and length scale λ_n. Therefore the vorticity equation for $\Omega = \nabla \times U$ becomes

$$\partial_\theta w_n = \Lambda w_n - (v_n \cdot \nabla) w_n + (w_n \cdot \nabla) v_n.$$

At $\theta = 0$,

$$w_n(\xi, 0) = \lambda_n^2 \Omega(\lambda_n \xi, a_n) = w_n^0(\xi),$$

and the Biot–Savart normalization in Definition 32.2 gives

$$v_n(\xi, 0) = \lambda_n U(\lambda_n \xi, a_n) = \mathrm{BS}[w_n^0](\xi).$$

Let $w_n^{GW}(\theta) = \Phi_\theta w_n^0$ be the Gallay–Wayne mild solution with initial data w_n^0, and let

$$v_n^{GW}(\theta) = \mathrm{BS}[w_n^{GW}(\theta)].$$

For

$$A_n(t) = \lambda_n^2 + t - a_n, \qquad \theta_n(t) = \log \frac{A_n(t)}{\lambda_n^2},$$

define the corresponding physical velocity

$$\widetilde{U}_n(x,t) = A_n(t)^{-1/2} v_n^{GW}\left(\frac{x}{\sqrt{A_n(t)}}, \theta_n(t)\right), \qquad a_n \leq t < 0.$$

For $\theta = 0$, the identity

$$w_n(\xi, 0) = w_n^{GW}(\xi, 0) = w_n^0(\xi)$$

holds by definition. Now fix

$$0 < \theta < \log \frac{\lambda_n^2 - a_n}{\lambda_n^2},$$

and set $T = t_n(\theta)$. Then $a_n < T < 0$, so by Lemma 32.5,

$$\widetilde{U}_n(x,t) = U(x,t)$$

for $a_n \leq t \leq T$. Evaluating the equality at time T and taking curls gives

$$w_n(\xi, \theta) = w_n^{GW}(\xi, \theta).$$

Since θ was arbitrary in the stated interval, w_n is exactly the restriction of the Gallay–Wayne mild solution launched from w_n^0. Therefore Theorem 32.1 with $\mu = 1$ yields

$$\|w_n(\theta)\|_m = \|w_n^{GW}(\theta)\|_m \leq K_m e^{-\theta}\|w_n^0\|_m \leq K_m \rho_m.$$

□

32.5. **Anchored perturbative exclusion.**

Proposition 32.7 (Anchored Gallay–Wayne perturbative branch vanishes)**.** *Let U be a smooth mild Type I ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$. If U satisfies the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2, then*

$$U \equiv 0.$$

Proof. Let

$$\Omega = \nabla \times U.$$

Fix $T < 0$. For all sufficiently large n, $a_n < T$. Define

$$A_n(T) = \lambda_n^2 + T - a_n, \qquad \theta_n(T) = \log \frac{A_n(T)}{\lambda_n^2}.$$

Then

$$A_n(T) \to +\infty$$

by the anchoring condition $\lambda_n^2 - a_n \to +\infty$.

At physical time T, the anchored transform gives

$$w_n(\xi, \theta_n(T)) = A_n(T)\, \Omega\big(\sqrt{A_n(T)}\, \xi, T\big).$$

By Lemma 32.6,

$$\|w_n(\theta_n(T))\|_m \le K_m \rho_m.$$

Let $\varphi \in C_c^\infty(\mathbb{R}^3)^3$. Changing variables $x = \sqrt{A_n(T)}\, \xi$, we obtain

$$\begin{aligned}\left| \int_{\mathbb{R}^3} \Omega(x, T) \cdot \varphi(x)\, dx \right| &= A_n(T)^{1/2} \left| \int_{\mathbb{R}^3} w_n(\xi, \theta_n(T)) \cdot \varphi(\sqrt{A_n(T)}\, \xi)\, d\xi \right| \\ &\le A_n(T)^{1/2} \|w_n(\theta_n(T))\|_m \left\| (1 + |\xi|)^{-m} \varphi(\sqrt{A_n(T)}\, \xi) \right\|_{L^2_\xi}.\end{aligned}$$

Since $(1 + |\xi|)^{-m} \le 1$,

$$\left\| (1 + |\xi|)^{-m} \varphi(\sqrt{A_n(T)}\, \xi) \right\|_{L^2_\xi} \le A_n(T)^{-3/4} \|\varphi\|_{L^2}.$$

Therefore

$$\left| \int_{\mathbb{R}^3} \Omega(x, T) \cdot \varphi(x)\, dx \right| \le K_m \rho_m A_n(T)^{-1/4} \|\varphi\|_{L^2}.$$

Letting $n \to \infty$ gives

$$\int_{\mathbb{R}^3} \Omega(x, T) \cdot \varphi(x)\, dx = 0.$$

Thus

$$\Omega(\cdot, T) = 0$$

in distributions. Since $T < 0$ was arbitrary,

$$\Omega \equiv 0 \qquad \text{on } \mathbb{R}^3 \times (-\infty, 0).$$

For each fixed $T < 0$, $U(\cdot, T)$ is smooth, bounded, divergence-free, and curl-free. Hence each component of $U(\cdot, T)$ is a bounded harmonic function on $\mathbb{R}^3$, so $U(\cdot, T)$ is spatially constant. Since this holds for every $T < 0$, write

$$U(x, t) = c(t).$$

By Lemma 30.2, $c(t)$ is constant in time, and then Lemma 30.4 gives $c = 0$. Therefore $U \equiv 0$. ☐

Remark 32.8. The anchored argument above is deliberately formulated in physical variables. It avoids treating the backward Type I centered vorticity as a single Gallay–Wayne forward orbit. Gallay–Wayne's estimates are used only on ordinary forward Cauchy problems launched from the anchor times a_n.

Proof of Theorem 28.2. This is Proposition 32.7. ☐

33. Summary of PDE consequences

33.1. Zero conclusions.

Theorem 33.1 (Zero conclusions for structured Type I ancient solutions). *Let U be a smooth mild Type I ancient solution in the sense of Definition 29.1. If U satisfies one of the following hypotheses:*

(i) *axisymmetry and $U_\theta \equiv 0$;*
(ii) *axisymmetry and the pointwise bound $|U(x,t)| \leq C/r$;*
(iii) *axisymmetry and finite-L^p control of the swirl scalar,*

$$\Gamma \in L_t^\infty L_x^p, \qquad 1 \leq p < \infty;$$

(iv) *axisymmetry, z-periodicity, and bounded Γ;*
(v) *U satisfies the anchored Gallay–Wayne perturbative vorticity condition of Definition 32.2.*

then $U \equiv 0$.

Proof. Items (i)–(iv) are Theorem 28.1. Item (v) is Proposition 32.7. ☐

Corollary 33.2 (Structured hypotheses cannot hold for nonzero Type I blow-up limits). *A nonzero Type I blow-up limit cannot satisfy any of the hypotheses in Theorem 33.1.*

Proof. If a nonzero Type I blow-up limit satisfied one of the hypotheses, then Theorem 33.1 would give $U \equiv 0$, contradicting Lemma 29.3. ☐

33.2. Consequences for Type I blow-up arguments.

Proof of Theorem 28.3. The assertion is Corollary 33.2. ☐

Theorem 33.3 (Combined zero conclusion). *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in Theorem 33.1, then $U \equiv 0$, a contradiction.*

Proof. The assertion is precisely the conclusion of Theorem 33.1. ☐

33.3. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS plus $r \to \infty$ constant removal
$\Gamma \in L_t^\infty L_x^p$	$U \equiv 0$	KNSS/LZZ split plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
anchored Gallay–Wayne perturbative vorticity	$U \equiv 0$	anchored Gallay–Wayne decay plus physical uniqueness comparison

The no-swirl, pointwise scale-invariant, finite-L^p-swirl, and periodic bounded-swirl cases first reduce, by the cited axisymmetric Liouville inputs and the theorem-level KNSS/LZZ split in the finite-L^p branch, to spatial constants or axial drifts in the physical Type I frame; the manuscript then removes those constants internally. The perturbative weighted-vorticity case follows from the anchored Gallay–Wayne forward-rescaling argument together with the physical uniqueness comparison. Thus every zero conclusion used in the Type I application rests on one of the explicit PDE hypotheses in the table.

Part V. Final Assembly: Local Type I Exclusion

Overview. This part collects the preceding results and proves the local pointwise Type I exclusion. The argument is linear: local concentration, local endpoint weak-Serrin no-escape, admissible local Type I concentration sequence, raw Seregin extraction, passage to the raw generated state space with invariant nonvanishing, exhaustion into Liouville classes plus residual class, exclusion of the four non-residual classes, and residual closure.

34. Final assembly

Theorem 34.1 (Local Type I regularity)**.** *Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$, and let $z_* = (x_*, T)$ be a terminal point satisfying the local pointwise Type I bound, for some $\rho > 0$,*

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))} < \infty.$$

Then z_* is regular.

Proof. Assume for contradiction that $z_* = (x_*, T)$ is singular.

Step 1: concentration. By Corollary 6.8, there are $\eta_v > 0$ and $r_n \downarrow 0$ such that

$$C(z_*, r_n) \geq \eta_v.$$

Step 2: terminal no-escape. The local pointwise Type I bound gives

$$u \in L_t^{2,\infty} L_x^\infty(Q_\rho(z_*)).$$

By Lemma 9.3, for each $0 < \theta < 1$ the terminal local A-quantity is uniformly bounded along the chosen small-scale sequence. Hence Theorem 9.4 gives that the rescaled sequence satisfies

$$\lim_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 = 0.$$

Step 3: nontrivial ancient extraction. Corollary 9.5 promotes the sequence to an admissible local Type I concentration sequence. The concentration and no-escape properties give, by Lemma 9.6, a fixed compact negative-time cylinder carrying positive L^3 mass. Then Lemma 9.7 yields a raw extracted normalized Seregin ancient limit V satisfying

$$\iint_K |V|^3 > 0$$

for some compact $K \Subset \mathbb{R}^3 \times \mathbb{R}$. By Lemma 14.12, V also satisfies the time-translation invariant local L^3 nonvanishing condition (14.1) for some data (R_0, η_*).

Step 4: class decomposition. Let

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

be the raw generated descendant hull formed with the fixed invariant normalization data obtained from the extracted profile V. By construction, the extracted profile V belongs to $\mathcal{S}$. By Proposition 15.11,

$$V \in \mathcal{C}_{\text{small}} \cup \mathcal{C}_{\text{stat},L^3} \cup \mathcal{C}_{L^3-\text{tight}} \cup \mathcal{C}_{\text{sd}} \cup \mathcal{R}(\mathcal{S}).$$

Step 5: contradiction. The first four alternatives are excluded by Proposition 16.10; each class definition includes exactly the hypotheses needed by its corresponding Liouville theorem. The residual alternative is excluded by Theorem 17.1, applied to the same raw generated descendant hull, because it states literally that $\mathcal{R}(\mathcal{S}) = \varnothing$. No alternative is possible, contradicting the invariant nonvanishing normalization of Corollary 14.19. This contradiction proves that z_* is regular. $\square$

Remark 34.2 (Unconditional conclusion). Theorem 34.1 gives the local pointwise Type I exclusion within the stated raw generated local pointwise Type I extraction class. The residual input used in its proof is the companion theorem Theorem 17.1, not an added hypothesis.

A note on human–AI collaboration

This paper grew out of a collaboration between the author and several AI systems, principally Claude Fable 5, GPT-5.4, GPT-5.5, and GPT-5.6-Sol.

This project began as a personal effort by the author to learn partial differential equations, to understand what singularities are and which structural mechanisms permit them to form, and to study how large language models fail when reasoning about mathematics. Because the subject was initially unfamiliar to him, part of the aim was to develop a working protocol for detecting hallucinations and avoiding reliance on arguments that he could not independently follow. The author asked GPT-5.4 to identify difficult problems together with recent progress on them and selected the Navier–Stokes regularity problem because it is a paradigmatic example of failure of compactness.

The work proceeded through an iterative dialogue. The author proposed possible approaches in natural language and asked the models to explain why they were insufficient, how they related to methods in the literature, and what singular or noncompact scenario would obstruct them. These obstruction scenarios became the input to subsequent rounds of questions. The author repeatedly requested elementary explanations or formulations in terms familiar to an AI and machine-learning engineer, with emphasis on the purpose of each analytic technique and on the structural features of a possible singularity that the current argument had not yet controlled.

The dialogue was then converted into mathematical manuscripts through repeated rounds of expansion and revision with GPT-5.5, Claude Opus 4.8, Claude Fable 5, and GPT-5.6-Sol. The author asked the models to replace informal descriptions with PDE statements, expand every step that he did not understand, and produce proof-flow diagrams, hypothesis ledgers, and audit tables that made the dependencies of the argument visible. He subsequently developed a red-teaming protocol that searched systematically for missing hypotheses, unsupported compactness passages, unclosed residual profiles, and other errors. When a possible gap was identified, the repair protocol introduced a separate branch for the missing case and attempted to close it through further literature-based reductions and structural refinements. The point at which these reviews stopped producing new objections was treated as a reason to move to stricter node-by-node auditing, rather than as a certificate of correctness.

The resulting proof architecture was encoded in an interactive proof explorer and a node-by-node audit framework. These tools treat the proof as a data-processing pipeline whose nodes record the hypotheses, conclusions, and retained structural facts available at each point in the argument. Their purpose is to make compactness passages, case splits, and transfers between local and global statements explicit, and to allow each step to be inspected independently.

The underlying framework was originally designed for combinatorial problems and was subsequently redesigned as reusable infrastructure for PDE arguments. Formalization is the next stage of the project. The analysis is consistently expressed through local PDE statements and explicit transfers between local states; this locality simplifies the formalization by giving individual nodes modular interfaces and permits the resulting code to be reused across the three Navier–Stokes manuscripts.

The author's role throughout was to select the problem and research direction, propose and revise ideas, request explanations and obstruction scenarios, design the review and repair protocols, construct the proof-tracking framework, and decide how identified gaps should be routed through the argument. The AI systems supplied literature-oriented explanations, counterexamples and obstruction profiles, mathematical formulations, proof drafts, diagrams, reviews, and repairs. This division of labor reflects the actual development of the manuscript: the author did not begin as an expert in PDE, and the argument emerged through repeated attempts to make every step explicit enough for him to question, organize, and check. The author also abstracted this process into a replicable protocol for decomposing difficult problems into sequences of standard moves drawn from the literature. The protocol is designed to use the models' strengths in searching, reformulating, and iterating over possible arguments while mitigating recurrent failures in hypothesis tracking, global consistency, compactness arguments, and the verification of individual steps.

References

[1] G. Duran Ballester, *Residual Type I Seregin Profiles for the Navier–Stokes Equations: Decomposition and Exclusion*, companion manuscript, 2026.

[2] D. Albritton and T. Barker, *On local Type I singularities of the Navier–Stokes equations and Liouville theorems*, J. Math. Fluid Mech. **21** (2019), Article 43.

[3] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Comm. Pure Appl. Math. **35** (1982), no. 6, 771–831, doi: `10.1002/cpa.3160350604`.

[4] L. Escauriaza, G. A. Seregin, and V. Šverak, *$L_{3,\infty}$-solutions of the Navier–Stokes equations and backward uniqueness*, Russian Math. Surveys **58** (2003), no. 2, 211–250, doi: `10.1070/RM2003v058n02ABEH000609`.

[5] Y. Giga, *Solutions for semilinear parabolic equations in L^p and regularity of weak solutions of the Navier–Stokes system*, J. Differential Equations **62** (1986), no. 2, 186–212, doi: `10.1016/0022-0396(86)90096-3`.

[6] S. Gustafson, K. Kang, and T.-P. Tsai, *Interior regularity criteria for suitable weak solutions of the Navier–Stokes equations*, Comm. Math. Phys. **273** (2007), no. 1, 161–176, doi: `10.1007/s00220-007-0214-6`.

[7] J. Leray, *Sur le mouvement d'un liquide visqueux emplissant l'espace*, Acta Math. **63** (1934), no. 1, 193–248, doi: `10.1007/BF02547354`.

[8] F.-H. Lin, *A new proof of the Caffarelli–Kohn–Nirenberg theorem*, Comm. Pure Appl. Math. **51** (1998), no. 3, 241–257, doi: `10.1002/(SICI)1097-0312(199803)51:3<241::AID-CPA2>3.0.CO;2-A`.

[9] V. Scheffer, *Partial regularity of solutions to the Navier–Stokes equations*, Pacific J. Math. **66** (1976), no. 2, 535–552, doi: `10.2140/pjm.1976.66.535`.

[10] G. Seregin, *A certain necessary condition of potential blow up for Navier–Stokes equations*, Comm. Math. Phys. **312** (2012), no. 3, 833–845, doi: `10.1007/s00220-011-1391-x`.

[11] J. Serrin, *On the interior regularity of weak solutions of the Navier–Stokes equations*, Arch. Rational Mech. Anal. **9** (1962), 187–195, doi: `10.1007/BF00253344`.

[12] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton Mathematical Series, No. 30, Princeton University Press, 1970.

[13] J.-P. Aubin, *Un theoreme de compacite*, C. R. Acad. Sci. Paris **256** (1963), 5042–5044.

[14] T. Kato, *Strong L^p-solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions*, Math. Z. **187** (1984), no. 4, 471–480, doi:`https://doi.org/10.1007/BF01174182`.

[15] G. Koch, N. Nadirashvili, G. Seregin, and V. Šverak, *Liouville theorems for the Navier–Stokes equations and applications*, Acta Math. **203** (2009), no. 1, 83–105.

[16] J.-L. Lions and E. Magenes, *Non-homogeneous boundary value problems and applications. Vol. I*, Springer-Verlag, 1972.

[17] J. Necas, M. Ruzicka, and V. Šverak, *On Leray's self-similar solutions of the Navier–Stokes equations*, Acta Math. **176** (1996), no. 2, 283–294.

[18] J. Simon, *Compact sets in the space $L^p(0,T;B)$*, Ann. Mat. Pura Appl. (4) **146** (1987), 65–96, doi:`https://doi.org/10.1007/BF01762360`.

[19] H. Sohr, *The Navier–Stokes equations. An elementary functional analytic approach*, Birkhauser, 2001.

[20] P. Billingsley, *Convergence of Probability Measures*, second ed., Wiley, New York, 1999.

[21] E. B. Fabes, B. F. Jones, and N. M. Riviere, *The initial value problem for the Navier–Stokes equations with data in L^p*, Arch. Rational Mech. Anal. 45 (1972), 222–240.

[22] Y. Giga and T. Miyakawa, *Solutions in L_r of the Navier–Stokes initial value problem*, Arch. Rational Mech. Anal. 89 (1985), 267–281.

[23] N. Kryloff and N. Bogoliouboff, *La theorie generale de la mesure dans son application a l'etude des systemes dynamiques de la mecanique non lineaire*, Ann. of Math. (2) 38 (1937), 65–113.

[24] O. A. Ladyzhenskaya, V. A. Solonnikov, and N. N. Ural'tseva, *Linear and Quasilinear Equations of Parabolic Type*, American Mathematical Society, Providence, 1968.

[25] T. Gallay and C. E. Wayne, *Long-time asymptotics of the Navier–Stokes and vorticity equations on $\mathbb{R}^3$*, Philos. Trans. Roy. Soc. London Ser. A **360** (2002), no. 1799, 2155–2188, doi:10.1098/rsta.2002.1068.

[26] Z. Lei, X. Ren, and Q. S. Zhang, *A Liouville theorem for axially symmetric Navier–Stokes equations on $\mathbb{R}^2 \times \mathbb{T}^1$*, Math. Ann. **383** (2022), no. 1–2, 415–431, doi:10.1007/s00208-020-02128-9.

[27] Z. Lei, Q. Zhang, and N. Zhao, *Improved Liouville theorems for axially symmetric Navier–Stokes equations*, Sci. Sin. Math. **47** (2017), no. 10, 1183–1198, doi:10.1360/N012016-00149.

Email address: guillem@fragile.tech